Information-Directed Selection for Top-Two Algorithms

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Abstract

We consider the best-k-arm identification problem for multi-armed bandits, where the objective is to select the *exact* set of k arms with the highest mean rewards by sequentially allocating measurement effort. We characterize the necessary and sufficient conditions for the optimal allocation using dual variables. Remarkably these optimality conditions lead to the extension of *top-two algorithm* design principle (Russo, 2020), initially proposed for best-arm identification. Furthermore, our optimality conditions induce a simple and effective selection rule dubbed *information-directed selection* (IDS) that selects one of the top-two candidates based on a measure of information gain. As a theoretical guarantee, we prove that integrated with IDS, *top-two Thompson sampling* is (asymptotically) optimal for Gaussian best-arm identification, solving a glaring open problem in the pure exploration literature (Russo, 2020). As a by-product, we show that for $k > 1$, top-two algorithms cannot achieve optimality even when the algorithm has access to the unknown "optimal" tuning parameter. Numerical experiments show the superior performance of the proposed top-two algorithms with IDS and considerable improvement compared with algorithms without adaptive selection.

Keywords: multi-armed bandits, best-k-arm identification, pure exploration, top-two algorithms

1 Introduction

This paper studies the best-k-arm identification problem in a stochastic multi-arm bandit with K arms. The goal is to identify precisely the set of k arms with the largest mean by sequentially allocating measurement effort. It serves as a fundamental model with rich applications. For example, in the COVID-19 vaccine race, the goal is to identify a set of best-performing vaccines as fast and accurately as possible and push them to mass production. The best-performing alternatives are determined based on costly sampling in this type of application. Algorithms that require fewer samples to reach the desired confidence level or accuracy guarantee are invaluable.

We focus on the simple top-two algorithm design principle (Russo, 2020), originally mainly proposed for the best-arm identification (BAI) problem with $k = 1$. A top-two algorithm follows a two-step procedure to measure an arm at each time step. It first identifies a pair of top-two candidates, usually dubbed as the *leader* and the *challenger*, and then flips a potentially biased coin to decide which of the candidates to sample. We refer to the second step as the *selection step*. Various top-two algorithms have been proposed for BAI, including top-two Thompson sampling (TTTS) (Russo, 2020), top-two expected improvement (TTEI) (Qin et al., 2017) and top-two transportation cost (T3C) (Shang et al., 2020). The literature on top-two algorithms predominately focuses on studying the first step of determining the top-two candidates, whereas the second step of selecting among the top-two candidates is primarily simplified. Existing top-two algorithms mostly require a tuning parameter β and sample the leader with a fixed probability β. Despite enjoying the good empirical performance, top-two algorithms' theoretical analyses are usually tailored to a weaker

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notion called β-optimality, namely, achieving the optimal problem complexity under the additional constraint that the best arm receives a β proportion of the sampling effort. It is worth mentioning that Russo (2020) shows the robustness of choosing $\beta = 1/2$, showing that the problem complexity for a subset of algorithms that asymptotically allocate $1/2$ proportion to the best arm is at most two times (achievable) that of the (optimal) problem complexity. Existing literature on top-two algorithms further proposes adaptive procedures to adjust the tuning parameter β by solving the instance complexity optimization problem with plug-in mean estimators. We refer to this procedure of adjusting β as adaptive β-*tuning*.

Moreover, although we can extend the top-two algorithms (designed for BAI) to tackle best-k-arm identification, we provide an example showing that for $k > 1$, top-two algorithms can fail to achieve the optimality even if the value of β is set to the *unknown* optimal value, let alone with the potential generalization of the aforementioned adaptive β-tuning procedures for $k > 1$. Indeed, we present a structural analysis of best-k-arm identification and show that $k > 1$ is surprisingly much more complicated than $k = 1$. Consequently, the optimality conditions widely used to design BAI algorithms, e.g., those in Garivier and Kaufmann (2016); Chen and Ryzhov (2023), are no longer sufficient for $k > 1$, so how to optimally select among the top-two candidates remains open.

Our contributions. We reformulate KKT conditions of the instance complexity optimization problem to characterize asymptotic optimality. The crucial feature of our approach is the inclusion of *dual variables* in the optimality conditions, which allows us to overcome the challenges due to the much more complex optimality conditions for $k > 1$. Based on the *complementary slackness conditions*, we provide a novel interpretation of top-two design principle, which leads us to extending the existing top-two algorithms and designing new ones. We propose an adaptive selection rule dubbed *information-directed selection* (IDS) that wisely selects among the top-two candidates based on the *stationarity conditions*. We show that integrated with IDS, *top-two Thompson sampling* is optimal for Gaussian BAI, which solves a glaring open problem in Russo (2020). A key feature of IDS is its adaptivity to the proposed top-two candidates. This differs from adaptive β-tuning procedures, which use the same value of β regardless of the proposed candidates, even though it may be updated over time. As a by-product, we show that (surprisingly) top-two algorithms with adaptive β-tuning cannot achieve the notion of β-optimality for $k > 1$. Finally, we demonstrate the superior performance of top-two algorithms with IDS in extensive numerical experiments.

Related work. Various algorithms for BAI have been developed, including Track-and-Stop (Garivier and Kaufmann, 2016) and the aforementioned TTTS (Russo, 2020), TTEI (Qin et al., 2017) and T3C (Shang et al., 2020). Algorithms dedicated to the best-k-arm identification problem includes OCBA-m (Chen et al., 2008), OCBAss (Gao and Chen, 2015), LUCB (Kalyanakrishnan et al., 2012), KL-LUCB (Kaufmann and Kalyanakrishnan, 2013), LUCB++ (Simchowitz et al., 2017), SAR (Bubeck et al., 2013), and unified gap-based exploration (Gabillon et al., 2012).

Some algorithms can be adapted to the best-k-arm identification problem, including gradient-ascent-based (Ménard, 2019), elimination-based (Tirinzoni and Degenne, 2022), and algorithms tailored to linear bandits (Réda et al., 2021a,b). Specifically, Réda et al. (2021a) proposes several statistically efficient Gap-Index Focused Algorithms (GIFA), which indeed share a similar spirit with the top-two design principle. Due to the combinatorial structure, most of the proposed algorithms in Réda et al. (2021a) are not computationally efficient, while we focus instead on computationally efficient top-two algorithms. Degenne and Koolen (2019) and Degenne et al. (2019) consider general pure exploration problems and propose meta-algorithms that require oracle solvers. Specifically, Degenne and Koolen (2019) necessitates an optimal allocation solver, while Degenne et al. (2019) takes a game theoretical view of pure exploration and requires a best response oracle algorithm for the game's players. Wang et al. (2021) proposes Frank-Wolfe-based Sampling (FWS), an algorithm that can be adapted to a wide range of pure exploration problems. Our approach is considerably different in the following aspects: 1. Each step of FWS requires solving a time-consuming linear programming; 2. We utilize the detailed structure of the optimality conditions to design intuitive algorithms, while FWS treats it as a generic maximin problem; and 3. Our algorithms can be easily implemented, while FWS requires characterization of the r-subdifferential subspace, where r is a tuning parameter.

2 Preliminary

Problem formulation. We study the *best-k-arm identification* problem a stochastic multi-armed bandit with $[K] \triangleq \{1, \ldots, K\}$ arms. The goal of a decision maker is to confidently identify among these K arms the best k arms where $k \in [K]$ is fixed and known to her. At each timestep $t \in \mathbb{N}_0 \triangleq \{0, 1, \ldots\}$, the decision maker selects an arm $I_t \in [K]$, and observes a reward Y_{t+1,I_t}. The rewards from arm $i \in [K]$ are independent and identically distributed random variables drawn from a distribution $p(\cdot|\theta_i)$ where $\theta_i \in \mathbb{R}$, and rewards from different arms are mutually independent. Throughout the paper, let $p(\cdot|\theta_i)$ be a one-parameter exponential family, parameterized by its mean θ_i. A problem instance is denoted as $\boldsymbol{\theta} = (\theta_1, \ldots, \theta_K) \in \mathbb{R}^K$, which is *fixed but unknown* to the decision maker and can only be estimated through the so-called *bandit feedback* in the sense that only the rewards associated with the selected arms are observed. In this paper, we focus on problem instances with a *unique* set of the best k arms,

$$\mathcal{I}(\boldsymbol{\theta}) \triangleq \operatorname*{argmax}_{\mathcal{A} \subseteq [K], |\mathcal{A}| = k} \left\{ \sum_{i \in \mathcal{A}} \theta_i \right\}, \tag{1}$$

where $|\mathcal{A}|$ is the cardinality of $\mathcal{A}$. The problem space consisting of all such problem instances is

$$\Theta \triangleq \left\{ \boldsymbol{\theta} \in \mathbb{R}^K \,:\, \min_{i \in \mathcal{I}(\boldsymbol{\theta})} \theta_i > \max_{j \in \mathcal{I}^c(\boldsymbol{\theta})} \theta_j \right\}.$$

We refer to arms in $\mathcal{I}(\boldsymbol{\theta})$ as the top arms and arms in $\mathcal{I}^c(\boldsymbol{\theta}) = [K] \backslash \mathcal{I}(\boldsymbol{\theta})$ as the bottom arms.

Generic top-two algorithm. This paper focuses on top-two algorithms with adaptive selection in the form of Algorithm 1. At each time step, the algorithm first identifies a pair of candidates and then flips a biased coin to decide which candidate to sample.

Algorithm 1 Top-two algorithm

Input: Sampling sub-routine and tuning sub-routine.

1: **for** $t = 0, 1, \ldots$ **do**
2: Run sampling sub-routine to propose top-two candidates $\left(I_t^{(1)}, I_t^{(2)}\right)$.
3: Run selection sub-routine obtain selection probability h_t.
4: Play arm $I_t = \begin{cases} I_t^{(1)} & \text{with probability } h_t, \\ I_t^{(2)} & \text{with probability } 1 - h_t. \end{cases}$
5: Observe reward Y_{t+1,I_t} and update mean parameters.
6: **end for**

3 Optimality conditions for best-k-arm identification

In this section, we provide a structural analysis of the optimal allocation problem related to the problem complexity of the best-k-arm identification problem. We propose KKT-based optimality conditions that directly incorporate the dual variables, which serve as a basis for our unified framework for top-two algorithms bundled with adaptive selection.

Problem complexity. In pure exploration literature, a classical setting is called *fixed-confidence setting* where a confidence level δ is given to the player. Besides a *sampling rule* and a *decision rule*, she needs to specify a *stopping rule* τ_δ to ensure that the probability of the recommendation $\hat{\mathcal{I}}_{\tau_\delta}$ being incorrect is no more than δ and to simultaneously minimize the expected number of samples $\mathbb{E}_{\boldsymbol{\theta}}[\tau_\delta]$. An algorithm is said to be δ-correct if $\mathbb{P}_{\boldsymbol{\theta}}(\tau_\delta < \infty, \hat{\mathcal{I}}_{\tau_\delta} \neq \mathcal{I}(\boldsymbol{\theta})) \leq \delta$ for any problem instance $\boldsymbol{\theta} \in \Theta$. Denote by $\mathcal{A}$ the class of δ-correct algorithms. We define the problem complexity of $\boldsymbol{\theta}$ as

$$\kappa(\boldsymbol{\theta}) \triangleq \inf_{\texttt{Alg} \in \mathcal{A}} \limsup_{\delta \to 0} \frac{\mathbb{E}_{\boldsymbol{\theta}}[\tau_\delta]}{\log(1/\delta)}. \tag{2}$$

Generalized Chernoff information. The key to our analysis is a generalization of Chernoff information, a classical quantity in hypothesis testing (Cover and Thomas, 2006). Let $\mathcal{S}_K \subset \mathbb{R}^K$ denote the probability simplex and let $\boldsymbol{p} \in \mathcal{S}_K$ denote a generic allocation vector, indicating the proportion of samples allocated to each arm. Let $d(\cdot, \cdot)$ denote the Kullback-Leibler (KL) divergence of two reward distributions. The *generalized Chernoff information* measures the strength of evidence of distinguishing θ_i and θ_j, defined as[1]

$$C_{i,j}(\boldsymbol{p}) = C_{i,j}(\boldsymbol{p}; \boldsymbol{\theta}) \triangleq p_i d(\theta_i, \bar{\theta}_{i,j}) + p_j d(\theta_j, \bar{\theta}_{i,j}), \quad \text{where} \quad \bar{\theta}_{i,j} = \bar{\theta}_{i,j}(\boldsymbol{p}) \triangleq \frac{p_i \theta_i + p_j \theta_j}{p_i + p_j}. \tag{3}$$

Given an allocation $\boldsymbol{p} \in \mathcal{S}_K$, the minimal strength of evidence of distinguishing an arm in the top set $\mathcal{I}$ and another arm in the bottom set $\mathcal{I}^c$ can be calculated as

$$\Gamma_{\boldsymbol{\theta}}(\boldsymbol{p}) \triangleq \min_{(i,j) \in \mathcal{I} \times \mathcal{I}^c} C_{i,j}(\boldsymbol{p}).$$

The decision maker aims to optimize this quantity over the possible allocations, i.e.,

$$\Gamma^*_{\boldsymbol{\theta}} \triangleq \max_{\boldsymbol{p} \in \mathcal{S}_K} \Gamma_{\boldsymbol{\theta}}(\boldsymbol{p}). \tag{4}$$

The following result shows that the inverse of this value is exactly the fixed-confidence problem complexity, which is closely related to those studied in Garivier and Kaufmann (2016).

Theorem 1 (Problem complexity)**.** *For best-k-arm identification,* $\kappa(\boldsymbol{\theta}) = (\Gamma^*_{\boldsymbol{\theta}})^{-1}$.

We defer the detailed proof to Appendix D.

3.1 Structure of the optimal solution and common approaches in the literature

To start, we show that the optimal solution to (4) is unique. To the best of our knowledge, this is the first uniqueness result for best-k-arm identification[2]. We postpone the proof to Appendix B.4. Additionally, we report two additional properties of the optimal solution, positivity and monotonicity, in Appendix B.1.

Lemma 2 (Uniqueness)**.** *The optimal solution to* (4) *is unique.*

Next, we show that common approaches for BAI cannot be easily extended to address the best-k-arm identification problem. We collect additional supporting materials in Appendix B.

Design principles based on optimality conditions only dependent on $\boldsymbol{p}$. The optimal allocation problem (4) has been widely studied in BAI. Sufficient conditions for optimality are known in the fixed-confidence setting, e.g., Garivier and Kaufmann (2016, Lemma 4), and in the large deviation rate formulation[3], e.g., Glynn and Juneja (2004, Theorem 1), and Chen and Ryzhov (2023, Lemma 1). These conditions are elegant and popular because (i) they are sufficient and necessary for BAI, (ii) they involve only two types of balancing equations, i.e., balancing allocations among individual sub-optimal arms and balancing between the best arm and the sub-optimal arms; and (iii) they involve only the allocation vector $\boldsymbol{p}$, making it straightforward to track the balancing conditions.

Given their simplicity and popularity, it may be intriguing to consider a direct generalization of these conditions for $k > 1$ and to hope that they remain sufficient, as in Gao and Chen (2015, Theorem 2) and Zhang et al. (2021, Theorem 3). Unfortunately, these conditions fail to remain sufficient.

We now summarize the generalized conditions. The first proposition is an extension of a well-known result for BAI, e.g., Russo (2020, Proposition 8).

[1] When the context is clear, we suppress $\boldsymbol{\theta}$ in the notation.

[2] Degenne and Koolen (2019) shows that it is possible to have non-unique optimal solutions for general pure exploration problems. For example, Degenne and Koolen (2019) and Jedra and Proutiere (2020) imply that for best-arm identification in linear bandits, there may be multiple optimal allocations. Our result indicates that this is impossible for the best-k-arm identification problem in unstructured bandits.

[3] There is a switch of arguments in the KL divergence under the large deviation rate formulation, but the structure of the problem remains the same, and our analysis extends.

Proposition 3 (A necessary condition – information balance)**.** *For general reward distributions, the optimal solution* $\boldsymbol{p}^*$ *to (4) satisfies*

$$\Gamma^*_{\boldsymbol{\theta}} = \min_{j' \in \mathcal{I}^c} C_{i,j'}(\boldsymbol{p}^*) \quad \forall i \in \mathcal{I} \quad \textit{and} \quad \Gamma^*_{\boldsymbol{\theta}} = \min_{i' \in \mathcal{I}} C_{i',j}(\boldsymbol{p}^*) \quad \forall j \in \mathcal{I}^c. \tag{5}$$

The next necessary condition states that the optimal solution must balance the overall allocation to the top-arm and bottom-arm sets. For ease of exposition, we present Proposition 4 for Gaussian bandits, which recovers the observations in Garivier and Kaufmann (2016). We note that our result can also be extended to the general exponential family, which extends the results in Glynn and Juneja (2004); see Appendix B.3.

Proposition 4 (A necessary condition for Gaussian – overall balance)**.** *For Gaussian bandits, the optimal solution* $\boldsymbol{p}^*$ *to (4) satisfies*

$$\sum_{i \in \mathcal{I}} (p^*_i)^2 = \sum_{j \in \mathcal{I}^c} (p^*_j)^2. \tag{6}$$

For BAI (i.e., $k = 1$), $\mathcal{I} = \{I^*\}$, where $I^* = I^*(\boldsymbol{\theta}) \triangleq \operatorname{argmax}_{i \in [K]} \theta_i$ is the best arm, and (5) reduces to $\Gamma^*_{\boldsymbol{\theta}} = C_{I^*,j}(\boldsymbol{p}^*)$ for all $j \neq I^*$. The intuition behind the information balance property is that the optimal allocation should gather *equal evidence* to distinguish any sub-optimal arms from the best arm.

However, the balancing of evidence becomes significantly more complex when $k > 1$. The information balance (5) suggests that the optimal allocation should balance the statistical evidence gathered to distinguish any top arm i from the most confusing bottom arm

$$j_i \triangleq \operatorname*{argmin}_{j' \in \mathcal{I}^c} C_{i,j'}(\boldsymbol{p}^*),$$

and similarly, any bottom arm from the most confusing top arm. But, it is a priori unclear which pairs of (i, j) are the most confusing under the optimal allocation; see Appendix C.

Indeed, these simplistic balancing conditions fail to be sufficient when $k > 1$; see Remark 5 and Example 4. Our result indicates that algorithms that rely only on overall balance is highly likely to fail. We also discuss the complete characterization of the necessary and sufficient condition (that depends only on $\boldsymbol{p}$) for general reward distributions in Appendix C. Our analysis shows that to obtain a sufficient condition, the overall balance property (6) needs to be decomposed into finer balancing conditions among subgroups of arms (Propositions 29 and 30). Unfortunately, the optimal decomposition relies on the true $\boldsymbol{\theta}$, which is unavailable. Due to the *combinatorial* number of scenarios, we caution that these conditions are too complicated to yield any meaningful algorithm.

Adaptive adjustment of the tuning parameter in top-two algorithms. An alternative approach to designing an optimal algorithm is to adaptively update the tuning parameter β used in top-two algorithms, e.g., Russo (2020). For BAI, this is relatively straightforward: (1) the optimal tuning parameter β^* equals the optimal allocation to the best arm, which can be solved using standard bisection search as in Garivier and Kaufmann (2016); and (2) the optimal solution is uniquely determined by (5) and $p_{I^*} = \beta^*$.

However, such an approach does not work for the best-k-arm problem. First, β^* is now interpreted as the total allocation to the *set* of best-k-arm[4]. Second, solving the optimal allocation no longer reduces to a one-dimensional optimization. Most importantly, (5) and $\sum_{i \in \mathcal{I}} p_i = \beta^*$ do not guarantee the uniqueness of the solution. To demonstrate the failure of top-two algorithms with a fixed tuning parameter, we present a counterexample in Figure 1; see Example 3 for details. This example suggests that top-two algorithms with adaptive β-tuning (but non-adaptive to the current candidates) are not guaranteed to be optimal even if the tuning parameter β is set to the optimal value $\beta^* = \sum_{i \in \mathcal{I}} p^*_i$.

3.2 Necessary and sufficient conditions via KKT

The major challenge in extending results from $k = 1$ to $k > 1$ lies in the significantly more complex structure in the information balance (5). To address these challenges, we propose including dual variables, which

[4]Assuming sufficient exploration, the top set $\mathcal{I}$ is accurately estimated. Then precisely one candidate will be selected from each of $\mathcal{I}$ and $\mathcal{I}^c$. Hence, β^* is asymptotically the total allocation to the top set $\mathcal{I}$.

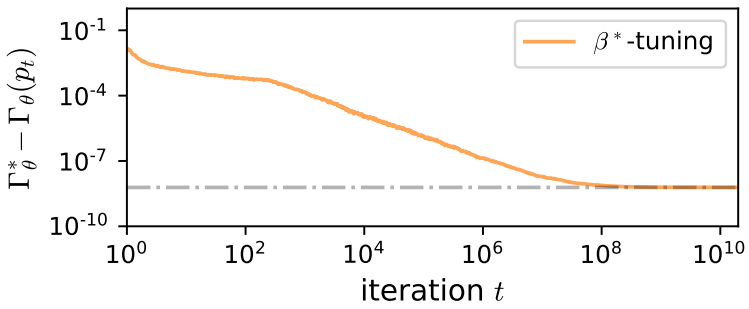


Figure 1: Gaussian bandits with $\boldsymbol{\theta} = (.51, .5, 0, -.01, -.092)$, $\sigma = 0.5$ and $k = 2$. We plot the sample path of $\Gamma_{\boldsymbol{\theta}}(\boldsymbol{p}_t)$, averaged across 100 replications, where $\boldsymbol{p}_t$ is the allocation under TS-KKT(0) (Algorithm 5) with β^*-tuning, i.e., let $h_t = \beta^* \triangleq p_1^* + p_2^*$ for all t. For demonstration, we let the algorithm have access to $\boldsymbol{\theta}$, i.e., we replace empirical/sampled mean vectors by the true $\boldsymbol{\theta}$. Surprisingly, it fails to achieve the optimal $\Gamma^*_{\boldsymbol{\theta}}$.

encode the structure of the optimal information balance. We present a novel necessary and sufficient condition for optimality based on a reformulated Karush–Kuhn–Tucker (KKT) conditions for (4). With a proper interpretation of the dual variables, we will see that our optimality conditions naturally induce top-two algorithms bundled with adaptive selection.

To start, note that the optimal allocation problem (4) can be reformulated as

$$\Gamma^*_{\boldsymbol{\theta}} = \max_{\phi, \boldsymbol{p}} \quad \phi \tag{7a}$$

$$\text{s.t.} \quad \sum_{i \in [K]} p_i - 1 = 0, \tag{7b}$$

$$p_i \geq 0, \quad \forall i \in [K], \tag{7c}$$

$$\phi - C_{i,j}(\boldsymbol{p}) \leq 0, \quad \forall (i,j) \in \mathcal{I} \times \mathcal{I}^c. \tag{7d}$$

It can be verified that (7) is a convex optimization problem. Slater's condition holds since $\boldsymbol{p} = \mathbf{1}/K$ and $\phi = 0$ is a feasible solution such that all inequality constraints hold strictly, where $\mathbf{1}$ is a vector of 1's. Hence, the KKT conditions are necessary and sufficient for global optimality.

Let $\lambda \in \mathbb{R}$, $\boldsymbol{\iota} = \{\iota_i : i \in [K]\}$ and $\boldsymbol{\mu} = \{\mu_{i,j} : i \in \mathcal{I}, j \in \mathcal{I}^c\}$ be the Lagrangian multipliers corresponding to (7b), (7c) and (7d), respectively. Define the following *selection functions*

$$h_{i,j}(\boldsymbol{p}) = h_{i,j}(\boldsymbol{p}; \boldsymbol{\theta}) \triangleq \frac{p_i d(\theta_i, \bar{\theta}_{i,j})}{C_{i,j}(\boldsymbol{p})} \quad \text{and} \quad h_{j,i}(\boldsymbol{p}) = h_{j,i}(\boldsymbol{p}; \boldsymbol{\theta}) \triangleq \frac{p_j d(\theta_j, \bar{\theta}_{i,j})}{C_{i,j}(\boldsymbol{p})}. \tag{8}$$

For Gaussian bandits, we have $h_{i,j}(\boldsymbol{p}) = p_j/(p_i + p_j)$ and $h_{j,i}(\boldsymbol{p}) = p_i/(p_i + p_j)$. The next theorem presents our reformulated KKT conditions, which guide our algorithm design. The key step here is to eliminate the dual variables λ and $\boldsymbol{\iota}$, while leaving $\boldsymbol{\mu}$ as is.

Theorem 5 (Necessary and sufficient conditions)**.** *A feasible solution $(\phi, \boldsymbol{p})$ to (7) is optimal if and only if $\phi = \min_{(i',j') \in \mathcal{I} \times \mathcal{I}^c} C_{i',j'}(\boldsymbol{p})$ and there exist dual variables $\boldsymbol{\mu} \in \mathcal{S}_{k(K-k)}$ such that*

$$p_i = \sum_{j' \in \mathcal{I}^c} \mu_{i,j'} h_{i,j'}(\boldsymbol{p}), \quad \forall i \in \mathcal{I}, \quad p_j = \sum_{i' \in \mathcal{I}} \mu_{i',j} h_{j,i'}(\boldsymbol{p}), \quad \forall j \in \mathcal{I}^c, \quad \textit{and} \tag{9a}$$

$$\mu_{i,j}(\phi - C_{i,j}(\boldsymbol{p})) = 0, \quad \forall (i,j) \in \mathcal{I} \times \mathcal{I}^c. \tag{9b}$$

4 Top-two algorithms and information-directed selection

Note that the KKT conditions consist of two parts, the stationarity conditions (9a) and the complementary slackness conditions (9b). These two sets of conditions serve distinctive roles in algorithm design. In Section 4.1, we discuss the connection between (9b) and top-two algorithms. In Section 4.2, we discuss how (9a) leads to information-directed selection in top-two algorithms.

4.1 Top-two algorithms and complementary slackness conditions

We show that complementary slackness conditions lead to top-two algorithms, including TTTS (Russo, 2020) and T3C (Shang et al., 2020). The key here is interpreting the dual variables $\boldsymbol{\mu}$ as the long-run proportion of time (i, j) are selected as the candidates.

Let $N_{t,i}$ denote the number of samples drawn from arm i before time t and define $p_{t,i} = N_{t,i}/t$. Furthermore, let $\mu_{t,i,j}$ be the proportion of time that (i, j) is selected as candidates before time t. The values of $\boldsymbol{p}_0$ and $\boldsymbol{\mu}_0$ do not matter, and can be set to any values. For top-two algorithms to be optimal, a necessary condition is that $(\boldsymbol{p}_t, \boldsymbol{\mu}_t)$ asymptotically solves (9).

Complementary slackness (9b) states that the optimal solution $(\boldsymbol{p}^*, \boldsymbol{\mu}^*)$ satisfies $\mu^*_{i,j} = 0$ if $C_{i,j}(\boldsymbol{p}^*) > \Gamma^*_{\boldsymbol{\theta}}$. This suggests that, to design an optimal top-two algorithm, we should propose (i, j) as candidates asymptotically if $C_{i,j}(\boldsymbol{p}_t; \boldsymbol{\theta}_t) = \Gamma_{\boldsymbol{\theta}_t}(\boldsymbol{p}_t)$. This is indeed the case for many existing top-two algorithms, as the following example shows.

Example 1 (Connection to T3C and TTTS). *T3C selects the first candidate by Thompson Sampling and then selects the second candidate by searching for the pair with the smallest transportation cost. With sufficient exploration, the first candidate coincides with the best arm with high probability, and their transportation cost is asymptotically equivalent to our $C_{i,j}(\boldsymbol{p})$ here. As a result, the pair of candidates selected by T3C asymptotically satisfies (9b).*

TTTS samples from the posterior distribution of the mean rewards until two distinct best-arm candidates are sampled. One concern for TTTS is that it might take enormous samples until the challenger differs from the leader when the posterior is concentrated. In that case, approximations can be used to avoid repeated sampling. For example, we can sample second arm $I_t^{(2)} = j$ with probability proportional to $\mathbb{P}_{\tilde{\boldsymbol{\theta}} \sim \Pi_t}\left(\tilde{\theta}_j > \tilde{\theta}_{I_t^{(1)}}\right)$, where Π_t is the posterior distribution of $\boldsymbol{\theta}$. This roughly equals sampling with probability proportional to $\exp\left(-tC_{t,I_t^{(1)},j}\right)$ in the asymptotic sense, see Russo (2020, Proposition 5). As $t \to \infty$, this becomes a hard minimum as in (9b).

4.2 Adaptive selection and stationarity conditions

To completely specify a top-two algorithm (Algorithm 1), we need a selection step. Previous top-two algorithms require a non-adaptive tuning parameter β and set $h_t = \beta$. Recall our discussions in Section 3.1 and Example 3, such algorithms are not guaranteed to converge to the optimal allocation even if β is tuned so that it converges to the optimal β^*.

We now discuss information-directed selection based on the stationarity conditions (9a). The key is to interpret the selection functions $h_{i,j}$ as the probability that arm i is played given candidates (i, j). For simplicity of demonstration, we consider a top-two algorithm that guarantees sufficient exploration so that the estimation of the set of the best-k-arm is accurate. We further assume[5] that the sample allocation $\boldsymbol{p}_t$ converges to some $\boldsymbol{p}_\infty$ and $\boldsymbol{\mu}_t$ converges to $\boldsymbol{\mu}_\infty$. We then focus on adaptive selection such that $(\boldsymbol{p}_\infty, \boldsymbol{\mu}_\infty)$ satisfies the stationarity conditions (9a). Recall that $\mu_{t,i,j}$ is the proportion of time that (i, j) is selected as candidates before time t. By the convergence of $(\boldsymbol{p}_t, \boldsymbol{\mu}_t)$, we have $p_{\infty,i} = \sum_{j \in \mathcal{I}^c} \mu_{\infty,i,j} h_{i,j}(\boldsymbol{p}_\infty)$. Similarly, for any arm $j \in \mathcal{I}^c$, we have $p_{\infty,j} = \sum_{i \in \mathcal{I}} \mu_{\infty,i,j} h_{j,i}(\boldsymbol{p}_\infty)$. Hence, the stationarity conditions (9a) hold asymptotically.

Consequently, we propose the information-directed selection for top-two algorithms in Algorithm 2. Here, we explicitly allow $h_{i,j}$ to depend on the mean vector $\boldsymbol{\theta}$ because the decision maker does not know $\boldsymbol{\theta}$ and the selection function must be estimated using an estimate of $\boldsymbol{\theta}$.

Algorithm 2 Information-directed selection

Input: Sample allocation $\boldsymbol{p}_t$, mean estimator $\boldsymbol{\theta}_t$ and candidates $\left(I_t^{(1)}, I_t^{(2)}\right)$.

1: Return $h_t = h_{I_t^{(1)}, I_t^{(2)}}(\boldsymbol{p}_t; \boldsymbol{\theta}_t)$.

[5] For BAI, we only need $\boldsymbol{p}_\infty$ since $\boldsymbol{\mu}_\infty$ can be explicitly expressed by $\boldsymbol{p}_\infty$.

Remark 1 (Information-directed selection)**.** *In interpreting our adaptive selection, we write*

$$C_{i,j}(\boldsymbol{p}) = p_i \frac{\partial C_{i,j}(\boldsymbol{p})}{\partial p_i} + p_j \frac{\partial C_{i,j}(\boldsymbol{p})}{\partial p_j}, \tag{10}$$

which follows from (3) and the fact that $\frac{\partial C_{i,j}(\boldsymbol{p})}{\partial p_i} = d(\theta_i, \bar{\theta}_{i,j})$ *and* $\frac{\partial C_{i,j}(\boldsymbol{p})}{\partial p_j} = d(\theta_j, \bar{\theta}_{i,j})$*. Note that* $C_{i,j}(\boldsymbol{p})$ *measures the amount of information collected per unit sample under allocation vector* $\boldsymbol{p}$ *to assert whether* $\theta_i \geq \theta_j$ *or not. Hence,* $\frac{\partial C_{i,j}(\boldsymbol{p})}{\partial p_i}$ *denotes the information gain per unit allocation to arm* i *and* $p_i \frac{\partial C_{i,j}(\boldsymbol{p})}{\partial p_i}$ *denotes the information gain contributed by allocating* p_i *proportion samples to arm* i*. The selection function* $h_{i,j}$ *suggests that we should select proportional to the information gain when deciding between a pair of candidates. Note that* $h_{i,j}$ *involves unknown oracle quantities. In the implementation, we estimate it using empirical quantities.*

To prevent any confusion, we note that our selection rule used in top-two algorithms (for pure exploration problems) shares the same abbreviation with the algorithm design principle proposed in Russo and Van Roy (2018) (for regret minimization problems), but they are not related.

Remark 2 (Selection versus tuning)**.** *We remark that our information-directed selection is fundamentally different from the standard approach of adaptive* β*-tuning where the same parameter* β *is used regardless of the proposed top-two candidates, even though the value of* β *may be updated over time. Hence, we regard this method as hyperparameter tuning. In contrast, our information-directed selection selects from the proposed candidates using the respective selection function* $h_{I_t^{(1)}, I_t^{(2)}}(\boldsymbol{p}_t; \boldsymbol{\theta}_t)$*. Given different choices of top-two candidates, the selection functions are determined differently. Hence, the selection functions cannot be regarded as the same parameter.*

4.3 The TTTS algorithm for best-k-arm identification

This section presents a TTTS sub-routine for best-k-arm identification in Algorithm 3, whose essential idea has appeared in Russo (2020). Our TTTS algorithm, when applied with IDS (Algorithm 2), follows the essential algorithm design principles from our KKT analysis. As a sanity check of IDS, we prove the fixed-confidence optimality of the proposed algorithm for Gaussian BAI. This solves an open problem in the pure exploration literature, e.g., Section 8 of Russo (2020).

We will propose several Bayesian algorithms for best-k-arm identification that start with a prior distribution Π_0 with density π_0 over a set of parameters $\tilde{\Theta}$ that contains $\boldsymbol{\theta}$. Conditional on the history $\mathcal{H}_t \triangleq \sigma\left(I_0, Y_{1,I_0}, \ldots, I_{t-1}, Y_{t,I_{t-1}}\right)$, the posterior distribution Π_t has density

$$\pi_t(\boldsymbol{\vartheta}) = \frac{\pi_0(\boldsymbol{\vartheta}) \prod_{\ell=0}^{t-1} p(Y_{\ell+1,I_\ell} | \vartheta_{I_\ell})}{\int_{\tilde{\Theta}} \pi_0(\tilde{\boldsymbol{\vartheta}}) \prod_{\ell=0}^{t-1} p(Y_{\ell+1,I_\ell} | \tilde{\vartheta}_{I_\ell}) \mathrm{d}\tilde{\boldsymbol{\vartheta}}}, \quad \forall \boldsymbol{\vartheta} = (\vartheta_1, \ldots, \vartheta_K) \in \tilde{\Theta}.$$

At each time step, our TTTS sub-routine repeatedly samples $\tilde{\boldsymbol{\theta}}$ from the posterior distribution Π_t until two distinct empirical best-k-arm sets appear. The top-two candidates are then selected from the symmetric difference of the two distinct best-k arm sets, breaking tie arbitrarily. By the same argument in Remark 1, TTTS asymptotically follows from the complementary slackness condition (9b), i.e., (i, j) is selected as the candidates if and only if the empirical estimation $C_{i,j}(\boldsymbol{p}_t; \boldsymbol{\theta}_t)$ is the smallest in the asymptotic sense as $t \to \infty$.

Algorithm 3 TTTS (for best-k-arm identification) sub-routine

Input: Posterior distribution Π_t.

1: Sample $\tilde{\boldsymbol{\theta}} \sim \Pi_t$ and define $\mathcal{I}_t = \mathcal{I}(\tilde{\boldsymbol{\theta}})$

2: Repeatedly sample $\tilde{\boldsymbol{\theta}}' \sim \Pi_t$ until $\mathcal{I}_t' = \mathcal{I}(\tilde{\boldsymbol{\theta}}')$ differs from $\mathcal{I}_t$

3: Return any $\left(I_t^{(1)}, I_t^{(2)}\right) \in (\mathcal{I}_t \backslash \mathcal{I}_t') \times (\mathcal{I}_t' \backslash \mathcal{I}_t)$.

Combining the TTTS sub-routine (Algorithm 3) and the information-directed selection sub-routine (Algorithm 2), we have the TTTS-IDS algorithm.

Stopping and recommendation rule. For notational convenience, we define $C_{t,i,j} \triangleq C_{i,j}(\boldsymbol{p}_t; \boldsymbol{\theta}_t)$ for any $(i,j) \in [K] \times [K]$. As shown Garivier and Kaufmann (2016), the Chernoff stopping rule is given by

$$\tau_\delta = \inf \left\{ t : Z_t \triangleq \min_{(i,j) \in \hat{\mathcal{I}}_t \times \hat{\mathcal{I}}_t^c} t \cdot C_{t,i,j} > \gamma(t, \delta) \right\}, \tag{11}$$

where $\hat{\mathcal{I}}_t = \mathcal{I}(\boldsymbol{\theta}_t)$ and $\gamma(t, \delta)$ is a threshold determined by the decision maker. Upon stopping, we recommend $\hat{\mathcal{I}}_{\tau_\delta} = \mathcal{I}(\boldsymbol{\theta}_{\tau_\delta})$. We postpone the discussion of (11) to Appendix F.

Approximate TTTS algorithms. Despite its simplicity, the TTTS algorithm could be computationally inefficient when the posterior concentrates due to the resampling step . Hence, we implement two approximate versions of TTTS called TS-PPS (with PPS referring to Posterior Probability Sampling) and TS-KKT(ρ) (which takes ρ as a hyperparameter).

Here we follow and extend the naming convention in Jourdan et al. (2022) for best-k-arm identification. For an algorithm with name `(top_set)-(candidates)`, `(top_set)` is a sub-routine calculating the estimated top-set and `(candidates)` is a sub-routine proposing the top-two candidates.

Algorithm 4 TS-PPS sub-routine

Input: Posterior distribution Π_t.

1: Sample $\tilde{\boldsymbol{\theta}} \sim \Pi_t$ and define $\mathcal{I}_t = \mathcal{I}(\tilde{\boldsymbol{\theta}})$

2: Sample and return $\left(I_t^{(1)}, I_t^{(2)}\right) = (i,j) \in \mathcal{I}_t \times \mathcal{I}_t^c$ with probability proportional to $\mathbb{P}_{\tilde{\boldsymbol{\theta}} \sim \Pi_t}\left(\tilde{\theta}_j > \tilde{\theta}_i\right)$.

Algorithm 5 TS-KKT(ρ) sub-routine

Input: Posterior distribution Π_t, mean estimator $\boldsymbol{\theta}_t$, and hyperparameter ρ.

1: Sample $\tilde{\boldsymbol{\theta}} \sim \Pi_t$ and define $\mathcal{I}_t = \mathcal{I}(\tilde{\boldsymbol{\theta}})$

2: Return $\left(I_t^{(1)}, I_t^{(2)}\right) \in \arg\min_{(i,j) \in \mathcal{I}_t \times \mathcal{I}_t^c} \left(C_{t,i,j} - \frac{\rho}{t} \log(1/N_{t,i} + 1/N_{t,j})\right)$.

These two algorithms are closely related to TTTS. Essentially they recover the complementary-slackness conditions in the asymptotic sense. Given any $i \in \mathcal{I}_t$, for TTTS to sample a different arm $j \in \mathcal{I}_t^c$ into $\mathcal{I}_t'$, we need $\tilde{\theta}_j > \tilde{\theta}_i$, which occurs with probability $\mathbb{P}_{\tilde{\boldsymbol{\theta}} \sim \Pi_t}\left(\tilde{\theta}_j > \tilde{\theta}_i\right)$. This is the motivation behind TS-PPS. We note that $\mathbb{P}_{\tilde{\boldsymbol{\theta}} \sim \Pi_t}\left(\tilde{\theta}_j > \tilde{\theta}_i\right)$ is asymptotically equivalent to $\exp(-tC_{t,i,j})$ by Russo (2020, Proposition 5). As $t \to \infty$, this becomes a hard minimum in (9b). TS-PPS samples the top-two candidates proportional to posterior probabilities, naturally encouraging exploration. The TS-KKT(ρ) algorithm includes an additional term to encourage measuring under-sampled pairs of candidates to reduce the persistent effect of rare events at early stages. The design of proposing the top-two candidates in TS-KKT(ρ) is inspired by the challenger used in TTEI (Qin et al., 2017), and shares a similar spirit with the Transportation Cost Improved (TCI) challenger proposed in Jourdan et al. (2022).

4.4 Optimality for Gaussian best-arm identification

As a theoretical guarantee, we show that TTTS with IDS achieves the optimality in the fixed-confidence setting for Gaussian BAI.

Theorem 6 (Fixed-confidence optimality)**.** *Consider any problem instance $\boldsymbol{\theta}$ with unique arm means. Let prior distribution be uninformative and independent across arms. Let $\delta \in (0,1)$ and $\alpha > 1$. Using the Chernoff stopping rule (11) with[6] threshold $\gamma(t,\delta) = \log(Ct^\alpha/\delta)$, for TTTS integrated with IDS,*

$$\limsup_{\delta \to 0} \frac{\mathbb{E}_{\boldsymbol{\theta}}[\tau_\delta]}{\log(1/\delta)} \leq \frac{1}{\Gamma_{\boldsymbol{\theta}}^*}.$$

[6]The constant $C = C(\alpha, K)$ is given in Garivier and Kaufmann (2016, Proposition 12).

By Theorems 6 and 1, TTTS with IDS achieves the problem complexity defined in Equation (2). It also implies the optimal rate of posterior convergence studied in Russo (2020).

Remark 3. *The analysis of TTTS might be the most involved among the top-two algorithms since its leader and challenger are both random (given the history). Our proof strategy also works for other top-two algorithms coupled with IDS, e.g., TS-PPS, TS-KKT(ρ) and those in Russo (2020); Qin et al. (2017); Shang et al. (2020); Jourdan et al. (2022).*

5 Numerical experiments

In this section, we numerically validate the convergence to optimal allocation for $k > 1$, demonstrate the advantage of IDS over β-tuning, and present extensive comparisons with existing algorithms[7].

For numerical experiments, we consider the problem instances listed below; some of them are also used in Russo (2020). For cases 1 to 4, we consider Bernoulli bandits. For cases 1 to 6, we consider Gaussian bandits with variance $\sigma^2 = 1$. We adopt an uninformative prior for our algorithms.

ID	K, k	$\boldsymbol{\theta}$	ID	K, k	$\boldsymbol{\theta}$
1	$5, 2$	$(0.1, 0.2, 0.3, 0.4, 0.5)$	4	$50, 25$	$\theta_i = 1, \forall i \in [25]; \theta_i = 0.5, \forall i \in [40]\backslash[25]; \theta_i = 0, \forall i \in [50]\backslash[40]$
2	$15, 1$	$(0.3, \ldots, 0.3, 0.5)$	5	$10, 3$	$(-1.9, -0.6, -0.5, -0.4, -0.3, -0.1, 0, 0.1, 0.4, 1.8)$
3	$100, 5$	$\theta_i = 0.3, \forall i \in [95]; \theta_i = 0.7, \forall i \in [100]\backslash[95]$	6	$10, 5$	$(-1.9, -0.6, -0.5, -0.4, -0.3, -0.1, 0, 0.1, 0.4, 1.8)$

Convergence to the optimal allocation. We numerically validate that $\Gamma_{\boldsymbol{\theta}}(\boldsymbol{p}_t)$ under TS-PPS and TS-KKT(ρ) converges to $\Gamma^*_{\boldsymbol{\theta}}$. Figure 2 reports the optimality gap $\Gamma^*_{\boldsymbol{\theta}} - \Gamma_{\boldsymbol{\theta}}(\boldsymbol{p}_t)$ in logarithm scale under cases 1B, 1G, 3B, and 3G where B and G refer to Bernoulli and Gaussian, respectively.

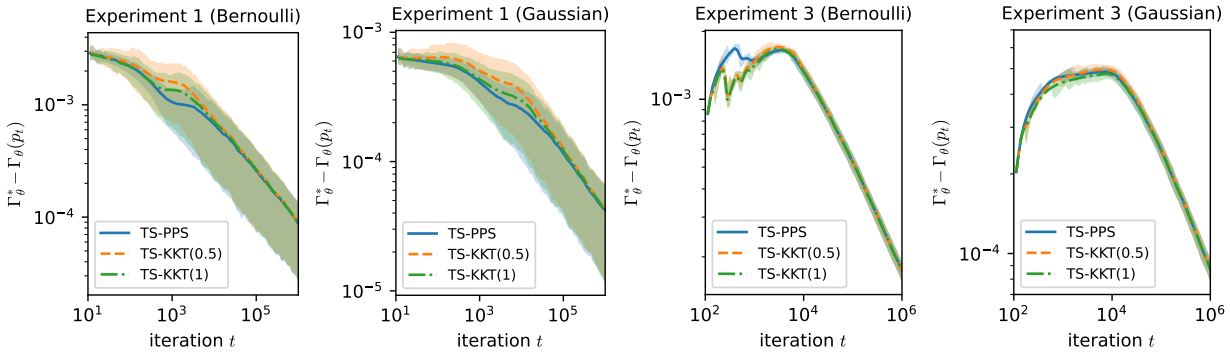


Figure 2: The optimality gap for TS-PPS and TS-KKT(ρ), averaged across 1000 replications. The shaded areas indicate the first and third quartiles.

Advantage of IDS over β-tuning. To isolate the effect of the selection rule, we fix the sampling subroutine to TS-PPS in Algorithm 5 and compare the selection subroutine IDS in Algorithm 2 with 0.5-tuning, i.e., set $\beta = 0.5$. We consider the slippage configuration where the mean $\boldsymbol{\theta}$ is given by $\theta_i = 0.75, i \in [k]$ and $\theta_j = 0.5, j \notin [k]$ for both Gaussian ($\sigma^2 = 1$) and Bernoulli bandits. We label the instances by (K, k)-distribution. For all fixed confidence experiments, we adopt the Chernoff stopping rule (11) with a heuristic threshold $\gamma(t, \delta) = \log\left((\log(t) + 1)/\delta\right)$. Table 1 shows a clear advantage of IDS in terms of fixed-confidence problem complexity.

[7] https://github.com/zihaophys/topk_colt23

(K,k)-dist	$(500,1)$-G	$(500,2)$-G	$(500,3)$-G	$(500,5)$-G	$(500,1)$-B	$(500,2)$-B	$(500,3)$-B	$(500,5)$-B
TS-PPS-IDS	199374	216668	223860	241869	44640	47044	50436	52767
TS-PPS-0.5	285823	285591	283401	283180	62596	61914	62660	62734
Increase	43.4%	31.8%	26.6%	17.0%	40.2%	31.6%	24.2%	18.9%

Table 1: Fixed-confidence problem complexity with $\delta = 0.001$. Averaged over 100 replications.

Fixed-confidence performance. We now compare our proposed algorithms with existing algorithms, which we briefly summarize in Appendix H. We use the m-LinGapE and MisLid algorithms implemented by Tirinzoni and Degenne (2022), which are designed for linear bandits and hence not implemented for Bernoulli bandits. To make fair comparisons, we choose the exploration rate as $\beta(t,\delta) = \log\left((\log(t)+1)/\delta\right)$ in UGapE and KL-LUCB, coupled with their own stopping rules. Table 2 reports the fixed confidence problem complexity, which shows the superior performance of our proposed algorithms. Comparing TS-KKT(0.5) and TS-KKT(1) provides intuitions for selecting the tuning parameter ρ. In particular, the larger the ρ, the more exploration is encouraged. More exploration helps correct the persistent effect of rare samples (e.g., in Example 3G, $\delta = 0.001$) at early stages but will slightly increase the average problem complexity. The FWS algorithm achieves a comparable performance at the cost of a much heavier computational complexity.

Case	δ	TS-KKT(0.5)	TS-KKT(1)	TS-PPS	FWS	m-LinGapE	MisLid	LMA	KL-LUCB	UGapE	Uniform
1 (Bernoulli)	0.1	**786**	816	892	879	–	–	1128	1653	1638	1254
	0.01	**1289**	1301	1398	1359	–	–	1673	2646	2651	2266
	0.001	1881	**1826**	1863	1871	–	–	2209	3594	3586	3324
2 (Bernoulli)	0.1	1146	1237	1410	**1102**	–	–	1895	2161	2199	2437
	0.01	1791	1844	2057	**1735**	–	–	2647	3398	3550	3599
	0.001	2408	2452	2636	**2340**	–	–	3435	4678	5020	4618
3 (Bernoulli)	0.1	**2301**	2401	3106	2380	–	–	6816	3393	4110	8238
	0.01	**3120**	3244	4060	3343	–	–	9046	5365	6615	10688
	0.001	**4108**	4232	5065	4378	–	–	11169	7400	9160	12931
1 (Gaussian)	0.1	**3699**	3747	3928	3763	4394	4395	4864	7547	7641	5885
	0.01	**5772**	6095	6159	5977	6732	6861	7246	11865	11791	9800
	0.001	**8206**	8441	8554	8219	9004	9525	9975	16152	16350	14574
2 (Gaussian)	0.1	**5082**	5359	6008	5406	7337	8471	7614	*12295	*11084	10876
	0.01	**7634**	7901	8604	8028	10422	12407	10922	14548	15404	15367
	0.001	**10257**	10513	11217	10607	13452	15807	14235	19688	21436	19975
3 (Gaussian)	0.1	10671	**10189**	12820	14789	17757	22308	22342	*65185	*58544	35139
	0.01	**13645**	13820	16977	18518	23394	28672	29471	*23694	*28765	45507
	0.001	18654	**17766**	21068	22884	28926	34696	35882	30825	38126	55412
4 (Gaussian)	0.1	**3680**	3786	4735	4911	5419	7594	7105	*14682	*19918	11223
	0.01	**4974**	4981	6265	6321	7156	9998	9598	*9657	*11708	14603
	0.001	**6366**	6530	7844	7563	8739	12096	11860	11502	11852	17753
5 (Gaussian)	0.1	3580	3618	3764	**3546**	4225	4724	6344	7203	7283	11818
	0.01	**5770**	5820	6049	5807	6412	6993	8906	11653	11691	20670
	0.001	7884	8029	8073	**7802**	8506	9582	11372	15692	15840	30608
6 (Gaussian)	0.1	**1267**	1271	1393	1294	1631	1933	1968	2332	2345	3490
	0.01	1943	1988	2114	**1921**	2362	2796	2817	3667	3699	5477
	0.001	**2636**	2707	2813	2652	3119	3590	3663	4950	5022	7830
	(K,k)	Execution time per iteration in milliseconds									
1 (Gaussian)	(5,2)	5.5	5.4	4.7	293.8	13.0	15.8	13.8	2.1	2.2	0.82
6 (Gaussian)	(10,5)	18.0	17.8	13.4	306.1	43.2	41.5	35.5	2.6	3.9	0.99
3 (Gaussian)	(100,5)	311.2	314.0	202.3	13782	4203	3392	1563	5.6	125.7	2.59

Table 2: Fixed-confidence problem complexity, averaged over 1000 replications. We marked with $*$ the cases where the algorithm is not stopped in 10^6 steps for some replications, in which cases the problem complexity is recorded as 10^6.

6 Conclusion

We provide optimality conditions for best-k-arm identification via KKT analysis and show that such conditions naturally lead to top-two algorithms coupled with the proposed information-directed selection rule that determines how to select from top-two candidates. As a theoretical guarantee, we prove that TTTS with IDS is optimal in the fixed-confidence setting for Gaussian BAI, solving an open problem in the pure exploration literature (Russo, 2020). Numerical experiments demonstrate convergence to the optimal allocation under top-two algorithms with IDS for best-k-arm identification, a considerable improvement over algorithms without adaptive selection and leading performance in the fixed-confidence setting. In future works, we plan to establish optimality for the case of $k > 1$. Furthermore, it would be interesting to apply our KKT analysis and algorithm design to broader pure exploration problems in unstructured and structured bandits that satisfy the assumptions in Wang et al. (2021).

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References

Stephen Boyd and Lieven Vandenberghe. *Convex Optimization*. Cambridge University Press, 2004.

Sébastian Bubeck, Tengyao Wang, and Nitin Viswanathan. Multiple identifications in multi-armed bandits. In *International Conference on Machine Learning*, pages 258–265. PMLR, 2013.

Chun-Hung Chen, Jianwu Lin, Enver Yücesan, and Stephen E Chick. Simulation budget allocation for further enhancing the efficiency of ordinal optimization. *Discrete Event Dynamic Systems*, 10:251–270, 2000.

Chun-Hung Chen, Donghai He, Michael Fu, and Loo Hay Lee. Efficient simulation budget allocation for selecting an optimal subset. *INFORMS Journal on Computing*, 20(4):579–595, 2008.

Ye Chen and Ilya O Ryzhov. Balancing optimal large deviations in sequential selection. *Management Science*, 69(6):3457–3473, 2023.

Thomas M. Cover and Joy A. Thomas. *Elements of Information Theory*. Wiley, 2006.

Rémy Degenne and Wouter M. Koolen. Pure exploration with multiple correct answers. *Advances in Neural Information Processing Systems*, 32, 2019.

Rémy Degenne, Wouter M. Koolen, and Pierre Ménard. Non-asymptotic pure exploration by solving games. *Advances in Neural Information Processing Systems*, 32, 2019.

Victor Gabillon, Mohammad Ghavamzadeh, and Alessandro Lazaric. Best arm identification: A unified approach to fixed budget and fixed confidence. *Advances in Neural Information Processing Systems*, 25, 2012.

Siyang Gao and Weiwei Chen. A note on the subset selection for simulation optimization. In *2015 Winter Simulation Conference (WSC)*, pages 3768–3776. IEEE, 2015.

Aurélien Garivier and Emilie Kaufmann. Optimal best arm identification with fixed confidence. In *Conference on Learning Theory*, pages 998–1027. PMLR, 2016.

Peter Glynn and Sandeep Juneja. A large deviations perspective on ordinal optimization. In *Proceedings of the 2004 Winter Simulation Conference*, 2004.

Yassir Jedra and Alexandre Proutiere. Optimal best-arm identification in linear bandits. *Advances in Neural Information Processing Systems*, 33:10007–10017, 2020.

Marc Jourdan, Rémy Degenne, Dorian Baudry, Rianne de Heide, and Emilie Kaufmann. Top two algorithms revisited. *arXiv preprint arXiv:2206.05979*, 2022.

Shivaram Kalyanakrishnan, Ambuj Tewari, Peter Auer, and Peter Stone. PAC subset selection in stochastic multi-armed bandits. In *ICML*, volume 12, pages 655–662, 2012.

Emilie Kaufmann and Shivaram Kalyanakrishnan. Information complexity in bandit subset selection. In *Conference on Learning Theory*, pages 228–251. PMLR, 2013.

Emilie Kaufmann, Olivier Cappé, and Aurélien Garivier. On the complexity of best-arm identification in multi-armed bandit models. *The Journal of Machine Learning Research*, 17(1):1–42, 2016.

Pierre Ménard. Gradient ascent for active exploration in bandit problems. *arXiv preprint arXiv:1905.08165*, 2019.

Chao Qin and Daniel Russo. Electronic companion to adaptivity and confounding in multi-armed bandit experiments. *Posted on SSRN*, 2022.

Chao Qin, Diego Klabjan, and Daniel Russo. Improving the expected improvement algorithm. In *Advances in Neural Information Processing Systems*, volume 30, 2017.

Clémence Réda, Emilie Kaufmann, and Andrée Delahaye-Duriez. Top-m identification for linear bandits. In *International Conference on Artificial Intelligence and Statistics*, pages 1108–1116. PMLR, 2021a.

Clémence Réda, Andrea Tirinzoni, and Rémy Degenne. Dealing with misspecification in fixed-confidence linear top-m identification. *Advances in Neural Information Processing Systems*, 34:25489–25501, 2021b.

Daniel Russo. Simple Bayesian algorithms for best-arm identification. *Operations Research*, 68(6):1625–1647, 2020.

Daniel Russo and Benjamin Van Roy. Learning to optimize via information-directed sampling. *Operations Research*, 66(1):230–252, 2018.

Xuedong Shang, Rianne de Heide, Pierre Menard, Emilie Kaufmann, and Michal Valko. Fixed-confidence guarantees for Bayesian best-arm identification. In *International Conference on Artificial Intelligence and Statistics*, pages 1823–1832. PMLR, 2020.

Max Simchowitz, Kevin Jamieson, and Benjamin Recht. The Simulator: Understanding adaptive sampling in the moderate-confidence regime. In *Conference on Learning Theory*, pages 1794–1834. PMLR, 2017.

Andrea Tirinzoni and Rémy Degenne. On elimination strategies for bandit fixed-confidence identification. In *Advances in Neural Information Processing Systems*, 2022.

Po-An Wang, Ruo-Chun Tzeng, and Alexandre Proutiere. Fast pure exploration via Frank-Wolfe. *Advances in Neural Information Processing Systems*, 34, 2021.

Gongbo Zhang, Yijie Peng, Jianghua Zhang, and Enlu Zhou. Asymptotically optimal sampling policy for selecting top-m alternatives. *arXiv preprint arXiv:2111.15172*, 2021.

Appendix

Table of Contents

A Proof of Theorem 6 **16**
- A.1 Preliminary . . . 16
- A.2 Proof outline . . . 18
- A.3 Empirical overall balance under TTTS with IDS . . . 19
- A.4 Sufficient exploration under TTTS with IDS . . . 23
- A.5 Strong convergence to optimal proportions: completing the proof of Proposition 15 . . . 25

B Supporting results for Section 3.1: structure of the optimal solution for best-k-arm identification **29**
- B.1 Properties of the optimal solution . . . 29
- B.2 Proof of Proposition 3 (A necessary condition – information balance) . . . 29
- B.3 Overall balance in connected subgraphs: additional necessary conditions . . . 30
- B.4 Proofs of properties of the optimal solution in Appendix B.1 . . . 34

C Optimality conditions based only on the allocation vector **37**
- C.1 A (conditional) sufficient condition . . . 37
- C.2 A Case study for 5 choose 2: A pathway to complete characterization of optimality based on the allocation vector . . . 37
- C.3 Proof of Proposition 31 . . . 39

D Proof of Theorem 1: fixed-confidence sample complexity **41**

E Proof of Theorem 5 **43**

F Stopping rule for best-k-arm identification **45**

G Facts about the exponential family and properties of $C_{i,j}(\cdot)$ **46**
- G.1 Properties of $C_{i,j}(\cdot)$ . . . 46

H Existing algorithms **49**

A Proof of Theorem 6

A.1 Preliminary

A.1.1 Notation

For Gaussian BAI with reward distributions $N(\theta_i, \sigma^2)$ for $i \in [K]$, the unique best arm is denoted by $I^* = I^*(\boldsymbol{\theta}) = \mathrm{argmax}_{i\in[K]}\, \theta_i$. The generalized Chernoff information can be calculated as follows,

$$C_{i,j}(\boldsymbol{p}) = \frac{(\theta_i - \theta_j)^2}{2\sigma^2 \left(p_i^{-1} + p_j^{-1}\right)}, \quad \forall j \neq i. \tag{12}$$

and the selection functions specialize to

$$h_{i,j} = h_{i,j}(\boldsymbol{p}) \triangleq \frac{p_i^{-1}}{p_i^{-1} + p_j^{-1}} = \frac{p_j}{p_i + p_j}, \quad \forall j \neq i. \tag{13}$$

The empirical versions of $C_{i,j}(\cdot)$ and $h_{i,j}$ can be obtained by plugging in sample proportion $\boldsymbol{p}_t$ and mean estimators $\boldsymbol{\theta}_t$.

Minimum and maximum gaps. We define the minimum and maximum values between the expected rewards of two different arms:

$$\Delta_{\min} \triangleq \min_{i\neq j} |\theta_i - \theta_j| \quad \text{and} \quad \Delta_{\max} \triangleq \max_{i,j\in[K]} (\theta_i - \theta_j)\,.$$

By the assumption that the arm means are unique, we have $\Delta_{\max} \geq \Delta_{\min} > 0$.

Another measure of cumulative effort and average effort. We have introduced one measure of cumulative effort and average effort: the number of samples and the proportion of total samples allocated allocated to arm $i \in [K]$ before time $t \in \mathbb{N}_0$ are denoted as

$$N_{t,i} = \sum_{\ell=0}^{t-1} \mathbb{1}(I_\ell = i) \quad \text{and} \quad p_{t,i} = \frac{N_{t,i}}{t}.$$

To analyze randomized algorithms (such as TTTS), we introduce the probability of sampling arm i at time t as

$$\psi_{t,i} \triangleq \mathbb{P}(I_t = i | \mathcal{H}_t),$$

where the history $\mathcal{H}_t$ is the sigma-algebra generated by the observations $\left(I_0, Y_{1,I_0}, \ldots, I_{t-1}, Y_{t,I_{t-1}}\right)$ available up to time t. Then we define another measure of cumulative effort and average effort:

$$\Psi_{t,i} \triangleq \sum_{\ell=0}^{t-1} \psi_{\ell,i} \quad \text{and} \quad w_{t,i} \triangleq \frac{\Psi_{t,i}}{t}.$$

Posterior means and variances under independent uninformative priors. Under the independent uninformative prior, the posterior mean and variance of the unknown mean of arm $i \in [K]$ are

$$\theta_{t,i} = \frac{1}{N_{t,i}} \sum_{\ell=0}^{t-1} \mathbb{1}(I_\ell = i) Y_{\ell+1,I_\ell} \quad \text{and} \quad \sigma_{t,i}^2 = \frac{\sigma^2}{N_{t,i}}.$$

A.1.2 Definition and properties of strong convergence

The proof of our Theorem 6 requires bounding time until quantities reaches their limits. Following Qin and Russo (2022), we first introduce a class of random variables that are "light-tailed". Let the space $\mathcal{L}^p$ consist of all measurable X with $\|X\|_p < \infty$ where $\|X\|_p \triangleq (\mathbb{E}[|X|^p])^{1/p}$ is the p-norm.

Definition 1 (Definition 1 in Qin and Russo (2022)). *For a real valued random variable X, we say $X \in \mathbb{M}$ if and only if $\|X\|_p < \infty$ for all $p \geq 1$. Equivalently, $\mathbb{M} = \cap_{p\geq 1}\mathcal{L}_p$.*

Then we introduce the so-called *strong convergence* for random variables, which is a stronger notion of convergence than almost sure convergence.

Definition 2 (Definition 2 in Qin and Russo (2022)). *For a sequence of real valued random variables $\{X_t\}_{t\in\mathbb{N}_0}$ and a scalar $x \in \mathbb{R}$, we say $X_t \xrightarrow{\mathbb{M}} x$ if*

$$\text{for all } \epsilon > 0 \text{ there exists } T \in \mathbb{M} \text{ such that for all } t \geq T,\ |X_t - x| \leq \epsilon.$$

We say $X_t \xrightarrow{\mathbb{M}} \infty$ if

$$\text{for all } c > 0 \text{ there exists } T \in \mathbb{M} \text{ such that for all } t \geq T,\ X_t \geq c,$$

and similarly, we say $X_t \xrightarrow{\mathbb{M}} -\infty$ if $-X_t \xrightarrow{\mathbb{M}} \infty$.
For a sequence of random vectors $\{\boldsymbol{X}_t\}_{t\in\mathbb{N}_0}$ taking values in $\mathbb{R}^d$ and a vector $\boldsymbol{x} \in \mathbb{R}^d$ where $\boldsymbol{X}_t \triangleq (X_{t,1}, \ldots, X_{t,d})$ and $\boldsymbol{x} = (x_1, \ldots, x_d)$, we say $\boldsymbol{X}_t \xrightarrow{\mathbb{M}} \boldsymbol{x}$ if $X_{t,i} \xrightarrow{\mathbb{M}} x_i$ for all $i \in [d]$.

Qin and Russo (2022) also introduces a number of properties of the class $\mathbb{M}$ (Definition 1) and strong convergence (Definition 2).

Lemma 7 (Lemma 1 in Qin and Russo (2022): Closedness of $\mathbb{M}$). *Let X and Y be non-negative random variables in $\mathbb{M}$.*

1. *$aX + bY \in \mathbb{M}$ for any scalars $a, b \in \mathbb{R}$.*
2. *$XY \in \mathbb{M}$.*
3. *$\max\{X, Y\} \in \mathbb{M}$.*
4. *$X^q \in \mathbb{M}$ for any $q \geq 0$.*
5. *If $g : \mathbb{R} \to \mathbb{R}$ satisfies $\sup_{x\in-[c,c]} |g(x)| < \infty$ for all $c \geq 0$ and $|g(x)| = O(|x|^q)$ as $|x| \to \infty$ for some $q \geq 0$, then $g(X) \in \mathbb{M}$.*
6. *If $g : \mathbb{R} \to \mathbb{R}$ is continuous and $|g(x)| = O(|x|^q)$ as $|x| \to \infty$ for some $q \geq 0$, then $g(X) \in \mathbb{M}$.*

As mentioned in Qin and Russo (2022), many of the above properties can be extended to any finite collection of random variables in $\mathbb{M}$. A notable one is that if $X_i \in \mathbb{M}$ for each $i \in [d]$ then

$$X_1 + \cdots + X_d \in \mathbb{M} \quad \text{and} \quad \max\{X_1, \ldots, X_d\} \in \mathbb{M}.$$

Qin and Russo (2022) also shows an equivalence between pointwise convergence and convergence in the maximum norm and a continuous mapping theorem under strong convergence (Definition 2).

Lemma 8 (Lemma 2 in Qin and Russo (2022)). *For a sequence of random vectors $\{X_n\}_{n\in\mathbb{N}}$ taking values in $\mathbb{R}^d$ and a vector $x \in \mathbb{R}^d$, if $X_n \xrightarrow{\mathbb{M}} x$, then $\|X_n - x\|_\infty \xrightarrow{\mathbb{M}} 0$. Equivalently, if $X_{n,i} \xrightarrow{\mathbb{M}} x_i$ for all $i \in [d]$, then for all $\epsilon > 0$, there exists $N \in \mathbb{M}$ such that $n \geq N$ implies $|X_{n,i} - x_i| \leq \epsilon$ for all $i \in [d]$.*

Lemma 9 (Lemma 3 in Qin and Russo (2022): Continuous Mapping). *For any x taking values in a normed vector space and any random sequence $\{X_n\}_{n\in\mathbb{N}}$:*

1. *If g is continuous at x, then $X_n \xrightarrow{\mathbb{M}} x$ implies $g(X_n) \xrightarrow{\mathbb{M}} g(x)$.*
2. *If the range of function g belongs to $\mathbb{R}$ and $g(y) \to \infty$ as $y \to \infty$, then $X_n \xrightarrow{\mathbb{M}} \infty$ implies $g(X_n) \xrightarrow{\mathbb{M}} \infty$.*

A.1.3 Maximal inequalities

Following Qin et al. (2017); Shang et al. (2020), we introduce the following path-dependent random variable to control the impact of observation noises:

$$W_1 = \sup_{(t,i)\in\mathbb{N}_0\times[K]} \sqrt{\frac{N_{t,i}+1}{\log(N_{t,i}+e)}}\frac{|\theta_{t,i}-\theta_i|}{\sigma}. \tag{14}$$

In addition, to control the impact of algorithmic randomness, we introduce the other path-dependent random variable:

$$W_2 = \sup_{(t,i)\in\mathbb{N}_0\times[K]} \frac{|N_{t,i}-\Psi_{t,i}|}{\sqrt{(t+1)\log(t+e^2)}}. \tag{15}$$

The next lemma ensures that these maximal deviations have light tails.

Lemma 10 (Lemma 6 of Qin et al. (2017) and Lemma 4 of Shang et al. (2020))**.** *For any $\lambda > 0$,*

$$\mathbb{E}[e^{\lambda W_1}] < \infty \quad \textit{and} \quad \mathbb{E}[e^{\lambda W_2}] < \infty.$$

The definition of W_2 in Equation (15) and Lemma 10 imply the following corollary.

Corollary 11. *There exists $T \in \mathbb{M}$ such that*[8] *for any $t \geq T$,*

$$N_{t,i} - \Psi_{t,i} \leq W_2 t^{0.6}, \quad \forall i \in [K].$$

This implies

$$p_{t,i} - w_{t,i} \xrightarrow{\mathbb{M}} 0, \quad \forall i \in [K].$$

A.2 Proof outline

Stopping rule and δ-correct guarantee. When $k = 1$, our Chernoff stopping rule specializes to the one used in Garivier and Kaufmann (2016) for BAI. The δ-correct guarantee is provided in Garivier and Kaufmann (2016, Proposition 12).

Proposition 12 (Proposition 12 in Garivier and Kaufmann (2016))**.** *Consider the best-arm identification problem. Let $\delta \in (0,1)$ and $\alpha > 1$. There exists a constant*[9] *$C = C(\alpha, K)$ such that for any sampling rule, using the Chernoff stopping rule (11) with threshold $\gamma(t,\delta) = \log(Ct^\alpha/\delta)$ and the recommendation rule $\hat{\mathcal{I}}_{\tau_\delta} = \mathcal{I}(\boldsymbol{\theta}_{\tau_\delta})$ ensures that for any $\boldsymbol{\theta} \in \Theta$, $\mathbb{P}_{\boldsymbol{\theta}}(\tau_\delta < \infty, \hat{\mathcal{I}}_{\tau_\delta} \neq \mathcal{I}(\boldsymbol{\theta})) \leq \delta$.*

Sufficient conditions for optimality in fixed-confidence setting. Theorem 3 in Qin et al. (2017) shows a sufficient condition of sampling rules that ensures the optimality in fixed-confidence setting.

Theorem 13 (Theorem 3 in Qin et al. (2017): A sufficient condition)**.** *Consider any problem instance $\boldsymbol{\theta} \in \Theta$. Let $\delta \in (0,1)$ and $\alpha > 1$. Using the Chernoff stopping rule (11) with threshold $\gamma(t,\delta) = \log(Ct^\alpha/\delta)$, for any sampling rule,*

$$\boldsymbol{p}_t \xrightarrow{\mathbb{M}} \boldsymbol{p}^* \quad \Longrightarrow \quad \limsup_{\delta\to 0} \frac{\mathbb{E}_{\boldsymbol{\theta}}[\tau_\delta]}{\log(1/\delta)} \leq \frac{1}{\Gamma^*_{\boldsymbol{\theta}}}.$$

Sometimes it is easier to study the following alternative sufficient conditions to prove optimality of sampling rules.

Proposition 14 (Alternative sufficient conditions)**.** *The following conditions are equivalent:*

1. $\boldsymbol{p}_t \xrightarrow{\mathbb{M}} \boldsymbol{p}^*$;
2. $\frac{p_{t,j}}{p_{t,I^*}} \xrightarrow{\mathbb{M}} \frac{p^*_j}{p^*_{I^*}}$ *for any* $j \neq I^*$;

[8] The exponent 0.6 is not essential in the sense that it can be any constant in $(0.5, 1)$.
[9] The constant $C = C(\alpha, K)$ is given in the proof of Garivier and Kaufmann (2016, Proposition 12).

3. $\boldsymbol{w}_t \xrightarrow{\mathbb{M}} \boldsymbol{p}^*$; *and*

4. $\frac{w_{t,j}}{w_{t,I^*}} \xrightarrow{\mathbb{M}} \frac{p_j^*}{p_{I^*}^*}$ *for any* $j \neq I^*$.

Proof. By Corollary 11, conditions 1 and 3 are equivalent.

Now we are going to show conditions 1 and 2 are equivalent. By Lemma 7, condition 1 implies condition 2. On the other hand, applying Lemma 7 to condition 3 gives

$$\frac{1-p_{t,I^*}}{p_{t,I^*}} = \sum_{j\neq I^*} \frac{p_{t,j}}{p_{t,I^*}} \xrightarrow{\mathbb{M}} \sum_{j\neq I^*} \frac{p_j^*}{p_{I^*}^*} = \frac{1-p_{I^*}^*}{p_{I^*}^*},$$

Using Lemma 7 again yields $p_{t,I^*} \xrightarrow{\mathbb{M}} p_{I^*}^*$, and then condition 2 gives $p_{t,j} \xrightarrow{\mathbb{M}} p_j^*$ for any $j \neq I^*$ Hence, condition 2 implies condition 1, and thus they are equivalent. Similarly, conditions 2 and 4 are equivalent. This completes the proof. □

To prove our Theorem 6, we are going to show that TTTS with IDS satisfies the last sufficient condition in Proposition 14:

Proposition 15. *Under TTTS with IDS,*

$$\frac{w_{t,j}}{w_{t,I^*}} \xrightarrow{\mathbb{M}} \frac{p_j^*}{p_{I^*}^*}, \quad \forall j \neq I^*.$$

The rest of this appendix is devoted to proving Proposition 15.

A.3 Empirical overall balance under TTTS with IDS

Previous papers (Qin et al., 2017; Shang et al., 2020; Jourdan et al., 2022) establishes β-optimality by proving either $w_{t,I^*} \xrightarrow{\mathbb{M}} \beta$ or $p_{t,I^*} \xrightarrow{\mathbb{M}} \beta$ where β is a fixed tuning parameter taken as an input to the algorithms, while the optimal $\beta^* =$ (i.e., $p_{I^*}^*$) is unknown a priori. In our information-directed selection rule, at each time step, h_t is adaptive to the top-two candidates, so novel proof techniques are required. In this subsection, we first show that the "empirical" version of overall balance (6) holds.

Proposition 16. *Under TTTS with IDS,*

$$w_{t,I^*}^2 - \sum_{j\neq I^*} w_{t,j}^2 \xrightarrow{\mathbb{M}} 0 \quad \textit{and} \quad p_{t,I^*}^2 - \sum_{j\neq I^*} p_{t,j}^2 \xrightarrow{\mathbb{M}} 0.$$

To prove this proposition, we need the following supporting result.

Lemma 17. *Under TTTS with IDS, for any* $t \in \mathbb{N}_0$,

$$\alpha_{t,I^*}\left(\sum_{\ell\neq I^*} \frac{\alpha_{t,\ell}}{1-\alpha_{t,I^*}} \frac{p_{t,\ell}}{p_{t,I^*}+p_{t,\ell}}\right) \leq \psi_{t,I^*} \leq \alpha_{t,I^*}\left(\sum_{\ell\neq I^*} \frac{\alpha_{t,\ell}}{1-\alpha_{t,I^*}} \frac{p_{t,\ell}}{p_{t,I^*}+p_{t,\ell}}\right) + (1-\alpha_{t,I^*}),$$

and for any $j \neq I^*$,

$$\alpha_{t,I^*} \frac{\alpha_{t,j}}{1-\alpha_{t,I^*}} \frac{p_{t,I^*}}{p_{t,I^*}+p_{t,j}} \leq \psi_{t,j} \leq \alpha_{t,I^*} \frac{\alpha_{t,j}}{1-\alpha_{t,I^*}} \frac{p_{t,I^*}}{p_{t,I^*}+p_{t,j}} + (1-\alpha_{t,I^*}).$$

Proof. Fix $t \in \mathbb{N}_0$. For a fixed $j \in [K]$ and any $\ell \neq j$, TTTS obtains the top-two candidates

$$\left(I_t^{(1)}, I_t^{(2)}\right) = \begin{cases} (j,\ell), & \text{with probability } \alpha_{t,j}\frac{\alpha_{t,\ell}}{1-\alpha_{t,j}}, \\ (\ell,j), & \text{with probability } \alpha_{t,\ell}\frac{\alpha_{t,j}}{1-\alpha_{t,\ell}}, \end{cases}$$

and when TTTS is applied with IDS,

$$\psi_{t,j} = \alpha_{t,j}\left(\sum_{\ell\neq j}\frac{\alpha_{t,\ell}}{1-\alpha_{t,j}}\frac{p_{t,\ell}}{p_{t,j}+p_{t,\ell}}\right) + \sum_{\ell\neq j}\alpha_{t,\ell}\frac{\alpha_{t,j}}{1-\alpha_{t,\ell}}\frac{p_{t,\ell}}{p_{t,j}+p_{t,\ell}}.$$

For $j = I^*$, we have

$$\psi_{t,I^*} = \alpha_{t,I^*}\left(\sum_{\ell\neq I^*}\frac{\alpha_{t,\ell}}{1-\alpha_{t,I^*}}\frac{p_{t,\ell}}{p_{t,I^*}+p_{t,\ell}}\right) + \sum_{\ell\neq I^*}\alpha_{t,\ell}\frac{\alpha_{t,I^*}}{1-\alpha_{t,\ell}}\frac{p_{t,\ell}}{p_{t,I^*}+p_{t,\ell}},$$

so

$$\psi_{t,I^*} \geq \alpha_{t,I^*}\left(\sum_{\ell\neq I^*}\frac{\alpha_{t,\ell}}{1-\alpha_{t,I^*}}\frac{p_{t,\ell}}{p_{t,I^*}+p_{t,\ell}}\right)$$

and

$$\begin{aligned}\psi_{t,I^*} &\leq \alpha_{t,I^*}\left(\sum_{\ell\neq I^*}\frac{\alpha_{t,\ell}}{1-\alpha_{t,I^*}}\frac{p_{t,\ell}}{p_{t,I^*}+p_{t,\ell}}\right) + \sum_{\ell\neq I^*}\alpha_{t,\ell}\\ &= \alpha_{t,I^*}\left(\sum_{\ell\neq I^*}\frac{\alpha_{t,\ell}}{1-\alpha_{t,I^*}}\frac{p_{t,\ell}}{p_{t,I^*}+p_{t,\ell}}\right) + (1-\alpha_{t,I^*}),\end{aligned}$$

where the inequality uses that for any $\ell \neq I^*$, $\frac{\alpha_{t,I^*}}{1-\alpha_{t,\ell}}\frac{p_{t,\ell}}{p_{t,I^*}+p_{t,\ell}} \leq \frac{\alpha_{t,I^*}}{1-\alpha_{t,\ell}} = \frac{\alpha_{t,I^*}}{\sum_{\ell'\neq\ell}\alpha_{t,\ell'}} \leq 1$.

For $j \neq I^*$, we have

$$\begin{aligned}\psi_{t,j} &= \sum_{\ell\neq j}\alpha_{t,\ell}\frac{\alpha_{t,j}}{1-\alpha_{t,\ell}}\frac{p_{t,\ell}}{p_{t,j}+p_{t,\ell}} + \alpha_{t,j}\left(\sum_{\ell\neq j}\frac{\alpha_{t,\ell}}{1-\alpha_{t,j}}\frac{p_{t,\ell}}{p_{t,j}+p_{t,\ell}}\right)\\ &= \alpha_{t,I^*}\frac{\alpha_{t,j}}{1-\alpha_{t,I^*}}\frac{p_{t,I^*}}{p_{t,j}+p_{t,I^*}} + \sum_{\ell\neq j,I^*}\alpha_{t,\ell}\frac{\alpha_{t,j}}{1-\alpha_{t,\ell}}\frac{p_{t,\ell}}{p_{t,\ell}+p_{t,j}} + \alpha_{t,j}\left(\sum_{\ell\neq j}\frac{\alpha_{t,\ell}}{1-\alpha_{t,j}}\frac{p_{t,\ell}}{p_{t,j}+p_{t,\ell}}\right),\end{aligned}$$

so

$$\psi_{t,j} \geq \alpha_{t,I^*}\frac{\alpha_{t,j}}{1-\alpha_{t,I^*}}\frac{p_{t,I^*}}{p_{t,j}+p_{t,I^*}}$$

and

$$\begin{aligned}\psi_{t,j} &\leq \alpha_{t,I^*}\frac{\alpha_{t,j}}{1-\alpha_{t,I^*}}\frac{p_{t,I^*}}{p_{t,j}+p_{t,I^*}} + \left(\sum_{\ell\neq j,I^*}\alpha_{t,\ell}\right) + \alpha_{t,j}\\ &= \alpha_{t,I^*}\frac{\alpha_{t,j}}{1-\alpha_{t,I^*}}\frac{p_{t,I^*}}{p_{t,j}+p_{t,I^*}} + (1-\alpha_{t,I^*}),\end{aligned}$$

where the inequality uses that

1. for any $\ell \neq j, I^*$, $\frac{\alpha_{t,j}}{1-\alpha_{t,\ell}}\frac{p_{t,\ell}}{p_{t,\ell}+p_{t,j}} \leq \frac{\alpha_{t,j}}{1-\alpha_{t,\ell}} = \frac{\alpha_{t,j}}{\sum_{\ell'\neq\ell}\alpha_{t,\ell'}} \leq 1$; and
2. $\sum_{\ell\neq j}\frac{\alpha_{t,\ell}}{1-\alpha_{t,j}}\frac{p_{t,\ell}}{p_{t,j}+p_{t,\ell}} \leq \sum_{\ell\neq j}\frac{\alpha_{t,\ell}}{1-\alpha_{t,j}} = 1$.

This completes the proof. □

Now we are ready to complete the proof of Proposition 16.

Proof of Proposition 16. By Corollary 11, it suffices to only show

$$w_{t,I^*}^2 - \sum_{j\neq I^*} w_{t,j}^2 \xrightarrow{\mathbb{M}} 0.$$

For any $t \in \mathbb{N}_0$, we let

$$G_t \triangleq \Psi_{t,I^*}^2 - \sum_{j\neq I^*} \Psi_{t,j}^2,$$

and we are going to show $\frac{G_t}{t^2} \xrightarrow{\mathbb{M}} 0$.

We argue that it suffices to show that there exists $T \in \mathbb{M}$ such that for any $t \geq T$,

$$|G_{t+1} - G_t| \leq 2W_2 t^{0.6} + 2. \tag{16}$$

The reason is that given (16), for any $t \geq T$

$$\begin{aligned}
|G_t| = \left|G_T + \sum_{\ell=T}^{t-1}(G_{\ell+1} - G_\ell)\right| &\leq |G_T| + \sum_{\ell=T}^{t-1} |G_{\ell+1} - G_\ell| \\
&\leq |G_T| + \sum_{\ell=T}^{t-1} \left(2W_2 \ell^{0.6} + 2\right) \\
&\leq |G_T| + 2(t-T) + 2W_2 \int_{T+1}^{t+1} x^{0.6} \mathrm{d}x \\
&= |G_T| + 2(t-T) + \frac{2W_2}{1.6}\left[(t+1)^{1.6} - (T+1)^{1.6}\right] \\
&\leq T^2 + 2(t-T) + \frac{2W_2}{1.6}\left[(t+1)^{1.6} - (T+1)^{1.6}\right],
\end{aligned}$$

where in the last inequality, $|G_T| \leq T^2$ holds since $G_T \geq -\sum_{j\neq I^*} \Psi_{T,j}^2 \geq -\left(\sum_{j\neq I^*} \Psi_{T,j}\right)^2 \geq -T^2$, and $G_T \leq \Psi_{T,I^*}^2 \leq T^2$. This gives $\frac{|G_t|}{t^2} \xrightarrow{\mathbb{M}} 0$.

Now we are going to prove the sufficient condition (16). We can calculate

$$\begin{aligned}
G_{t+1} - G_t &= \left[\left(\Psi_{t,I^*} + \psi_{t,I^*}\right)^2 - \Psi_{t,I^*}^2\right] - \sum_{j\neq I^*}\left[\left(\Psi_{t,j} + \psi_{t,j}\right)^2 - \Psi_{t,j}^2\right] \\
&= 2\left(\psi_{t,I^*}\Psi_{t,I^*} - \sum_{j\neq I^*} \psi_{t,j}\Psi_{t,j}\right) + \left(\psi_{t,I^*}^2 - \sum_{j\neq I^*} \psi_{t,j}^2\right).
\end{aligned}$$

Then by Lemma 17, we have

$$\begin{aligned}
&G_{t+1} - G_t \\
\geq& 2\left[\alpha_{t,I^*}\sum_{j\neq I^*}\frac{\alpha_{t,j}}{1-\alpha_{t,I^*}}\frac{p_{t,j}\Psi_{t,I^*}}{p_{t,I^*}+p_{t,j}} - \alpha_{t,I^*}\sum_{j\neq I^*}\frac{\alpha_{t,j}}{1-\alpha_{t,I^*}}\frac{p_{t,I^*}\Psi_{t,j}}{p_{t,I^*}+p_{t,j}} - (1-\alpha_{t,I^*})\sum_{j\neq I^*}\Psi_{t,j}\right] - 1 \\
\geq& 2\alpha_{t,I^*}\sum_{j\neq I^*}\frac{\alpha_{t,j}}{1-\alpha_{t,I^*}}\frac{p_{t,j}\Psi_{t,I^*} - p_{t,I^*}\Psi_{t,j}}{p_{t,I^*}+p_{t,j}} - 2(1-\alpha_{t,I^*})t - 1,
\end{aligned}$$

where the last inequality uses $\sum_{j\neq I^*}\Psi_{t,j} \leq t$, and

$$\begin{aligned}
&G_{t+1} - G_t \\
\leq& 2\left[\alpha_{t,I^*}\sum_{j\neq I^*}\frac{\alpha_{t,j}}{1-\alpha_{t,I^*}}\frac{p_{t,j}\Psi_{t,I^*}}{p_{t,I^*}+p_{t,j}} + (1-\alpha_{t,I^*})\Psi_{t,I^*} - \alpha_{t,I^*}\sum_{j\neq I^*}\frac{\alpha_{t,j}}{1-\alpha_{t,I^*}}\frac{p_{t,I^*}\Psi_{t,j}}{p_{t,I^*}+p_{t,j}}\right] + 1 \\
\leq& 2\alpha_{t,I^*}\left(\sum_{j\neq I^*}\frac{\alpha_{t,j}}{1-\alpha_{t,I^*}}\frac{p_{t,j}\Psi_{t,I^*} - p_{t,I^*}\Psi_{t,j}}{p_{t,I^*}+p_{t,j}}\right) + 2(1-\alpha_{t,I^*})t + 1,
\end{aligned}$$

where the last inequality uses $\Psi_{t,I^*} \le t$. Hence,

$$|G_{t+1} - G_t| \le 2\alpha_{t,I^*} \sum_{j \neq I^*} \frac{\alpha_{t,j}}{1-\alpha_{t,I^*}} \left| \frac{p_{t,j}\Psi_{t,I^*} - p_{t,I^*}\Psi_{t,j}}{p_{t,I^*} + p_{t,j}} \right| + 2(1-\alpha_{t,I^*})t + 1.$$

By Lemma 23, there exists $T_1 \in \mathbb{M}$ such that for any $t \ge T_1$, $(1-\alpha_{t,I^*})t \le 1/2$, and thus

$$\begin{aligned}
|G_{t+1} - G_t| &\le 2\alpha_{t,I^*} \sum_{j \neq I^*} \frac{\alpha_{t,j}}{1-\alpha_{t,I^*}} \max_{j \neq I^*} \left| \frac{p_{t,j}\Psi_{t,I^*} - p_{t,I^*}\Psi_{t,j}}{p_{t,I^*} + p_{t,j}} \right| + 2 \\
&\le 2 \max_{j \neq I^*} \left| \frac{p_{t,j}\Psi_{t,I^*} - p_{t,I^*}\Psi_{t,j}}{p_{t,I^*} + p_{t,j}} \right| + 2 \\
&= 2 \max_{j \neq I^*} \left| \frac{N_{t,j}\Psi_{t,I^*} - N_{t,I^*}\Psi_{t,j}}{N_{t,I^*} + N_{t,j}} \right| + 2,
\end{aligned}$$

where the last inequality applies $\alpha_{t,I^*} \le 1$ and $\sum_{j \neq I^*} \frac{\alpha_{t,j}}{1-\alpha_{t,I^*}} = 1$. By Corollary 11, there exists $T_2 \in \mathbb{M}$ such that for any $t \ge T_2$ and $i \in [K]$, $|N_{t,i} - \Psi_{t,i}| \le W_2 t^{0.6}$, and thus for $j \neq I^*$

$$\left| \frac{N_{t,j}\Psi_{t,I^*} - N_{t,I^*}\Psi_{t,j}}{N_{t,I^*} + N_{t,j}} \right| \le \left| \frac{N_{t,j}\left(N_{t,I^*} + W_2 t^{0.6}\right) - N_{t,I^*}\left(N_{t,j} - W_2 t^{0.6}\right)}{N_{t,I^*} + N_{t,j}} \right| \le W_2 t^{0.6}.$$

Hence, for $t \ge T \triangleq \max\{T_1, T_2\}$, $|G_{t+1} - G_t| \le 2W_2 t^{0.6} + 2$. Note that $T \in \mathbb{M}$ by Lemma 7. This completes the proof of the sufficient condition (16). □

A.3.1 Implication of empirical overall balance

Here we present some results that are implied by empirical overall balance (Proposition 16) and needed for proving the sufficient condition for optimality in fixed-confidence setting (Proposition 15). The first one shows that when t is large enough, two measures of average effort are bounded away from 0 and 1.

Lemma 18. *Let* $b_1 \triangleq \frac{1}{\sqrt{32(K-1)}}$ *and* $b_2 \triangleq \frac{3}{4}$. *Under TTTS with IDS, there exists a random time* $T \in \mathbb{M}$ *such that for any* $t \ge T$,

$$b_1 \le w_{t,I^*} \le b_2 \quad \text{and} \quad b_1 \le p_{t,I^*} \le b_2.$$

Since w_{t,I^*} and p_{t,I^*} are bounded away from 0 when t is large enough, we obtain the following corollary by dividing the equations in Proposition 16 by w_{t,I^*} and p_{t,I^*}, respectively,

Corollary 19. *Under TTTS with IDS,*

$$\sum_{j \neq I^*} \frac{w_{t,j}^2}{w_{t,I^*}^2} \xrightarrow{\mathbb{M}} 1 \quad \text{and} \quad \sum_{j \neq I^*} \frac{p_{t,j}^2}{p_{t,I^*}^2} \xrightarrow{\mathbb{M}} 1.$$

Proof of Lemma 18. We only show the upper and lower bounds for w_{t,I^*} since proving those for p_{t,I^*} is the same.

By Proposition 16, there exists $T_1 \in \mathbb{M}$ such that for any $t \ge T_1$,

$$w_{t,I^*}^2 - \sum_{j \neq I^*} w_{t,j}^2 \le \frac{1}{2}.$$

This implies

$$\frac{1}{2} \ge w_{t,I^*}^2 - \left(\sum_{j \neq I^*} w_{t,j} \right)^2 = w_{t,I^*}^2 - (1 - w_{t,I^*})^2 = 2w_{t,I^*} - 1,$$

which gives $w_{t,I^*} \le b_2 \triangleq \frac{3}{4}$.

Using Proposition 16 again, there exists $T_2 \in \mathbb{M}$ such that for any $t \geq T_2$,

$$w_{t,I^*}^2 - \sum_{j \neq I^*} w_{t,j}^2 \geq -\frac{1}{32(K-1)}.$$

By Cauchy-Schwarz inequality,

$$\begin{aligned} w_{t,I^*}^2 &\geq -\frac{1}{32(K-1)} + \frac{\left(\sum_{j \neq I^*} w_{t,j}\right)^2}{K-1} \\ &= -\frac{1}{32(K-1)} + \frac{(1-w_{t,I^*})^2}{K-1} \\ &\geq -\frac{1}{32(K-1)} + \frac{1}{16(K-1)} = \frac{1}{32(K-1)}, \end{aligned}$$

where the last inequality uses $w_{t,I^*} \leq b_2 = \frac{3}{4}$. This gives $w_{t,I^*} \geq b_1 \triangleq \frac{1}{\sqrt{32(K-1)}}$. Taking $T \triangleq \max\{T_1, T_2\}$ completes the proof since $T \in \mathbb{M}$ by Lemma 7. □

A.4 Sufficient exploration under TTTS with IDS

In this subsection, we show that applied with information-directed selection, TTTS sufficiently explores all arms:

Proposition 20. *Under TTTS with IDS, there exists $T \in \mathbb{M}$ such that for any $t \geq T$,*

$$\min_{i \in [K]} N_{t,i} \geq \sqrt{t/K}.$$

The previous analyses in Qin et al. (2017); Shang et al. (2020); Jourdan et al. (2022) only work for any sequence of tuning parameters $\{h_t\}_{t \in \mathbb{N}_0}$ that is *uniformly strictly bounded*, i.e., there exists $h_{\min} > 0$ such that with probability one,

$$\inf_{t \in \mathbb{N}_0} \min\{h_t, 1-h_t\} \geq h_{\min}.$$

Our IDS is adaptive to the algorithmic randomness in TTTS in the sense that it depends on the algorithmic randomness in picking top-two candidates, so it does not enjoy the property above. We need some novel analysis to prove the sufficient exploration of TTTS with IDS.

Following Shang et al. (2020) of analyzing TTTS, we define the following auxiliary arms, the two "most promising arms":

$$J_t^{(1)} \in \operatorname*{argmax}_{i \in [K]} \alpha_{t,i} \quad \text{and} \quad J_t^{(2)} \in \operatorname*{argmax}_{i \neq J_t^{(1)}} \alpha_{t,i}. \tag{17}$$

We further define the one that is less sampled as:

$$J_t \triangleq \operatorname*{argmin}_{i \in \left\{J_t^{(1)}, J_t^{(2)}\right\}} N_{t,i} = \operatorname*{argmin}_{i \in \left\{J_t^{(1)}, J_t^{(2)}\right\}} p_{t,i}.$$

Note that the identity of J_t can change over time. Next we show that IDS allocates a decent amount of effort to J_t.

Lemma 21. *Under TTTS with IDS, for any $t \in \mathbb{N}_0$,*

$$\psi_{t,J_t} \geq \frac{1}{2K(K-1)}.$$

Proof. Fix $t \in \mathbb{N}_0$. We have

$$\left(I_t^{(1)}, I_t^{(2)}\right) = \left(J_t^{(1)}, J_t^{(2)}\right) \quad \text{with probability} \quad \alpha_{t,J_t^{(1)}} \frac{\alpha_{t,J_t^{(2)}}}{1-\alpha_{t,J_t^{(1)}}} \geq \frac{1}{K(K-1)},$$

where the inequality follows from the definition of $J_t^{(1)}$ and $J_t^{(2)}$. Note that

$$h_{J_t^{(1)},J_t^{(2)}}(\boldsymbol{p}_t;\boldsymbol{\theta}_t) = \frac{p_{t,J_t^{(2)}}}{p_{t,J_t^{(1)}} + p_{t,J_t^{(2)}}}.$$

If $J_t = \operatorname{argmin}_{i\in\left\{J_t^{(1)},J_t^{(2)}\right\}} p_{t,i} = J_t^{(1)}$,

$$\psi_{t,J_t} \geq \frac{1}{K(K-1)} \frac{p_{t,J_t^{(2)}}}{p_{t,J_t^{(1)}} + p_{t,J_t^{(2)}}} \geq \frac{1}{2K(K-1)}.$$

Otherwise, $J_t = \operatorname{argmin}_{i\in\left\{J_t^{(1)},J_t^{(2)}\right\}} p_{t,i} = J_t^{(2)}$ and

$$\psi_{t,J_t} \geq \frac{1}{K(K-1)} \frac{p_{t,J_t^{(1)}}}{p_{t,J_t^{(1)}} + p_{t,J_t^{(2)}}} \geq \frac{1}{2K(K-1)}.$$

This completes the proof. □

Following Qin et al. (2017); Shang et al. (2020), we define an insufficiently sampled set for any $t \in \mathbb{N}_0$ and $s \geq 0$,:

$$U_t^s \triangleq \{i \in [K] \,:\, N_{t,i} < s^{1/2}\}.$$

Proposition 20 can be rephrased as that there exists $T \in \mathbb{M}$ such that for all $t \geq T$, $U_t^{t/K}$ is empty.

Lemma 11 in Shang et al. (2020) shows a nice property of TTTS: if TTTS with a selection or tuning rule allocates a decent amount of effort to J_ℓ at any time ℓ, the insufficiently sampled set U_t^s is empty when t is large enough.

Lemma 22 (Lemma 11 in Shang et al. (2020)). [10] *Under TTTS with any selection rule (e.g., IDS) such that*

$$\exists \psi_{\min} > 0 \quad \forall t \in \mathbb{N}_0 \quad \psi_{t,J_t} \geq \psi_{\min},$$

there exists $S \in \mathbb{M}$ such that for any $s \geq S$, $U_{\lfloor Ks \rfloor}^s = \emptyset$.

Now we can complete the proof of Proposition 20.

Proof of Proposition 20. By Lemma 21, IDS satisfies the condition in Lemma 22. Take the corresponding S in Lemma 22 for IDS, and let $T \triangleq KS$. For any $t \geq T$, we let $s = t/K \geq S$, and then by Lemma 22, we have $U_{\lfloor Ks \rfloor}^s = U_t^{t/K}$ is empty. □

A.4.1 Implication of sufficient exploration

Here we present some results that are implied by sufficient exploration (Proposition 20) and needed for proving the sufficient condition for optimality in fixed-confidence setting (Proposition 15). With sufficient exploration, the posterior means strongly converge to the unknown true means, and the probability of any sub-optimal arm being the best decays exponentially.

Lemma 23 (Lemmas 6 and 12 in Shang et al. (2020)). [11] *Under TTTS with any selection rule that ensures sufficient exploration (e.g., IDS), i.e.,*

$$\exists \tilde{T} \in \mathbb{M} \quad \forall t \geq \tilde{T} \quad \min_{i\in[K]} N_{t,i} \geq \sqrt{t/K},$$

the following properties hold:

1. $\boldsymbol{\theta}_t \xrightarrow{\mathbb{M}} \boldsymbol{\theta}$; *and*
2. *there exists $T \in \mathbb{M}$ such that for any $t \geq T$ and $j \neq I^*$,*

$$\alpha_{t,j} \leq \exp\left(-ct^{1/2}\right), \quad \textit{where} \quad c \triangleq \frac{\Delta_{\min}^2}{16\sigma^2 K^{1/2}}.$$

[10] While the original result focuses on β-tuning, the same proof applies to any selection rule that satisfies the condition.
[11] While the original result focuses on β-tuning, the same proof applies to any selection rule that ensures sufficient exploration.

A.5 Strong convergence to optimal proportions: completing the proof of Proposition 15

In this subsection, we are going to complete the proof of Proposition 15, the sufficient condition for optimality for fixed-confidence setting, which requires a sequence of results.

Lemma 24. *Under TTTS with IDS, for any arm $j \neq I^*$,*

$$\frac{C_{t,I^*,j}}{p_{t,I^*} \cdot f_j\left(\frac{p_{t,j}}{p_{t,I^*}}\right)} \xrightarrow{\mathbb{M}} 1, \quad \text{where} \quad f_j(x) \triangleq \frac{(\theta_{I^*} - \theta_j)^2}{2\sigma^2\left(1 + \frac{1}{x}\right)} \quad \text{and} \quad C_{t,I^*,j} = \frac{(\theta_{t,I^*} - \theta_{t,j})^2}{2\sigma^2\left(\frac{1}{p_{t,I^*}} + \frac{1}{p_{t,j}}\right)}.$$

Proof. Fix $j \neq I^*$. We have

$$p_{t,I^*} f_j\left(\frac{p_{t,j}}{p_{t,I^*}}\right) = p_{t,I^*} \frac{(\theta_{I^*} - \theta_j)^2}{2\sigma^2\left(1 + \frac{p_{t,I^*}}{p_{t,j}}\right)} = \frac{(\theta_{I^*} - \theta_j)^2}{2\sigma^2\left(\frac{1}{p_{t,I^*}} + \frac{1}{p_{t,j}}\right)}.$$

By Proposition 20, TTTS with IDS ensures sufficient exploration, and thus by Lemma 23, $\boldsymbol{\theta}_t \xrightarrow{\mathbb{M}} \boldsymbol{\theta}$. Then applying the continuous mapping theorem (Lemma 9), we have

$$\frac{C_{t,I^*,j}}{p_{t,I^*} \cdot f_j\left(\frac{p_{t,j}}{p_{t,I^*}}\right)} \xrightarrow{\mathbb{M}} 1.$$

This completes the proof. □

Lemma 25. *Under TTTS with IDS, for any $\epsilon > 0$, there exists a deterministic constant $c_\epsilon > 0$ and a random time $T_\epsilon \in \mathbb{M}$ such that for any $t \geq T_\epsilon$ and $j \neq I^*$,*

$$\frac{w_{t,j}}{w_{t,I^*}} > \frac{p_j^* + \epsilon}{p_{I^*}^*} \quad \Longrightarrow \quad \psi_{t,j} \leq \exp\left(-c_\epsilon t\right) + (K-1)\exp\left(-ct^{1/2}\right),$$

where c is defined in Lemma 23.

Proof. Fix $\epsilon > 0$. It suffices to show that for any $j \neq I^*$, there exist a deterministic constant $c_\epsilon^{(j)} > 0$ and a random time $T_\epsilon^{(j)} \in \mathbb{M}$ such that for any $t \geq T_\epsilon^{(j)}$,

$$\frac{w_{t,j}}{w_{t,I^*}} > \frac{p_j^* + \epsilon}{p_{I^*}^*} \quad \Longrightarrow \quad \psi_{t,j} \leq \exp\left(-c_\epsilon^{(j)} t\right) + (K-1)\exp\left(-ct^{1/2}\right),$$

since this completes the proof by taking $T_\epsilon \triangleq \max_{j \neq I^*} T_\epsilon^{(j)}$ and $c_\epsilon \triangleq \min_{j \neq I^*} c_\epsilon^{(j)}$.

From now on, we fix $j \neq I^*$. By Corollary 11, there exists $T_{\epsilon,1}^{(j)} \in \mathbb{M}$ such that for any $t \geq T_{\epsilon,1}^{(j)}$,

$$\frac{w_{t,j}}{w_{t,I^*}} = \frac{\Psi_{t,j}}{\Psi_{t,I^*}} > \frac{p_j^* + \epsilon}{p_{I^*}^*} \quad \Longrightarrow \quad \frac{p_{t,j}}{p_{t,I^*}} = \frac{N_{t,j}}{N_{t,I^*}} > \frac{p_j^* + \epsilon/2}{p_{I^*}^*}.$$

Then by Corollary 19, there exists $T_{\epsilon,2}^{(j)} \in \mathbb{M}$ such that for any $t \geq T_{\epsilon,2}^{(j)}$,

$$\frac{p_{t,j}}{p_{t,I^*}} > \frac{p_j^* + \epsilon/2}{p_{I^*}^*} \quad \Longrightarrow \quad \exists A_t \neq I^* \ : \ \frac{p_{t,A_t}}{p_{t,I^*}} \leq \frac{p_{A_t}^*}{p_{I^*}^*}.$$

From now on, we consider $t \geq \max\left\{T_{\epsilon,1}^{(j)}, T_{\epsilon,2}^{(j)}\right\}$ and we have

$$\frac{w_{t,j}}{w_{t,I^*}} = \frac{\Psi_{t,j}}{\Psi_{t,I^*}} > \frac{p_j^* + \epsilon}{p_{I^*}^*} \quad \Longrightarrow \quad \frac{p_{t,j}}{p_{t,I^*}} = \frac{N_{t,j}}{N_{t,I^*}} > \frac{p_j^* + \epsilon/2}{p_{I^*}^*} \quad \text{and} \quad \exists A_t \neq I^* \ : \ \frac{p_{t,A_t}}{p_{t,I^*}} \leq \frac{p_{A_t}^*}{p_{I^*}^*}. \tag{18}$$

By Lemmas 17 and 23,

$$\psi_{t,j} \leq \alpha_{t,I^*} \frac{\alpha_{t,j}}{1-\alpha_{t,I^*}} \frac{p_{t,I^*}}{p_{t,I^*}+p_{t,j}} + (1-\alpha_{t,I^*}) \leq \frac{\alpha_{t,j}}{1-\alpha_{t,I^*}} + (K-1)\exp\left(-ct^{1/2}\right).$$

We can upper bound the numerator $\alpha_{t,j}$ as follows,

$$\begin{aligned}\alpha_{t,j} \leq \mathbb{P}_t\left(\tilde{\theta}_j \geq \tilde{\theta}_{I^*}\right) &= \mathbb{P}\left(\frac{\tilde{\theta}_{I^*} - \tilde{\theta}_j - (\theta_{t,I^*} - \theta_{t,j})}{\sigma\sqrt{1/N_{t,I^*} + 1/N_{t,j}}} \leq \frac{-(\theta_{t,I^*} - \theta_{t,j})}{\sigma\sqrt{1/N_{t,I^*} + 1/N_{t,j}}}\right) \\ &= \Phi\left(-\sqrt{2tC_{t,I^*,j}}\right),\end{aligned}$$

where $\tilde{\boldsymbol{\theta}} = \left(\tilde{\theta}_1, \ldots, \tilde{\theta}_K\right)$ is a sample drawn from the posterior distribution Π_t and $\Phi(\cdot)$ is Gaussian cumulative distribution function. Similarly, we can lower bound the denominator $1-\alpha_{t,I^*}$ as follows,

$$1-\alpha_{t,I^*} = \mathbb{P}_t\left(\exists a \neq I^* : \tilde{\theta}_a \geq \tilde{\theta}_{I^*}\right) \geq \mathbb{P}_t\left(\tilde{\theta}_{A_t} \geq \tilde{\theta}_{I^*}\right) = \Phi\left(-\sqrt{2tC_{t,I^*,A_t}}\right).$$

Hence,

$$\psi_{t,j} \leq \frac{\Phi\left(-\sqrt{2tC_{t,I^*,j}}\right)}{\Phi\left(-\sqrt{2tC_{t,I^*,A_t}}\right)} + (K-1)\exp\left(-ct^{1/2}\right).$$

The remaining task is to show that there exist a deterministic constant $c_\epsilon^{(j)} > 0$ and a random time $T_{\epsilon,3}^{(j)} \in \mathbb{M}$ such that for any $t \geq T_{\epsilon,3}^{(j)}$,

$$\frac{\Phi\left(-\sqrt{2tC_{t,I^*,j}}\right)}{\Phi\left(-\sqrt{2tC_{t,I^*,A_t}}\right)} \leq \exp\left(-c_\epsilon^{(j)} t\right). \tag{19}$$

By Lemma 24, for any $\delta > 0$,

$$\begin{aligned}\frac{\Phi\left(-\sqrt{2tC_{t,I^*,j}}\right)}{\Phi\left(-\sqrt{2tC_{t,I^*,A_t}}\right)} &\leq \frac{\Phi\left(-\sqrt{2tp_{t,I^*}f_j\left(\frac{p_{t,j}}{p_{t,I^*}}\right)(1-\delta)}\right)}{\Phi\left(-\sqrt{2tp_{t,I^*}f_{A_t}\left(\frac{p_{t,A_t}}{p_{t,I^*}}\right)(1+\delta)}\right)} \\ &\leq \frac{\Phi\left(-\sqrt{2tp_{t,I^*}f_j\left(\frac{p_j^*+\epsilon/2}{p_{I^*}^*}\right)(1-\delta)}\right)}{\Phi\left(-\sqrt{2tp_{t,I^*}f_{A_t}\left(\frac{p_{A_t}^*}{p_{I^*}^*}\right)(1+\delta)}\right)} \\ &= \frac{\Phi\left(-\sqrt{2tp_{t,I^*}f_j\left(\frac{p_j^*+\epsilon/2}{p_{I^*}^*}\right)(1-\delta)}\right)}{\Phi\left(-\sqrt{2tp_{t,I^*}f_j\left(\frac{p_j^*}{p_{I^*}^*}\right)(1+\delta)}\right)},\end{aligned}$$

where the second inequality follows from (18) the monotonicity of f_j and f_{A_t} and the final equality uses the information balance (12).

We can pick a sufficiently small δ as a function of ϵ such that

$$c_1 \triangleq f_j\left(\frac{p_j^*+\epsilon/2}{p_{I^*}^*}\right)(1-\delta) > f_j\left(\frac{p_j^*}{p_{I^*}^*}\right)(1+\delta) \triangleq c_2.$$

Then (19) follows from Gaussian tail upper and lower bounds (or $\frac{1}{t}\log\Phi\left(-\sqrt{t}x\right) \to -x^2/2$ as $t \to \infty$) and that p_{t,I^*} is bounded away from 0 and 1 when t is large (Lemma 18). This completes the proof. □

Lemma 26. *Under TTTS with IDS, for any $\epsilon > 0$, there exists $T_\epsilon \in \mathbb{M}$ such that for any $t \geq T_\epsilon$,*

$$\frac{w_{t,j}}{w_{t,I^*}} \leq \frac{p_j^*+\epsilon}{p_{I^*}^*}, \quad \forall j \neq I^*.$$

Proof. Fix $\epsilon > 0$ and $j \neq I^*$. By Lemma 25, there exists $\tilde{T}_\epsilon \in \mathbb{M}$ such that for any $t \geq \tilde{T}_\epsilon$ and $j \neq I^*$,

$$\frac{w_{t,j}}{w_{t,I^*}} > \frac{p_j^* + \epsilon/2}{p_{I^*}^*} \quad \Longrightarrow \quad \psi_{t,j} \leq \exp\left(-c_\epsilon t\right) + (K-1)\exp\left(-ct^{1/2}\right).$$

Define

$$\kappa_\epsilon \triangleq \sum_{\ell=0}^{\infty}\left[\exp\left(-c_\epsilon \ell\right) + (K-1)\exp\left(-c\ell^{1/2}\right)\right] < \infty.$$

From now on, we consider $t \geq \tilde{T}_\epsilon$. We need to consider two cases, and we provide an upper bound on $\frac{w_{t,j}}{w_{t,I^*}}$ for each case.

1. The first case supposes that

$$\forall \ell \in \left\{\tilde{T}_\epsilon, \tilde{T}_\epsilon + 1, \ldots, t-1\right\}: \quad \frac{w_{\ell,j}}{w_{\ell,I^*}} \geq \frac{p_j^* + \epsilon/2}{p_{I^*}^*}.$$

We have

$$\begin{aligned}
\Psi_{t,j} &= \Psi_{\tilde{T}_\epsilon,j} + \sum_{\ell=\tilde{T}_\epsilon}^{t-1} \psi_{\ell,j} \\
&= \Psi_{\tilde{T}_\epsilon,j} + \sum_{\ell=\tilde{T}_\epsilon}^{t-1} \psi_{\ell,j}\mathbf{1}\left(\frac{w_{\ell,j}}{w_{\ell,I^*}} \geq \frac{p_j^* + \epsilon/2}{p_{I^*}^*}\right) \\
&\leq \Psi_{\tilde{T}_\epsilon,j} + \sum_{\ell=\tilde{T}_\epsilon}^{t-1}\left[\exp\left(-c_\epsilon \ell\right) + (K-1)\exp\left(-c\ell^{1/2}\right)\right] \\
&\leq \Psi_{\tilde{T}_\epsilon,j} + \kappa_\epsilon.
\end{aligned}$$

Take $T \in \mathbb{M}$ in Lemma 18, so for any $t \geq T$, $\Psi_{t,I^*} \geq b_1 t$, and thus

$$\frac{w_{t,j}}{w_{t,I^*}} = \frac{\Psi_{t,j}}{\Psi_{t,I^*}} \leq \frac{\Psi_{\tilde{T}_\epsilon,j} + \kappa_\epsilon}{\Psi_{t,I^*}} \leq \frac{\tilde{T}_\epsilon + \kappa_\epsilon}{b_1 t}. \tag{20}$$

2. The alternative case supposes that

$$\exists \ell \in \left\{\tilde{T}_\epsilon, \tilde{T}_\epsilon + 1, \ldots, t-1\right\}: \quad \frac{w_{\ell,j}}{w_{\ell,I^*}} < \frac{p_j^* + \epsilon/2}{p_{I^*}^*}.$$

We define

$$L_t \triangleq \max\left\{\ell \in \left\{\tilde{T}_\epsilon, \tilde{T}_\epsilon + 1, \ldots, t-1\right\} : \frac{w_{\ell,j}}{w_{\ell,I^*}} < \frac{p_j^* + \epsilon/2}{p_{I^*}^*}\right\}.$$

We have

$$\begin{aligned}
\Psi_{t,j} &= \Psi_{L_t,j} + \psi_{L_t,j} + \sum_{\ell=L_t+1}^{t-1} \psi_{\ell,j} \\
&= \Psi_{L_t,j} + \psi_{L_t,j} + \sum_{\ell=L_t+1}^{t-1} \psi_{\ell,j}\mathbf{1}\left(\frac{w_{\ell,j}}{w_{\ell,I^*}} \geq \frac{p_j^* + \epsilon/2}{p_{I^*}^*}\right) \\
&\leq \frac{p_j^* + \epsilon/2}{p_{I^*}^*}\Psi_{L_t,I^*} + 1 + \sum_{\ell=L_t+1}^{t-1}\left[\exp\left(-c_\epsilon \ell\right) + (K-1)\exp\left(-c\ell^{1/2}\right)\right] \\
&\leq \frac{p_j^* + \epsilon/2}{p_{I^*}^*}\Psi_{L_t,I^*} + (\kappa_\epsilon + 1),
\end{aligned}$$

where the first inequality uses $\frac{\Psi_{L_t,j}}{\Psi_{L_t,I^*}} = \frac{w_{L_t,j}}{w_{L_t,I^*}} < \frac{p_j^*+\epsilon/2}{p_{I^*}^*}$. Then by Lemma 18, for any $t \geq T$,

$$\frac{w_{t,j}}{w_{t,I^*}} \leq \frac{p_j^*+\epsilon/2}{p_{I^*}^*}\frac{\Psi_{L_t,I^*}}{\Psi_{t,I^*}} + \frac{1+\kappa_\epsilon}{\Psi_{t,I^*}} \leq \frac{p_j^*+\epsilon/2}{p_{I^*}^*} + \frac{1+\kappa_\epsilon}{b_1 t}. \tag{21}$$

Putting (20) and (21) together, we find that for any $t \geq T_1$,

$$\frac{w_{t,j}}{w_{t,I^*}} = \frac{\Psi_{t,j}}{\Psi_{t,I^*}} \leq \max\left\{\frac{\tilde{T}_\epsilon+\kappa_\epsilon}{b_1 t}, \frac{p_j^*+\epsilon/2}{p_{I^*}^*} + \frac{\kappa_\epsilon+1}{b_1 t}\right\} \leq \frac{p_j^*+\epsilon/2}{p_{I^*}^*} + \frac{\tilde{T}_\epsilon+2\kappa_\epsilon+1}{t}.$$

Then we have

$$t \geq \frac{2\left(\tilde{T}_\epsilon+2\kappa_\epsilon+1\right)}{\epsilon} \quad \Longrightarrow \quad \frac{w_{t,j}}{w_{t,I^*}} \leq \frac{p_j^*+\epsilon}{p_{I^*}^*}.$$

Taking $T_\epsilon \triangleq \max\left\{\tilde{T}_\epsilon, T, \frac{2\left(\tilde{T}_\epsilon+2\kappa_\epsilon+1\right)}{\epsilon}\right\}$ completes the proof since Lemma 7 implies $T_\epsilon \in \mathbb{M}$. □

Now we are ready to complete the proof of Proposition 15.

Proof of Proposition 15. Corollary 19 and Lemma 26 give Proposition 15. □

B Supporting results for Section 3.1: structure of the optimal solution for best-k-arm identification

In this section, we present detailed study of the optimal allocation problem (4). In particular, we characterize the optimality conditions in terms of the allocation vector $\boldsymbol{p}$ *without the presence of the dual variable* $\boldsymbol{\mu}$. This type of conditions is frequently analyzed for BAI, see Garivier and Kaufmann (2016); Chen et al. (2000); Glynn and Juneja (2004); Chen and Ryzhov (2023). Conditions based only on $\boldsymbol{p}$ are attractive because they directly yield intuitive and effective algorithms.

The goal of this section is to show that sufficient and necessary conditions based *only on* $\boldsymbol{p}$ can present complicated structure and generalizing algorithms such as Chen and Ryzhov (2023) is a highly non-trivial task, if not impossible.

The key challenge in solving (4) is that $\Gamma_{\boldsymbol{\theta}}(\boldsymbol{p})$ is a minimum among a set of functions. Hence, there are multiple (i,j) pairs such that $C_{i,j}(\boldsymbol{p}^*)$ achieves the optimal value $\Gamma^*_{\boldsymbol{\theta}}$. Note that by complementary slackness, the dual variable $\mu_{i,j}$ can only be positive for the pairs (i,j) such that $C_{i,j}(\boldsymbol{p}^*)$ achieves the optimal value $\Gamma^*_{\boldsymbol{\theta}}$. Our KKT analysis greatly simplify the characterization of the optimal solution by directly encoding the solution structure in the complementary slackness conditions. However, if necessary and sufficient conditions independent of the dual variable $\boldsymbol{\mu}$ were to be derived, the discussion of the detailed structure cannot be bypassed.

B.1 Properties of the optimal solution

Before discussing the structure results, we present several properties of the optimal solution to (4). These properties generalize those for BAI in Garivier and Kaufmann (2016).

Lemma 2 (Uniqueness)**.** *The optimal solution to* (4) *is unique.*

Lemma 27 (Positivity)**.** *The optimal solution* $\boldsymbol{p}^* > \boldsymbol{0}$.

Lemma 28 (Monotonicity)**.** *The optimal solution* $\boldsymbol{p}^*$ *satisfies the following properties:*

$$\forall i,i' \in \mathcal{I}: \quad \theta_i \geq \theta_{i'} \implies p^*_i \leq p^*_{i'} \quad \textit{and} \quad \forall j,j' \in \mathcal{I}^c: \quad \theta_j \geq \theta_{j'} \implies p^*_j \geq p^*_{j'}.$$

B.2 Proof of Proposition 3 (A necessary condition – information balance)

Note that for any $(i,j) \in \mathcal{I} \times \mathcal{I}^c$, $C_{i,j}(\cdot)$ depends on $\boldsymbol{p}$ only through p_i, p_j. Throughout this section, we write $C_{i,j}(p_i,p_j) = C_{i,j}(\boldsymbol{p})$ with a slight abuse of notation.

Proof. We first prove the second statement of (5), i.e.,

$$\Gamma^*_{\boldsymbol{\theta}} = \min_{i' \in \mathcal{I}} C_{i',j}(\boldsymbol{p}^*), \quad \forall j \in \mathcal{I}^c. \tag{22}$$

We prove by contradiction. Suppose that (22) does not hold. Let $\mathcal{I}^c_{\min} \subset \mathcal{I}^c$ be the set of arms $j \in \mathcal{I}^c$ such that $\min_{i' \in \mathcal{I}} C_{i',j}(\boldsymbol{p}^*) = \Gamma^*_{\boldsymbol{\theta}}$. Then $\mathcal{I}^c_{\min} \neq \mathcal{I}^c$. By the definition of $\Gamma^*_{\boldsymbol{\theta}}$, $\mathcal{I}^c_{\min}$ is non-empty.

Since $C_{i,j}$'s are continuous functions strictly increasing in p_j, there exists a sufficiently small $\epsilon > 0$ such that

$$\min_{i \in \mathcal{I}} C_{i,j}\left(p^*_i, p^*_j - \epsilon/(|\mathcal{I}^c| - |\mathcal{I}^c_{\min}|)\right) > \Gamma^*_{\boldsymbol{\theta}}, \quad \forall j \in \mathcal{I}^c \backslash \mathcal{I}^c_{\min}$$

and

$$\min_{i \in \mathcal{I}} C_{i,j'}(p^*_i, p^*_{j'} + \epsilon/|\mathcal{I}^c_{\min}|) > \min_{i \in \mathcal{I}} C_{i,j'}(p^*_i, p^*_{j'}) = \Gamma^*_{\boldsymbol{\theta}}, \quad \forall j' \in \mathcal{I}^c_{\min}.$$

Hence, $\boldsymbol{p}^*$ cannot be an optimal solution, leading to a contradiction. Consequently, we must have (22).

With similar argument, we can show that

$$\Gamma^*_{\boldsymbol{\theta}} = \min_{j' \in \mathcal{I}^c} C_{i,j}(\boldsymbol{p}^*), \quad \forall i \in \mathcal{I},$$

using the fact that $C_{i,j}$'s are continuous functions strictly increasing in p_i. □

B.3 Overall balance in connected subgraphs: additional necessary conditions

Next, we identify set of necessary condition for optimality based on balancing the allocation between the sets of top arms and bottom arms. For general reward distributions, this condition appears to have an obscure expression. However, the result is significantly simplified for Gaussian bandits as we discuss now.

Recall from Proposition 3 that there are multiple (i,j) pairs such that $C_{i,j}(\boldsymbol{p}^*) = \Gamma^*_{\boldsymbol{\theta}}$. To facilitate discussion of the solution structure, we require the following definition.

Definition 3 (Bipartite graph induced by the optimal solution)**.** *The* bipartite graph *induced by the optimal solution $\boldsymbol{p}^*$ is denoted by $\mathcal{G}(\mathcal{I}, \mathcal{I}^c, \mathcal{E})$ where the set of edges $\mathcal{E}$ contains any $(i,j) \in \mathcal{I} \times \mathcal{I}^c$ such that $C_{i,j}(\boldsymbol{p}^*) = \Gamma^*_{\boldsymbol{\theta}}$.*

Remark 4 (Connected subgraphs)**.** *By Proposition 3, for any arm $i \in \mathcal{I} \cup \mathcal{I}^c$, there must exist an edge $e \in \mathcal{E}$ such that i is incident to e. Hence, the bipartite graph $\mathcal{G}(\mathcal{I}, \mathcal{I}^c, \mathcal{E})$ can be uniquely decomposed into a collection of L connected subgraphs $\{\mathcal{G}_\ell(\mathcal{I}_\ell, \mathcal{I}^c_\ell, \mathcal{E}_\ell)\}_{\ell \in [L]}$ such that*

1. *for any $\ell \in [L]$, $\mathcal{I}_\ell, \mathcal{I}^c_\ell, \mathcal{E}_\ell \neq \varnothing$;*
2. *$\{\mathcal{I}_\ell\}_{\ell \in [L]}$ are mutually exclusive with $\bigcup_{\ell \in [L]} \mathcal{I}_\ell = \mathcal{I}$; $\{\mathcal{I}^c_\ell\}_{\ell \in [L]}$ are mutually exclusive with $\bigcup_{\ell \in [L]} \mathcal{I}^c_\ell = \mathcal{I}^c$; $\{\mathcal{E}_\ell\}_{\ell \in [L]}$ are mutually exclusive with $\bigcup_{\ell \in [L]} \mathcal{E}_\ell = \mathcal{E}$.*

We give the following example 2 to illustrate the concept of the bipartite graph and connected subgraphs.

Example 2. *Consider Gaussian bandits with mean rewards $\boldsymbol{\theta} = \{0.51, 0.5, 0, -0.01, -0.092\}$, variance $\sigma^2 = 1/4$ and $k = 2$. The optimal allocation[12] is*

$$\boldsymbol{p}^* \approx (0.21851674, 0.23713642, 0.21851674, 0.20260818, 0.12322192).$$

We have $\Gamma^*_{\boldsymbol{\theta}} = C_{1,3}(\boldsymbol{p}^*) = C_{2,4}(\boldsymbol{p}^*) = C_{2,5}(\boldsymbol{p}^*) = 0.0568362041$. Hence, there are two connected subgraphs. Note that in each subgraph, the sum-of-squared allocations are balanced, i.e.,

$$(p^*_1)^2 = (p^*_3)^2 \quad \text{and} \quad (p^*_2)^2 = (p^*_4)^2 + (p^*_5)^2.$$

Now, we are ready to present the necessary condition for Gaussian bandits.

Proposition 29 (Overall balance in connected subgraphs – Gaussian bandits)**.** *For Gaussian bandits, each connected subgraph of the bipartite graph induced by the optimal solution balances the sum-of-squared allocations, i.e.,*

$$\sum_{i \in \mathcal{I}_\ell} (p^*_i)^2 = \sum_{j \in \mathcal{I}^c_\ell} (p^*_j)^2, \quad \forall \ell \in [L]. \tag{23}$$

Remark 5 (Proposition 29 implies overall balance in Proposition 4)**.** *Equation (23) is strictly stronger than Equation (6). Example 2 provides such an illustration.*

Example 3 (Failure of β-tuning)**.** *Consider the Gaussian bandits in Example 2. We now show that top-two algorithms with β^*-tuning can fail, where $\beta^* = p^*_1 + p^*_2$. As a counterexample,*

$$\boldsymbol{p}' \approx (0.21843623, 0.23721697, 0.21859726, 0.20254937, 0.12320016)$$

is a feasible solution such that $p'_1 + p'_2 = p^*_1 + p^*_2 \triangleq \beta^*$ but

$$\Gamma_{\boldsymbol{\theta}}(\boldsymbol{p}') = C_{1,3}(\boldsymbol{p}') = C_{1,4}(\boldsymbol{p}') = C_{2,4}(\boldsymbol{p}') = C_{2,5}(\boldsymbol{p}') = 0.056836198 < 0.0568362041 = \Gamma^*_{\boldsymbol{\theta}}.$$

This example show that information balance (5), together with $p_1 + p_2 = \beta^*$ does not guarantee the uniqueness of the solution. More importantly, we show with an numerical example in Figure 3 that two-two algorithms with β-tuning is not guaranteed to be optimal, even if supplied with the optimal β^*. In particular, we consider the Gaussian bandits in Example 2, assuming that the true mean vector is known. Figure 3 reports the sample

[12] The optimal allocation can be obtained by numerically solving the three scenarios listed in Remark 9 and selected the best solution among them.

path averaged across 100 *replications of* $\Gamma_{\boldsymbol{\theta}}(\boldsymbol{p}_t)$ *where* $\boldsymbol{p}_t$ *is the sample allocation under either* TS-KKT+(0)[13] *or an* β^**-tuning version of* TS-KKT+(0)*, i.e., let* $\beta_t = \beta^*$ *with* $\beta^* = p_1^* + p_2^*$*. It shows that the* TS-KKT+(0) *algorithm with* β^**-tuning converges to the solution* $\boldsymbol{p}'$*, which is sub-optimal. We further remark that* $O(1/t)$ *is the best achievable convergence rate, because we are only allowed to draw one arm in each step, effectively fixing the step size to* $1/t$*. The rational behind this counterexample is that the optimal solution induces two*

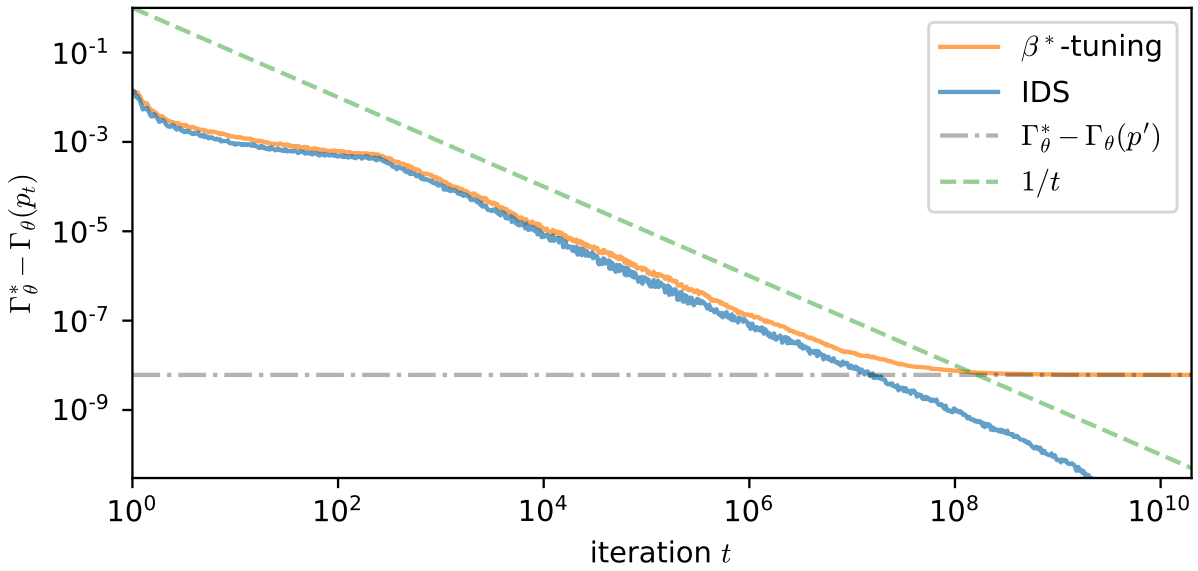


Figure 3: The sample path of $\Gamma_{\boldsymbol{\theta}}(\boldsymbol{p}_t)$ for $\boldsymbol{p}_t$ being the allocation under either TS-KKT+(0) or an β^*-tuning version of TS-KKT+(0), i.e., let $\beta_t \equiv \beta^* = p_1^* + p_2^*$. Averaged across 100 replications. For demonstration, we assume that $\boldsymbol{\theta}$ is known so that both algorithms are effectively deterministic. In this setting without learning, all empirical and sampled quantities are replaced by the true mean $\boldsymbol{\theta}$ and the TS-KKT+(0) algorithm is equivalent to the (best-k-arm version of) EB-TC algorithm in Jourdan et al. (2022).

connected subgraphs, and each subgraph need its dedicated balancing condition. Hence, an overall balance as such induced by β*-tuning (even if set to* β^**) is not sufficient to discern the optimal allocation.*

Example 4. *We point out that* (5) *and* (6) *together do not guarantee the uniqueness of the solution nor the optimality. Even for choosing the best-*2 *among* 4 *arms in a Gaussian bandit, counterexamples are ubiquitous. Consider the Gaussian bandits with mean rewards* $\boldsymbol{\theta} = (1, 0.7, 0, -0.5)$*, variance* $\sigma^2 = 1/4$ *and* $k = 2$*. The solution* $\boldsymbol{p}_1 \approx (0.0482, 0.459, 0.4603, 0.0325)$ *satisfies* (5) *and* (6) *with an objective value of* $\Gamma_{\boldsymbol{\theta}}(\boldsymbol{p}_1) \approx 0.0873$*. For* $\boldsymbol{p}_1$*, we have* $C_{1,3}(\boldsymbol{p}_1) = C_{1,4}(\boldsymbol{p}_1) = C_{2,4}(\boldsymbol{p}_1) < C_{2,3}(\boldsymbol{p}_1)$*. However, the optimal solution* $\boldsymbol{p}^* \approx (0.1277, 0.3894, 0.4016, 0.0813)$*, with an optimal objective value of* $\Gamma_{\boldsymbol{\theta}}^* \approx 0.1938 > \Gamma_{\boldsymbol{\theta}}(\boldsymbol{p}_1)$*. For the optimal solution, we have* $C_{1,3}(\boldsymbol{p}^*) = C_{2,4}(\boldsymbol{p}^*) = C_{2,3}(\boldsymbol{p}^*) < C_{1,4}(\boldsymbol{p}^*)$*.*

Now we formally prove Proposition 29.

Proof of Proposition 29 (Overall balance in connected subgraphs – Gaussian). Consider any connected subgraph $\mathcal{G}_\ell = \mathcal{G}(\mathcal{I}_\ell, \mathcal{I}_\ell^c, \mathcal{E}_\ell)$ of the bipartite graph induced by an optimal solution $\boldsymbol{p}$. We show that if

$$\sum_{i \in \mathcal{I}_\ell} p_i^2 \neq \sum_{j \in \mathcal{I}_\ell^c} p_j^2.$$

then we can find another $\boldsymbol{p}' \neq \boldsymbol{p}$ with higher objective value, and hence $\boldsymbol{p}$ cannot be an optimal solution.

[13] Since $\boldsymbol{\theta}$ is assumed to be known, the empirical best-k is just the true best-k. We focus on achieving the optimal allocation for this demonstration.

By Proposition 3, for any arm $i \in [K]$, there must exist an edge $e \in \mathcal{E}$ such that i is incident to e. Hence, we have $\mathcal{I}_\ell \neq \varnothing$ and $\mathcal{I}_\ell^c \neq \varnothing$. Take any $i_1 \in \mathcal{I}_\ell$ as an anchor point. By the connectedness of $\mathcal{G}_\ell$, for any $j \in \mathcal{I}_\ell^c$, there exist a path $p_{i_1,j} = (i_1, j_1, i_2, j_2, \ldots, j_r = j)$ connecting i_1 and j. Along the path, we perform Taylor's expansion

$$C_{i,j}(\boldsymbol{p} + \boldsymbol{\epsilon}) = C_{i,j}(\boldsymbol{p}) + \frac{\partial C_{i,j}(\boldsymbol{p})}{\partial p_i}\epsilon_i + \frac{\partial C_{i,j}(\boldsymbol{p})}{\partial p_j}\epsilon_j + o(\|\boldsymbol{\epsilon}\|).$$

Starting from the first edge (i_1, j_1), we define $\tilde{\epsilon}_{i_1}$ and $\tilde{\epsilon}_{j_1}$ such that the first order terms in the Taylor's series vanishes

$$\frac{\partial C_{i_1,j_1}(\boldsymbol{p})}{\partial p_{i_1}}\tilde{\epsilon}_{i_1} + \frac{\partial C_{i_1,j_1}(\boldsymbol{p})}{\partial p_{j_1}}\tilde{\epsilon}_{j_1} = 0. \tag{24}$$

Recall that, for Gaussian bandits with reward distribution $N(\theta_i, \sigma^2)$, we have

$$C_{i,j}(\boldsymbol{p}) = \frac{(\theta_i - \theta_j)^2}{2\sigma^2}\frac{p_i p_j}{p_i + p_j}$$

and

$$\frac{\partial C_{i,j}(\boldsymbol{p})}{\partial p_i} = \frac{(\theta_i - \theta_j)^2}{2\sigma^2}\frac{p_j^2}{(p_i + p_j)^2}, \quad \frac{\partial C_{i,j}(\boldsymbol{p})}{\partial p_j} = \frac{(\theta_i - \theta_j)^2}{2\sigma^2}\frac{p_i^2}{(p_i + p_j)^2}.$$

Hence, we can rewrite (24) as

$$\tilde{\epsilon}_{j_1} = -\tilde{\epsilon}_{i_1}\frac{p_{j_1}^2}{p_{i_1}^2}.$$

Repeat this procedure for (j_1, i_2) on the path, we have

$$\tilde{\epsilon}_{i_2} = -\tilde{\epsilon}_{j_1}\frac{p_{i_2}^2}{p_{j_1}^2} = \tilde{\epsilon}_{i_1}\frac{p_{j_1}^2}{p_{i_1}^2}\frac{p_{i_2}^2}{p_{j_1}^2} = \tilde{\epsilon}_{i_1}\frac{p_{i_2}^2}{p_{i_1}^2}.$$

Repeat for all edges on the path, we have

$$\tilde{\epsilon}_j = -\tilde{\epsilon}_{i_1}\frac{p_j^2}{p_{i_1}^2}, \quad j \in \mathcal{I}_\ell^c.$$

Similarly, for all $i \in \mathcal{I}_\ell$, we have

$$\tilde{\epsilon}_i = \tilde{\epsilon}_{i_1}\frac{p_i^2}{p_{i_1}^2}, \quad i \in \mathcal{I}_\ell.$$

Now, we claim that if $\sum_{i\in\mathcal{I}_\ell} p_i^2 \neq \sum_{j\in\mathcal{I}_\ell^c} p_j^2$, then we can find a $\boldsymbol{\epsilon}$ such that $\Gamma_{\boldsymbol{\theta}}(\boldsymbol{p}+\boldsymbol{\epsilon}) > \Gamma_{\boldsymbol{\theta}}(\boldsymbol{p})$. Consequently, $\boldsymbol{p}$ cannot be an optimal solution.

To this end, let

$$\delta \equiv \sum_{i\in\mathcal{I}_\ell}\tilde{\epsilon}_i + \sum_{j\in\mathcal{I}_\ell^c}\tilde{\epsilon}_j = \tilde{\epsilon}_{i_1}\left(\sum_{i\in\mathcal{I}_\ell}\frac{p_i^2}{p_{i_1}^2} - \sum_{j\in\mathcal{I}_\ell^c}\frac{p_j^2}{p_{i_1}^2}\right).$$

We consider two cases.

1. If $\sum_{i\in\mathcal{I}_\ell} p_i^2 > \sum_{j\in\mathcal{I}_\ell^c} p_j^2$, then let $\epsilon_{i_1} < 0$ be sufficiently small such that the first order terms dominates the remainder in the Taylor series. We have $\delta < 0$. Define

$$\epsilon_i = \tilde{\epsilon}_i - \frac{\delta}{|\mathcal{I}_\ell|} > \tilde{\epsilon}_i, \quad i \in \mathcal{I}_\ell,$$
$$\epsilon_j = \tilde{\epsilon}_j, \quad i \in \mathcal{I}_\ell.$$

   Note that $\frac{\partial C_{i,j}(\boldsymbol{p})}{\partial p_i}$ is strictly positive, then the first order term is strictly positive for any (i,j) pairs, and hence $\Gamma_{\boldsymbol{\theta}}(\boldsymbol{p}+\boldsymbol{\epsilon}) > \Gamma_{\boldsymbol{\theta}}(\boldsymbol{p})$.

2. If $\sum_{i\in\mathcal{I}_\ell} p_i^2 < \sum_{j\in\mathcal{I}_\ell^c} p_j^2$, then let $\epsilon_{i_1} > 0$ be sufficiently small such that the first order terms dominates the remainder in the Taylor series. We have $\delta < 0$. Define

$$\epsilon_i = \tilde{\epsilon}_i - \frac{\delta}{|\mathcal{I}_\ell|} > \tilde{\epsilon}_i, \quad i \in \mathcal{I}_\ell,$$
$$\epsilon_j = \tilde{\epsilon}_j, \quad i \in \mathcal{I}_\ell.$$

Then the first order term is strictly positive for any (i, j) pairs, and hence $\Gamma_{\boldsymbol{\theta}}(\boldsymbol{p}+\boldsymbol{\epsilon}) > \Gamma_{\boldsymbol{\theta}}(\boldsymbol{p})$.

Finally, we remark that it is straightforward to generalize the above result to Gaussian bandits with heterogeneous variances. To this end, note that

$$C_{i,j}(\boldsymbol{p}) = \frac{(\theta_i-\theta_j)^2}{2}\frac{p_i p_j}{\sigma_j^2 p_i + \sigma_i^2 p_j}$$

and

$$\frac{\partial C_{i,j}(\boldsymbol{p})}{\partial p_i} = \frac{(\theta_i-\theta_j)^2}{2}\frac{\sigma_i^2 p_j^2}{(\sigma_j^2 p_i + \sigma_i^2 p_j)^2}, \quad \frac{\partial C_{i,j}(\boldsymbol{p})}{\partial p_j} = \frac{(\theta_i-\theta_j)^2}{2}\frac{\sigma_j^2 p_i^2}{(\sigma_j^2 p_i + \sigma_i^2 p_j)^2}.$$

Consequently, we have

$$\frac{\frac{\partial C_{i,j}(\boldsymbol{p})}{\partial p_i}}{\frac{\partial C_{i,j}(\boldsymbol{p})}{\partial p_j}} = \frac{p_j^2/\sigma_j^2}{p_i^2/\sigma_i^2}.$$

The rest follows exactly as the case with equal variances. We omit the details. □

In Proposition 29, we show that for Gaussian bandits, the optimal solution must balance the top and bottom arms in all connected subgraphs in terms of the sum-of-squared allocations. For general reward distribution, we have similar results for each connected subgraph. However, the form of this necessary condition cannot be easily simplified.

To derive the general result, we mirror the proof of Proposition 29. Consider any connected subgraph $\mathcal{G}_\ell = \mathcal{G}(\mathcal{I}_\ell, \mathcal{I}_\ell^c, \mathcal{E}_\ell)$ of the bipartite graph induced by the optimal solution $\boldsymbol{p}^*$. By Proposition 3, for any arm $i \in \mathcal{I}\cup\mathcal{I}^c$, there must exist an edge $e \in \mathcal{E}$ such that i is incident to e. Hence, we have $\mathcal{I}_\ell \neq \varnothing$ and $\mathcal{I}_\ell^c \neq \varnothing$.

Take any $i_1 \in \mathcal{I}_\ell$ as an *anchor point*. By the connectedness of G_ℓ, for any $j \in \mathcal{I}_\ell^c$, there exists a path $x_{i_1,j} = (i_1, j_1, i_2, j_2, \ldots, j_r = j)$ connecting i_1 and j. Along the path, we perform Taylor's expansion and derive that

$$C_{i,j}(\boldsymbol{p}+\boldsymbol{\epsilon}) = C_{i,j}(\boldsymbol{p}) + \frac{\partial C_{i,j}(\boldsymbol{p})}{\partial p_i}\epsilon_i + \frac{\partial C_{i,j}(\boldsymbol{p})}{\partial p_j}\epsilon_j + o(\|\boldsymbol{\epsilon}\|).$$

By Lemma 37, we have

$$\frac{\partial C_{i,j}(\boldsymbol{p}^*)}{\partial p_i} = d(\theta_i, \bar{\theta}_{i,j}) \quad \text{and} \quad \frac{\partial C_{i,j}(\boldsymbol{p}^*)}{\partial p_j} = d(\theta_j, \bar{\theta}_{i,j}).$$

Define $\tilde{\epsilon}_{i_1}$ and $\tilde{\epsilon}_{j_1}$ such that the first order terms in the Taylor's series vanish

$$d(\theta_{i_1}, \bar{\theta}_{i_1,j_1})\tilde{\epsilon}_{i_1} + d(\theta_{j_1}, \bar{\theta}_{i_1,j_1})\tilde{\epsilon}_{j_1} = 0.$$

Hence, we can write

$$\tilde{\epsilon}_{j_1} = -\tilde{\epsilon}_{i_1}\frac{d(\theta_{i_1}, \bar{\theta}_{i_1,j_1})}{d(\theta_{j_1}, \bar{\theta}_{i_1,j_1})}.$$

Repeating this procedure along the path connecting i_1 and j, we have

$$\tilde{\epsilon}_j = \tilde{\epsilon}_{j_r} = -\tilde{\epsilon}_{i_1} S_{i_1\to j},$$

where

$$S_{i_1\to j} = \frac{d(\theta_{i_r}, \bar{\theta}_{i_r,j_r})}{d(\theta_{j_r}, \bar{\theta}_{i_r,j_r})} \cdots \frac{d(\theta_{j_1}, \bar{\theta}_{i_2,j_1})}{d(\theta_{i_2}, \bar{\theta}_{i_2,j_1})}\frac{d(\theta_{i_1}, \bar{\theta}_{i_1,j_1})}{d(\theta_{j_1}, \bar{\theta}_{i_1,j_1})}. \tag{25}$$

Similarly, we have for any $i \in \mathcal{I}_\ell$, there exists a path $p_{i_1,i} = (i_1, j'_1, i'_2, j'_2, \ldots, i'_{r'} = i)$ connecting i_1 and i. We can then derive

$$\tilde{\epsilon}_i = \tilde{\epsilon}_{i'_{r'}} = \tilde{\epsilon}_{i_1} S_{i_1 \to i},$$

where

$$S_{i_1 \to i} = \frac{d(\theta_{j'_{r'-1}}, \bar{\theta}_{i'_{r'}, j'_{r'-1}})}{d(\theta_{i'_{r'}}, \bar{\theta}_{i'_{r'}, j'_{r'-1}})} \cdots \frac{d(\theta_{j'_1}, \bar{\theta}_{i'_2, j'_1})}{d(\theta_{i'_2}, \bar{\theta}_{i'_2, j'_1})} \frac{d(\theta_{i'_1}, \bar{\theta}_{i'_1, j'_1})}{d(\theta_{j'_1}, \bar{\theta}_{i'_1, j'_1})}. \tag{26}$$

We adopt the convention that $S_{i_1 \to i_1} = 1$. For Gaussian bandits, this simplifies to

$$S_{i_1 \to j} = \frac{(p^*_j)^2}{(p^*_{i_1})^2} \quad \text{and} \quad S_{i_1 \to i} = \frac{(p^*_i)^2}{(p^*_{i_1})^2}.$$

For general distribution, we cannot easily simplify $S_{i_1 \to j}$ and $S_{i_1 \to i}$. With the same argument as in the proof of Proposition 29, we have the following proposition.

Proposition 30 (Overall balance in connected subgraphs)**.** *Consider a bandit with general exponential family rewards. Let $\mathcal{G}$ be the bipartite graph induced by the optimal solution $\boldsymbol{p}^*$ and let $\mathcal{G}_\ell$ be any connected subgraph of $\mathcal{G}$. For any anchor point $i_1 \in \mathcal{I}_\ell$, $\mathcal{G}_\ell$ balances the top and bottom arms*

$$\sum_{i \in \mathcal{I}_\ell} S_{i_1 \to i} = \sum_{j \in \mathcal{I}^c_\ell} S_{i_1 \to j}. \tag{27}$$

B.4 Proofs of properties of the optimal solution in Appendix B.1

Proof of Lemma 2 (Uniqueness). By Lemma 37, $C_{i,j}(\cdot)$ is a concave function. Hence the optimal allocation problem (4) is a concave programming. Consequently, the optimal set is convex, see Boyd and Vandenberghe (2004, Chapter 2).

We prove by contradiction. Suppose there exist two optimal solutions $\boldsymbol{p}' \neq \boldsymbol{p}''$, then

$$\boldsymbol{p}(\varepsilon) = (1-\varepsilon)\boldsymbol{p}' + \varepsilon \boldsymbol{p}''$$

is also an optimal solution for any $\varepsilon \in (0,1)$. To facilitate discussion, we define the index set where $\boldsymbol{p}'$ differs from $\boldsymbol{p}''$

$$\mathcal{D} = \{i \in [K] : p'_i \neq p''_i\} \neq \varnothing.$$

Now, consider the optimal solution $\boldsymbol{p}'$. By the information balance (5), there exist a set $\mathcal{B} \subset \mathcal{I} \times \mathcal{I}^c$ such that

$$C_{i,j}(\boldsymbol{p}') = \Gamma_{\boldsymbol{\theta}}(\boldsymbol{p}'), \quad \forall (i,j) \in \mathcal{B}, \quad \text{and} \quad C_{i',j'}(\boldsymbol{p}') > \Gamma_{\boldsymbol{\theta}}(\boldsymbol{p}'), \quad \forall (i',j') \notin \mathcal{B}.$$

For any collection $\mathcal{C}$ of (i,j) pairs, we define

$$\mathcal{C}_{\cap \mathcal{D}} \triangleq \{(i,j) \in \mathcal{C} : \{i,j\} \cap \mathcal{D} \neq \varnothing\}.$$

1. For any $(i,j) \in (\mathcal{B} \backslash \mathcal{B}_{\cap \mathcal{D}})$, we have

$$C_{i,j}(\boldsymbol{p}(\varepsilon)) = C_{i,j}(\boldsymbol{p}') = \Gamma_{\boldsymbol{\theta}}(\boldsymbol{p}'). \tag{28}$$

2. Next, consider any $(i,j) \in \mathcal{B}^c$. Since $|\mathcal{B}^c| < \infty$, we can define

$$\Delta_{\min}^{\Gamma_{\boldsymbol{\theta}}(\boldsymbol{p}')} \triangleq \min_{(i,j) \notin \mathcal{B}} C_{i,j}(\boldsymbol{p}') - \Gamma_{\boldsymbol{\theta}}(\boldsymbol{p}') > 0.$$

   By the continuity of $C_{i,j}(\cdot)$, there exists a sufficiently small ε such that

$$\min_{(i,j) \notin \mathcal{B}} C_{i,j}(\boldsymbol{p}(\varepsilon)) - \Gamma_{\boldsymbol{\theta}}(\boldsymbol{p}') > \Delta_{\min}^{\Gamma_{\boldsymbol{\theta}}(\boldsymbol{p}')}/2 > 0,$$

   then

$$\min_{(i,j) \notin \mathcal{B}} C_{i,j}(\boldsymbol{p}(\varepsilon)) > \Gamma_{\boldsymbol{\theta}}(\boldsymbol{p}') = \Gamma_{\boldsymbol{\theta}}(\boldsymbol{p}(\varepsilon)), \tag{29}$$

   where the last equality follows from the fact that both $\boldsymbol{p}'$ and $\boldsymbol{p}(\varepsilon)$ are optimal solutions.

3. Finally, consider any $(i,j) \in \mathcal{B}_{\cap\mathcal{D}}$. Invoking Lemma 38, we conclude that ε can be selected such that

$$C_{i,j}(\boldsymbol{p}(\varepsilon)) \neq C_{i,j}(\boldsymbol{p}'), \quad \forall (i,j) \in \mathcal{B}_{\cap\mathcal{D}}. \tag{30}$$

We have the following contradictions.

1. Suppose that there exists some $(i_1,j_1) \in \mathcal{B}_{\cap\mathcal{D}}$ such that

$$C_{i_1,j_1}(\boldsymbol{p}(\varepsilon)) < C_{i_1,j_1}(\boldsymbol{p}') = \Gamma_{\boldsymbol{\theta}}(\boldsymbol{p}').$$

Then

$$\Gamma_{\boldsymbol{\theta}}(\boldsymbol{p}(\varepsilon)) \leq C_{i_1,j_1}(\boldsymbol{p}(\varepsilon)) < \Gamma_{\boldsymbol{\theta}}(\boldsymbol{p}').$$

This contradicts with the fact that $\boldsymbol{p}(\varepsilon)$ is an optimal solution.

2. Suppose that for all $(i,j) \in \mathcal{B}_{\cap\mathcal{D}}$, we have

$$C_{i_1,j_1}(\boldsymbol{p}(\varepsilon)) > C_{i_1,j_1}(\boldsymbol{p}') = \Gamma_{\boldsymbol{\theta}}(\boldsymbol{p}').$$

Recall that $\mathcal{D} \neq \varnothing$, then there exist $i \in \mathcal{D}$ or $j \in \mathcal{D}$. Suppose $i_2 \in \mathcal{I} \cap \mathcal{D} \neq \varnothing$. Combining (28),(29) and (30), we have

$$\min_{j \in \mathcal{I}^c} C_{i_2,j}(\boldsymbol{p}(\varepsilon)) > \Gamma_{\boldsymbol{\theta}}(\boldsymbol{p}').$$

Invoking the information balance in Proposition 3 we conclude that

$$\Gamma_{\boldsymbol{\theta}}(\boldsymbol{p}(\varepsilon)) > \Gamma_{\boldsymbol{\theta}}(\boldsymbol{p}').$$

This contradicts with the fact that $\boldsymbol{p}'$ is an optimal solution. Suppose $j_2 \in \mathcal{I}^c \cap \mathcal{D} \neq \varnothing$, the proof follows similarly.

In summary, we have uniqueness of the optimal solution to (4). □

Proof of Lemma 27 (Positivity). Consider any fixed $i \in \mathcal{I}$ and any feasible solution $\boldsymbol{p}$. If $p_i = 0$ for some $i \in \mathcal{I}$, then $C_{i,j}(\boldsymbol{p}) = p_i d(\theta_i, \bar{\theta}_{i,j}) + p_j d\left(\theta_j, \frac{p_i\theta_i + p_j\theta_j}{p_i+p_j}\right) = 0$ holds for all $j \in \mathcal{I}^c$. Similarly, if $p_j = 0$ for some $j \in \mathcal{I}^c$, then $C_{i,j} = 0$ holds for all $i \in \mathcal{I}$. However, there exists $\boldsymbol{p} = (1/K, 1/K, \ldots, 1/K)$ such that $C_{i,j} = \frac{1}{K} d\left(\theta_i, \frac{\theta_i+\theta_j}{2}\right) + \frac{1}{K} d\left(\theta_j, \frac{\theta_i+\theta_j}{2}\right) > 0$ holds for all $i \in \mathcal{I}$ and $j \in \mathcal{I}^c$. Thus, the optimal solution cannot have $p_i = 0$ for any $i \in [K]$. □

Proof of Lemma 28 (Monotonicity). For the ease of exposition, we introduce the following function

$$g_{i,j}(x) = d\left(\theta_i, \frac{\theta_i + x\theta_j}{1+x}\right) + xd\left(\theta_j, \frac{\theta_i + x\theta_j}{1+x}\right), \quad i \in \mathcal{I}, j \in \mathcal{I}^c.$$

Then $C_{i,j}(\boldsymbol{p})$ can be written as

$$C_{i,j}(\boldsymbol{p}) = p_i g_{i,j}\left(\frac{p_j}{p_i}\right).$$

At the optimal solution $\boldsymbol{p}^*$, consider any j, $j' \in \mathcal{I}^c$ such that $\theta_{j'} \geq \theta_j$. Proposition 3 implies that there exist a $i \in \mathcal{I}$ such that

$$C_{i,j}(\boldsymbol{p}^*) = \Gamma^*_{\boldsymbol{\theta}} \leq C_{i,j'}(\boldsymbol{p}^*).$$

Note that we must have $p^*_i > 0$, the inequality above implies that

$$g_{i,j}\left(\frac{p^*_j}{p^*_i}\right) \leq g_{i,j'}\left(\frac{p^*_{j'}}{p^*_i}\right). \tag{31}$$

By Lemma 39, for any i, j and j' such that $\theta_i > \theta_{j'} \geq \theta_j$, we have

$$g_{i,j}(x) \geq g_{i,j'}(x) \quad \text{for all } x. \tag{32}$$

Plugging $x = \frac{p_j^*}{p_i^*}$ into (32), we have,

$$g_{i,j}\left(\frac{p_j^*}{p_i^*}\right) \geq g_{i,j'}\left(\frac{p_j^*}{p_i^*}\right). \tag{33}$$

Combining (31), (33) and $g_{i,j'}(x) > 0$, we conclude that $p_{j'}^* \geq p_j^*$. This shows the monotonicity of $\boldsymbol{p}^*$ within the bottom arms $\mathcal{I}^c$.

For the top arms $\mathcal{I}$, we introduce

$$g_{j,i}(x) = d\left(\theta_j, \frac{\theta_j + x\theta_i}{1+x}\right) + xd\left(\theta_i, \frac{\theta_j + x\theta_i}{1+x}\right).$$

Then $C_{i,j}$ can be written as

$$C_{i,j}(\boldsymbol{p}) = p_j g_{j,i}\left(\frac{p_i}{p_j}\right).$$

From the proof of Lemma 39 (See (66) and replace θ_i with θ_j in h), we have

$$g'_{j,i}(x) = d\left(\theta_i, \frac{\theta_j + x\theta_i}{1+x}\right) > 0$$

and, for any $\theta_i \geq \theta_{i'} > \theta_j$, we have

$$g_{j,i}(x) \geq g_{j,i'}(x) \quad \text{for all } x. \tag{34}$$

The rest of the proof mirrors the proof of the monotonicity within the bottom arms. At the optimal solution $\boldsymbol{p}^*$, consider any i, $i' \in \mathcal{I}$ such that $\theta_i \geq \theta_{i'}$. Proposition 3 also implies that there exists a $j \in \mathcal{I}^c$ such that

$$C_{i,j}(\boldsymbol{p}^*) = \Gamma_{\boldsymbol{\theta}}^* \leq C_{i',j}(\boldsymbol{p}^*).$$

Note that we must have $p_j^* > 0$, the inequality above implies that

$$g_{j,i}\left(\frac{p_i^*}{p_j^*}\right) \leq g_{j,i'}\left(\frac{p_{i'}^*}{p_j^*}\right). \tag{35}$$

Plugging $x = \frac{p_i^*}{p_j^*}$ into (34), we have,

$$g_{j,i}\left(\frac{p_i^*}{p_j^*}\right) \geq g_{j,i'}\left(\frac{p_i^*}{p_j^*}\right). \tag{36}$$

Combining (35), (36) and $g_{j,i'}(x) > 0$, we conclude that $p_{i'}^* \geq p_i^*$. This shows the monotonicity of $\boldsymbol{p}^*$ within the top arms $\mathcal{I}$. The proof is complete. □

C Optimality conditions based only on the allocation vector

Optimality conditions that depends only on the allocation vector $\boldsymbol{p}$ can be derived for the best-k-arm identification problem. However, we show in this section that these optimality conditions are complicated and impractical for algorithm design.

C.1 A (conditional) sufficient condition

To begin, we present a sufficient condition for a large set of problem instances $\boldsymbol{\theta}$. For the ease of notation, we re-label the arm with the ℓ-th largest mean as arm i_ℓ, for $\ell = 1, \ldots, k$ and re-label the arm with the $(\ell+k)$-th largest mean as arm j_ℓ, for $\ell = 1, \ldots, K-k$. It may be intriguing to think that the optimal solution should gather equal statistical evidence to distinguish (i) the arm i_k from any bottom arm j; and (ii) the arm j_1 from any top arm i. The next proposition show that this intuitive balancing rule (together with the balance of sum-of-squared allocation) is optimal under certain condition. To simplify notation, we define $\bar{\theta}^*_{i,j} \triangleq \bar{\theta}_{i,j}(\boldsymbol{p}^*)$ for all $(i,j) \in \mathcal{I} \times \mathcal{I}^c$.

Proposition 31. *Consider any problem instance such that the optimal solution $\boldsymbol{p}^*$ satisfies*

$$1 - \sum_{j \in \mathcal{I}^c \setminus \{j_1\}} \frac{d(\theta_{i_k}, \bar{\theta}^*_{i_k,j})}{d(\theta_j, \bar{\theta}^*_{i_k,j})} > 0. \tag{37}$$

Then $\boldsymbol{p}^*$ is the optimal solution to (4) if

$$\Gamma^*_{\boldsymbol{\theta}} = C_{i_k,j}(\boldsymbol{p}^*) = C_{i,j_1}(\boldsymbol{p}^*), \quad \forall i \in \mathcal{I}, \forall j \in \mathcal{I}^c, \quad \text{and} \tag{38}$$

$$\frac{d(\theta_{i_k}, \bar{\theta}^*_{i_k,j_1})}{d(\theta_{j_1}, \bar{\theta}^*_{i_k,j_1})} \sum_{i \in \mathcal{I}} \frac{d(\theta_{j_1}, \bar{\theta}^*_{i,j_1})}{d(\theta_i, \bar{\theta}^*_{i,j_1})} = \sum_{j \in \mathcal{I}^c} \frac{d(\theta_{i_k}, \bar{\theta}^*_{i_k,j})}{d(\theta_j, \bar{\theta}^*_{i_k,j})}. \tag{39}$$

Complete characterization of sufficient and necessary conditions based only on $\boldsymbol{p}$ can be derived following the lines of Proposition 31, see Appendix C.2. However, we caution that such results rely on assumptions similar to (37), which is non-trivial to check. Hence, developing a sampling rule based on tracking the sufficient condition can be challenging. We address this challenge by introducing the dual variable, and analyze the KKT conditions.

Remark 6. *Condition (38) implies that the bipartite graph is connected. By setting the anchor point as i_k in Proposition 30, we see that (39) is equivalent to the balancing of top and bottom arms.*

Remark 7. *For general bandits, condition (37) always holds when (i) $|\mathcal{I}| \leq 2$ and $|\mathcal{I}^c| \leq 2$; and (ii) $|\mathcal{I}| = 1$ or $|\mathcal{I}^c| = 1$. When condition (37) holds, there is only one connected subgraph. For Gaussian bandits with equal variance, the assumption (37) reduces to $(p^*_{i_k})^2 > \sum_{j \in \mathcal{I}^c \setminus \{j_1\}} (p^*_j)^2$; and (39) reduces to the balancing of the sum-of-squared allocations. Recall the monotonicity in Lemma 28, we have $p_{j_1} \geq p_j$ for $j \neq j_1$. In general, $p^*_{j_1} - p^*_j$ increases as the gap $(\theta_{j_1} - \theta_j)$ increases. Roughly speaking, when $(\theta_{j_1} - \theta_j)$ is large enough with respect to $(\theta_{i_k} - \theta_{j_1})$, condition (37) will hold. This suggest that problem instances closer to the slippage configuration are the most subtle instances in terms of the structure of the optimal solution.*

C.2 A Case study for 5 choose 2: A pathway to complete characterization of optimality based on the allocation vector

In Proposition 31, we characterized the sufficient conditions for optimality under the assumption of (37). We now discuss how to obtain complete characterization.

We assume that the mean rewards are mutually different without loss of generality (for the optimal allocation problem). Otherwise, we can combine arms with the same mean rewards and evenly distribute the allocation among them.

To facilitate our discussion, we present $C_{i,j}(\boldsymbol{p}^*)$ as a matrix in $\mathbb{R}^{|\mathcal{I}| \times |\mathcal{I}^c|}$, where the (i,j)-th entry is "=" if $C_{i,j}(\boldsymbol{p}^*) = \Gamma^*_{\boldsymbol{\theta}}$ and ">" if $C_{i,j}(\boldsymbol{p}^*) > \Gamma^*_{\boldsymbol{\theta}}$. The following matrix represents the case discussed in Proposition 31.

arms	j_1	j_2	$\cdots$	j_{K-k}
i_1	$=$	$>$	$\cdots$	$>$
i_2	$=$	$>$	$\cdots$	$>$
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
i_{k-1}	$=$	$>$	$\cdots$	$>$
i_k	$=$	$=$	$\cdots$	$=$

Recall the assumption in Proposition 31, i.e.,

$$1 - \sum_{j \in \mathcal{I}^c \setminus \{j_1\}} \frac{d(\theta_{i_k}, \bar{\theta}^*_{i_k,j})}{d(\theta_j, \bar{\theta}^*_{i_k,j})} > 0.$$

In general, the ='s locations can be quite different from the clean structure in the matrix above.

We illustrate with an example and discuss the sufficient conditions for it. Consider choosing the best-2 arms among 5 competing arms. Without loss of generality, we assume that the arms are sorted in descending order of the mean rewards, i.e., $\mathcal{I} = \{1, 2\}$ and $\mathcal{I}^c = \{3, 4, 5\}$. We may have the following scenario.

arms	3	4	5
1	$=$	$=$	$>$
2	$>$	$=$	$=$

Mirroring the proof of Proposition 31, we have the following sufficient conditions for this case.

Proposition 32. *Consider any problem instance such that the optimal solution $\boldsymbol{p}^*$ satisfies*

$$\begin{aligned} &1 - \frac{d(\theta_2, \bar{\theta}^*_{2,5})}{d(\theta_5, \bar{\theta}^*_{2,5})} > 0, \\ &1 - \frac{d(\theta_4, \bar{\theta}^*_{2,4})}{d(\theta_2, \bar{\theta}^*_{2,4})} + \frac{d(\theta_4, \bar{\theta}^*_{2,4})}{d(\theta_2, \bar{\theta}^*_{2,4})} \frac{d(\theta_2, \bar{\theta}^*_{2,5})}{d(\theta_5, \bar{\theta}^*_{2,5})} > 0. \end{aligned} \tag{40}$$

Then $\boldsymbol{p}^*$ is the optimal solution to (4) if

$$\Gamma^*_{\boldsymbol{\theta}} = C_{1,3}(\boldsymbol{p}^*) = C_{1,4}(\boldsymbol{p}^*) = C_{2,4}(\boldsymbol{p}^*) = C_{2,5}(\boldsymbol{p}^*) \tag{41}$$

and

$$1 + \frac{d(\theta_1, \bar{\theta}^*_{1,4})}{d(\theta_4, \bar{\theta}^*_{1,4})} \frac{d(\theta_4, \bar{\theta}^*_{2,4})}{d(\theta_2, \bar{\theta}^*_{2,4})} = \frac{d(\theta_1, \bar{\theta}^*_{1,3})}{d(\theta_3, \bar{\theta}^*_{1,3})} + \frac{d(\theta_1, \bar{\theta}^*_{1,4})}{d(\theta_4, \bar{\theta}^*_{1,4})} + \frac{d(\theta_1, \bar{\theta}^*_{1,4})}{d(\theta_4, \bar{\theta}^*_{1,4})} \frac{d(\theta_4, \bar{\theta}^*_{2,4})}{d(\theta_2, \bar{\theta}^*_{2,4})} \frac{d(\theta_2, \bar{\theta}^*_{2,5})}{d(\theta_5, \bar{\theta}^*_{2,5})}. \tag{42}$$

Remark 8. *For this scenario, the bipartite graph is connected, and the condition (42) is equivalent to the balancing of top and bottom arms (by setting arm 1 as the anchor point in Proposition 30).*

Remark 9 (Complete characterization of the case with $(K, k) = (5, 2)$)**.** *We consider Gaussian bandit with $(K, k) = (5, 2)$. For simplicity, we consider the problem instances where the mean rewards are mutually different. Recall that we have the balancing of top and bottom arms in the entire bipartite graph in (6), i.e.,*

$$(p^*_1)^2 + (p^*_2)^2 = (p^*_3)^2 + (p^*_4)^2 + (p^*_5)^2.$$

By the monotonicity in Lemma 28, we must have

$$(p^*_2)^2 > (p^*_5)^2,$$

otherwise, $(p^*_3)^2 \geq (p^*_4)^2 \geq (p^*_5)^2 \geq (p^*_2)^2 \geq (p^*_1)^2$, which contradicts with (6).

Note that the condition (40) is equivalent to

$$(p^*_2)^2 > (p^*_5)^2 \quad \textit{and} \quad (p^*_2)^2 < (p^*_4)^2 + (p^*_5)^2.$$

Since we must have $(p_2^*)^2 > (p_5^*)^2$, *condition* (40) *reduces to*

$$(p_2^*)^2 < (p_4^*)^2 + (p_5^*)^2.$$

Recall that condition (37) *reduces to*

$$(p_2^*)^2 > (p_4^*)^2 + (p_5^*)^2.$$

Hence the two scenarios are mutually exclusive. Finally, recall that Example 2 *corresponds to* $\Gamma_{\boldsymbol{\theta}}^* = C_{1,3}(p_1^*, p_3^*) = C_{2,4}(p_2^*, p_4^*) = C_{2,5}(p_2^*, p_5^*)$ *and*

$$(p_2^*)^2 = (p_4^*)^2 + (p_5^*)^2 \quad \textit{and} \quad (p_1^*)^2 = (p_3^*)^2, \tag{43}$$

which corresponds to

arms	3	4	5
1	$=$	$>$	$>$
2	$>$	$=$	$=$

Combining (37), (40) *and* (43), *we have now fully characterized the sufficient conditions for the case of choosing* 2 *arms among* 5, *which we summarize in Table* 3.

Condition	$(p_2^*)^2 > (p_4^*)^2 + (p_5^*)^2$	$(p_2^*)^2 < (p_4^*)^2 + (p_5^*)^2$	$(p_2^*)^2 = (p_4^*)^2 + (p_5^*)^2$
Information balance	arms | 3 4 5 1 | = > > 2 | = = =	arms | 3 4 5 1 | = = > 2 | > = =	arms | 3 4 5 1 | = > > 2 | > = =
Overall balance	$\sum_{i\in\mathcal{I}}(p_i^*)^2 = \sum_{j\in\mathcal{I}^c}(p_j^*)^2$	$\sum_{i\in\mathcal{I}}(p_i^*)^2 = \sum_{j\in\mathcal{I}^c}(p_j^*)^2$	$(p_1^*)^2 = (p_3^*)^2$ $(p_2^*)^2 = (p_4^*)^2 + (p_5^*)^2$

Table 3: Complete characterization of the optimal solution for Gaussian bandits, choose 2 from 5 arms, with mutually distinctive mean rewards.

We omit the discussion for general (K, k) pair and instances with arms sharing the same mean. A similar analysis can be carried out. However, we are cautious that any elegant presentation can be derived due to the *combinatorial* number of scenarios, even though conditions such as (37) and (40) can significantly reduce the number of scenarios to be discussed.

C.3 Proof of Proposition 31

Proof. We restate the KKT conditions from Section E here for easier reference:

$$-1 + \sum_{i\in\mathcal{I}} \sum_{j\in\mathcal{I}^c} \mu_{i,j} = 0, \tag{44a}$$

$$\lambda - \sum_{j\in\mathcal{I}^c} \mu_{i,j} d(\theta_i, \bar{\theta}_{i,j}) = 0, \quad \forall i \in \mathcal{I}, \tag{44b}$$

$$\lambda - \sum_{i\in\mathcal{I}} \mu_{i,j} d(\theta_j, \bar{\theta}_{i,j}) = 0, \quad \forall j \in \mathcal{I}^c, \tag{44c}$$

$$\sum_{i\in[K]} p_i - 1 = 0, \quad p_i \geq 0, \quad \forall i \in [K], \tag{44d}$$

$$\phi - C_{i,j}(\boldsymbol{p}) \leq 0, \quad \forall i \in \mathcal{I}, j \in \mathcal{I}^c, \tag{44e}$$

$$\mu_{i,j} \geq 0, \quad \forall i \in \mathcal{I}, j \in \mathcal{I}^c, \tag{44f}$$

$$\sum_{i\in\mathcal{I}} \sum_{j\in\mathcal{I}^c} \mu_{i,j}(\phi - C_{i,j}(\boldsymbol{p})) = 0. \tag{44g}$$

Let $\boldsymbol{p}$ be the optimal primal solution so that (44d) holds automatically. We aim to find a KKT solution such that

$$\mu_{i,j} = 0, \quad \text{if } i \neq i_k \text{ or } j \neq j_1. \tag{45}$$

In particular, under (45), we find that the solution to (44b) and (44c) satisfies

$$\begin{aligned}
\mu_{i,j} &= 0, \quad \text{if } i \neq i_k \text{ or } j \neq j_1, \\
\mu_{i_k,j} &= \frac{\lambda}{d(\theta_j, \bar{\theta}_{i_k,j})}, \quad \text{for } j \neq j_1, \\
\mu_{i,j_1} &= \frac{\lambda}{d(\theta_i, \bar{\theta}_{i,j_1})}, \quad \text{for } i \neq i_k, \\
\mu_{i_k,j_1} &= \lambda \left(1 - \sum_{j \in \mathcal{I}^c \setminus \{j_1\}} \frac{d(\theta_{i_k}, \bar{\theta}_{i_k,j})}{d(\theta_j, \bar{\theta}_{i_k,j})}\right) \Big/ d(\theta_{i_k}, \bar{\theta}_{i_k,j_1}) \\
&= \lambda \left(1 - \sum_{i \in \mathcal{I} \setminus \{i_k\}} \frac{d(\theta_{j_1}, \bar{\theta}_{i,j_1})}{d(\theta_i, \bar{\theta}_{i,j_1})}\right) \Big/ d(\theta_{j_1}, \bar{\theta}_{i_k,j_1}).
\end{aligned} \tag{46}$$

where the multiplier λ is determined by (44a), i.e.,

$$\sum_{j \neq j_1} \mu_{i_k,j} + \sum_{i \neq i_k} \mu_{i,j_1} + \mu_{i_k,j_1} = 1. \tag{47}$$

For the dual feasibility condition (44f) to hold, we require condition (37) to hold, i.e.,

$$1 - \sum_{j \in \mathcal{I}^c \setminus \{j_1\}} \frac{d(\theta_{i_k}, \bar{\theta}_{i_k,j})}{d(\theta_j, \bar{\theta}_{i_k,j})} > 0, \tag{48}$$

For (46) to hold, we require (39), i.e.,

$$\frac{d(\theta_{i_k}, \bar{\theta}_{i_k,j_1})}{d(\theta_{j_1}, \bar{\theta}_{i_k,j_1})} \sum_{i \in \mathcal{I}} \frac{d(\theta_{j_1}, \bar{\theta}_{i,j_1})}{d(\theta_i, \bar{\theta}_{i,j_1})} = \sum_{j \in \mathcal{I}^c} \frac{d(\theta_{i_k}, \bar{\theta}_{i_k,j})}{d(\theta_j, \bar{\theta}_{i_k,j})}. \tag{49}$$

We have now checked that equations (44a), (44b), (44c), (44d) and (44f) hold.
For the primal variable (equivalent to objective value) ϕ, we let

$$\phi = C_{i_k,j}(\boldsymbol{p}) = C_{i,j_1}(\boldsymbol{p}), \quad \forall (i,j) \in \mathcal{I} \times \mathcal{I}^c. \tag{50}$$

By noting that $\mu_{i,j} = 0$ for $i \neq i_k$ and $j \neq j_1$, we have (44g). At last, when $\Gamma^*_{\boldsymbol{\theta}} = \phi$, the inequality in (44e) also holds automatically.
Hence, all the conditions in (44) holds. The proof is complete. □

D Proof of Theorem 1: fixed-confidence sample complexity

For problem instance $\boldsymbol{\theta} \in \Theta$, we define the set of alternative problem instances with different best-k-arm sets,

$$\begin{aligned}\mathrm{Alt}(\boldsymbol{\theta}) &\triangleq \{\boldsymbol{\vartheta} \in \Theta : \mathcal{I}(\boldsymbol{\vartheta}) \neq \mathcal{I}(\boldsymbol{\theta})\} \\ &= \bigcup_{(i,j)\in\mathcal{I}(\boldsymbol{\theta})\times\mathcal{I}^c(\boldsymbol{\theta})} \{\boldsymbol{\vartheta} \in \Theta : \vartheta_i < \vartheta_j\}.\end{aligned}$$

For $(i,j) \in \mathcal{I}(\boldsymbol{\theta}) \times \mathcal{I}^c(\boldsymbol{\theta})$, let $\mathrm{Alt}_{i,j}(\boldsymbol{\theta}) \triangleq \{\boldsymbol{\vartheta} \in \Theta : \vartheta_i < \vartheta_j\}$, and thus

$$\mathrm{Alt}(\boldsymbol{\theta}) = \bigcup_{(i,j)\in\mathcal{I}(\boldsymbol{\theta})\times\mathcal{I}^c(\boldsymbol{\theta})} \mathrm{Alt}_{i,j}(\boldsymbol{\theta}).$$

We further define the *discriminative information* between $\boldsymbol{\theta}, \boldsymbol{\vartheta} \in \Theta$ under allocation $\boldsymbol{p} \in \mathcal{S}_K$

$$D_{\boldsymbol{p}}(\boldsymbol{\theta}, \boldsymbol{\vartheta}) = \sum_{i=1}^{K} p_i d(\theta_i, \vartheta_i).$$

Then we can write each generalized Chernoff information $C_{i,k}(\boldsymbol{p})$ defined in Equation (3) as follows,

Lemma 33. *For any $(i,j) \in \mathcal{I}(\boldsymbol{\theta}) \times \mathcal{I}^c(\boldsymbol{\theta})$ and $\boldsymbol{p} \in \mathcal{S}_K$,*

$$C_{i,j}(\boldsymbol{p}) = \inf_{\boldsymbol{\vartheta}\in\mathrm{Alt}_{i,j}(\boldsymbol{\theta})} D_{\boldsymbol{p}}(\boldsymbol{\theta}, \boldsymbol{\vartheta}).$$

We hence write $\Gamma^*_{\boldsymbol{\theta}}$ defined in Equation (4) as

$$\Gamma^*_{\boldsymbol{\theta}} = \max_{\boldsymbol{p}\in\mathcal{S}_K} \min_{(i,j)\in\mathcal{I}(\boldsymbol{\theta})\times\mathcal{I}^c(\boldsymbol{\theta})} \inf_{\boldsymbol{\vartheta}\in\mathrm{Alt}_{i,j}(\boldsymbol{\theta})} D_{\boldsymbol{p}}(\boldsymbol{\theta}, \boldsymbol{\vartheta}),$$

Proof of Lemma 33. Recall that

$$\inf_{\boldsymbol{\vartheta}:\vartheta_j>\vartheta_i} D_{\boldsymbol{p}}(\boldsymbol{\theta}, \boldsymbol{\vartheta}) = \inf_{\boldsymbol{\vartheta}:\vartheta_j>\vartheta_i} \sum_{i=1}^{K} p_i d(\theta_i, \vartheta_i).$$

We have

$$\begin{aligned}\inf_{\boldsymbol{\vartheta}:\vartheta_j>\vartheta_i} \sum_{\ell=1}^{K} p_\ell d(\theta_\ell, \vartheta_\ell) &= \inf_{\vartheta_j>\vartheta_i} \{p_i d(\theta_i, \vartheta_i) + p_j d(\theta_j, \vartheta_j)\} + \sum_{\ell\neq i,j} \inf_{\vartheta_\ell} p_\ell d(\theta_\ell, \vartheta_\ell) \\ &= \inf_{\vartheta_j>\vartheta_i} \{p_i d(\theta_i, \vartheta_i) + p_j d(\theta_j, \vartheta_j)\} \qquad [\text{setting } \vartheta_\ell = \theta_\ell] \\ &= \inf_{\tilde{\theta}} \left\{p_i d(\theta_i, \tilde{\theta}) + p_j d(\theta_j, \tilde{\theta})\right\}. \qquad [\text{monotonicity of KL-divergence}]\end{aligned}$$

Taking derivative of the function inside the infimum with respect to $\tilde{\theta}$ and equating it to 0, we have

$$\frac{\mathrm{d}\eta(\tilde{\theta})}{\mathrm{d}\theta}\left(p_i(\tilde{\theta} - \theta_i) + p_j(\tilde{\theta} - \theta_j)\right) = 0, \tag{51}$$

where we use $d(\theta_i, \tilde{\theta}) = A(\eta(\tilde{\theta})) - A(\eta(\theta_i)) - \theta_i(\eta(\tilde{\theta}) - \eta(\theta_i))$. We find that the infimum is achieved at $\bar{\theta}_{i,j} = (p_i\theta_i + p_j\theta_j)/(p_i + p_j)$. Hence,

$$\inf_{\boldsymbol{\vartheta}\in\mathrm{Alt}_{i,j}(\boldsymbol{\theta})} D_{\boldsymbol{p}}(\boldsymbol{\theta}, \boldsymbol{\vartheta}) = p_i d(\theta_i, \bar{\theta}_{i,j}) + p_j d(\theta_j, \bar{\theta}_{i,j}) = C_{i,j}(\boldsymbol{p}).$$

□

To complete the proof of Theorem 1, we also need to utilize Lemma 1 in Kaufmann et al. (2016), which is rephrased below.

Lemma 34 (Lemma 1 in Kaufmann et al. (2016)). *Let $\boldsymbol{\theta}$ and $\boldsymbol{\vartheta}$ be two bandit models with K arms such that for all $i \in [K]$, the distributions θ_i and ϑ_i are mutually absolutely continuous. For any almost-surely finite stopping time τ_σ with respect to $\mathcal{F}_t = \sigma\left(I_0, Y_{1,I_0}, \ldots, I_{t-1}, Y_{t,I_{t-1}}\right)$,*

$$\sum_{i=1}^{K} \mathbb{E}_{\boldsymbol{\theta}}[N_{\tau_\delta,i}] d(\theta_i, \vartheta_i) \geq \sup_{\mathcal{E} \in \mathcal{F}_{\tau_\delta}} \mathrm{kl}(\mathbb{P}_{\boldsymbol{\theta}}(\mathcal{E}), \mathbb{P}_{\boldsymbol{\vartheta}}(\mathcal{E})), \tag{52}$$

where $\mathrm{kl}(\cdot,\cdot)$ is the KL divergence of two Bernoulli distributions.

Proof of Theorem 1. The δ-correctness of an algorithm implies that, for any $\boldsymbol{\theta} \in \Theta$, $\boldsymbol{\vartheta} \in \mathrm{Alt}(\boldsymbol{\theta})$ and $\mathcal{E} \in \mathcal{F}_{\tau_\delta}$, we have $\mathbb{P}_{\boldsymbol{\theta}}(\mathcal{E}) > 1-\delta$ and $\mathbb{P}_{\boldsymbol{\vartheta}}(\mathcal{E}) \leq \delta$. Hence, (52) implies that

$$\sum_{i=1}^{K} \mathbb{E}_{\boldsymbol{\theta}}[N_{\tau_\delta,i}] d(\theta_i, \vartheta_i) \geq \mathrm{kl}(1-\delta, \delta),$$

where we use the monotonicity of the KL divergence of Bernoulli distribution, and thus

$$\mathbb{E}_{\boldsymbol{\theta}}[\tau_\delta] \inf_{\boldsymbol{\vartheta} \in \mathrm{Alt}(\boldsymbol{\theta})} \sum_{i=1}^{K} \frac{\mathbb{E}_{\boldsymbol{\theta}}[N_{\tau_\delta,i}]}{\mathbb{E}_{\boldsymbol{\theta}}[\tau_\delta]} d(\theta_i, \theta_i') \geq \mathrm{kl}(1-\delta, \delta).$$

the lower bound of sample complexity $\mathbb{E}_{\boldsymbol{\theta}}[\tau_\delta]$ is

$$\mathbb{E}_{\boldsymbol{\theta}}[\tau_\delta] \geq \mathrm{kl}(1-\delta, \delta) \left(\inf_{\boldsymbol{\vartheta} \in \mathrm{Alt}(\boldsymbol{\theta})} \sum_{i=1}^{K} p_i d(\theta_i, \vartheta_i) \right)^{-1}.$$

It can be verify that $\lim_{\delta \to 0} \frac{\mathrm{kl}(1-\delta,\delta)}{\log(1/\delta)} = 1$. Hence, we have the asymptotic lower bound

$$\liminf_{\delta \to 0} \frac{\mathbb{E}_{\boldsymbol{\theta}}[\tau_\delta]}{\log(1/\delta)} \geq \left(\inf_{\boldsymbol{\vartheta} \in \mathrm{Alt}(\boldsymbol{\theta})} \sum_{i=1}^{K} p_i d(\theta_i, \vartheta_i) \right)^{-1}.$$

Minimizing this lower bound is equivalent to the following optimization problem,

$$\begin{aligned}
\max_{\boldsymbol{p} \in \mathcal{S}_K} \inf_{\boldsymbol{\vartheta} \in \mathrm{Alt}(\boldsymbol{\theta})} \sum_{i=1}^{K} p_i d(\theta_i, \vartheta_i) &= \max_{\boldsymbol{p} \in \mathcal{S}_K} \inf_{\boldsymbol{\vartheta} \in \mathrm{Alt}(\boldsymbol{\theta})} D_{\boldsymbol{p}}(\boldsymbol{\theta}, \boldsymbol{\vartheta}) \\
&= \max_{\boldsymbol{p} \in \mathcal{S}_K} \min_{(i,j) \in \mathcal{I}(\boldsymbol{\theta}) \times \mathcal{I}^c(\boldsymbol{\theta})} \inf_{\boldsymbol{\vartheta} \in \mathrm{Alt}_{i,j}(\boldsymbol{\theta})} D_{\boldsymbol{p}}(\boldsymbol{\theta}, \boldsymbol{\vartheta}) \\
&= \Gamma^*_{\boldsymbol{\theta}}.
\end{aligned}$$

This completes the proof. □

E Proof of Theorem 5

Proof. Let $\lambda \in \mathbb{R}$, $\boldsymbol{\iota} = \{\iota_i : i \in [K]\}$ and $\boldsymbol{\mu} = \{\mu_{i,j} : i \in \mathcal{I}, j \in \mathcal{I}^c\}$ be the Lagrangian multipliers corresponding to the equality constraint (7b), inequality constraints (7c) and inequality constraints (7d), respectively. The Lagrangian is

$$\mathcal{L}(\phi, \boldsymbol{p}, \lambda, \boldsymbol{\iota}, \boldsymbol{\mu}) = \phi - \lambda \left(\sum_{i\in[K]} p_i - 1 \right) + \sum_{i\in[K]} \iota_i p_i - \sum_{i\in\mathcal{I}} \sum_{j\in\mathcal{I}^c} \mu_{i,j}(\phi - C_{i,j}(\boldsymbol{p})). \tag{53}$$

Lemma 27 simplifies the Lagrangian in the sense that $\boldsymbol{\iota} = \mathbf{0}$, which follows from the complementary slackness condition for (7c).

Using Lemma 37, the KKT conditions are given by

$$-1 + \sum_{i\in\mathcal{I}} \sum_{j\in\mathcal{I}^c} \mu_{i,j} = 0, \tag{54a}$$

$$\lambda - \sum_{j\in\mathcal{I}^c} \mu_{i,j} d(\theta_i, \bar{\theta}_{i,j}) = 0, \quad \forall i \in \mathcal{I}, \tag{54b}$$

$$\lambda - \sum_{i\in\mathcal{I}} \mu_{i,j} d(\theta_j, \bar{\theta}_{i,j}) = 0, \quad \forall j \in \mathcal{I}^c, \tag{54c}$$

$$\sum_{i\in[K]} p_i - 1 = 0, \quad p_i \geq 0, \quad \forall i \in [K], \tag{54d}$$

$$\phi - C_{i,j}(\boldsymbol{p}) \leq 0, \quad \forall i \in \mathcal{I}, j \in \mathcal{I}^c, \tag{54e}$$

$$\mu_{i,j} \geq 0, \quad \forall i \in \mathcal{I}, j \in \mathcal{I}^c, \tag{54f}$$

$$\sum_{i\in\mathcal{I}} \sum_{j\in\mathcal{I}^c} \mu_{i,j}(\phi - C_{i,j}(\boldsymbol{p})) = 0. \tag{54g}$$

The KKT conditions in (54) is not immediately insightful for the design of optimal adaptive selection in that the key stationarity condition (54b) and (54c) are not directly written in terms of the allocation proportion $\boldsymbol{p}$. To obtain insights on algorithmic design and optimal adaptive selection, we reformulate the KKT conditions into a set of necessary and sufficient conditions that is explicitly expressed by the dual variables $\boldsymbol{\mu}$ and primal variables $\boldsymbol{p}$.

To show the equivalent between (9) and (54), we first show that (54) implies (9).

We claim that there must exist a $i' \in \mathcal{I}$ and a $j' \in \mathcal{I}^c$ such that $\phi = C_{i',j'}(\boldsymbol{p})$ and hence $\phi = \min_{i\in\mathcal{I}} \min_{j\in\mathcal{I}^c} C_{i,j}(\boldsymbol{p})$. Together with (54g), this implies (9b). Suppose otherwise, we have $\phi < C_{i,j}(\boldsymbol{p})$ for all $i \in \mathcal{I}$ and $j \in \mathcal{I}^c$. By the complementary slackness condition (54g), we have $\mu_{i,j} = 0$ for all $i \in \mathcal{I}$ and $j \in \mathcal{I}^c$, which yields contradiction with (54a). Furthermore, by the complementary slackness condition (54g), $\phi = C_{i.j}(\boldsymbol{p})$ whenever $\mu_{i,j} > 0$. Hence, we can write the stationarity conditions (54b) and (54c) as

$$\frac{\lambda}{\phi} p_i = \sum_{j\in\mathcal{I}^c} \mu_{i,j} \frac{p_i d(\theta_i, \bar{\theta}_{i,j})}{C_{i,j}(\boldsymbol{p})} = \sum_j \mu_{i,j} h_{i,j}, \tag{55a}$$

$$\frac{\lambda}{\phi} p_j = \sum_{i\in\mathcal{I}} \mu_{i,j} \frac{p_j d(\theta_j, \bar{\theta}_{i,j})}{C_{i,j}(\boldsymbol{p})} = \sum_i \mu_{i,j} h_{j,i}. \tag{55b}$$

Summing the above over all $i \in \mathcal{I}$ and $j \in \mathcal{I}^c$ and using the fact that $h_{i,j} + h_{j,i} = 1$, we have $\lambda = \phi$. Plugging $\lambda = \phi$ into (55), we obtain (9a).

Next we show that (9) also implies (54). Notice that (54a), (54e), (54f) and (54g) holds automatically. Using (8) and (9b), we can rewrite (9a) as

$$\begin{aligned} p_i &= \sum_{j:\mu_{i,j}>0} \mu_{i,j} \frac{p_i d(\theta_i, \bar{\theta}_{i,j})}{C_{i,j}(\boldsymbol{p})} = \sum_{j:\mu_{i,j}>0} \mu_{i,j} \frac{p_i d(\theta_i, \bar{\theta}_{i,j})}{\phi}, \\ p_j &= \sum_{i:\mu_{i,j}>0} \mu_{i,j} \frac{p_j d(\theta_j, \bar{\theta}_{i,j})}{C_{i,j}(\boldsymbol{p})} = \sum_{i:\mu_{i,j}>0} \mu_{i,j} \frac{p_j d(\theta_j, \bar{\theta}_{i,j})}{\phi}. \end{aligned} \tag{56}$$

Thus, we have

$$\phi = \sum_{j \in \mathcal{I}^c} \mu_{i,j} d(\theta_i, \bar{\theta}_{i,j}) = \sum_{i \in \mathcal{I}} \mu_{i,j} d(\theta_j, \bar{\theta}_{i,j}). \tag{57}$$

Let $\lambda = \phi$, we have

$$\lambda - \sum_{j \in \mathcal{I}^c} \mu_{i,j} d(\theta_i, \bar{\theta}_{i,j}) = 0, \text{for all } i \in \mathcal{I},$$
$$\lambda - \sum_{i \in \mathcal{I}} \mu_{i,j} d(\theta_j, \bar{\theta}_{i,j}) = 0, \text{for all } j \in \mathcal{I}^c,$$

which coincide with (54b) and (54c). At last, (54d) can be seen from directly summing (9a) over all $i \in \mathcal{I}$ and $j \in \mathcal{I}^c$ and noting that $\boldsymbol{\mu} \in \mathcal{S}_{k(K-k)}$. This completes the proof. □

F Stopping rule for best-k-arm identification

We consider the Chernoff stopping rule from the perspective of generalized hypothesis testing. Recall that at the end of each round t, the information available to the player is $\mathcal{H}_t$. If $\tau_\delta = t$, conditional on $\mathcal{H}_t$, the player recommend top-k arm set $\hat{\mathcal{I}}_t$ as

$$\hat{\mathcal{I}}_t = \mathcal{I}(\boldsymbol{\theta}_t) = \operatorname*{argmax}_{\mathcal{I}\subset[K],|\mathcal{I}|=k} \left\{ \sum_{i\in\mathcal{I}} \theta_{t,i} \right\}.$$

Define the rejection region as

$$R_t = \left\{ \boldsymbol{\theta} \in \Theta : \min_{i\in\hat{\mathcal{I}}_t} \theta_i < \max_{j\notin\hat{\mathcal{I}}_t} \theta_j \right\}. \tag{58}$$

Then the generalized likelihood ratio test statistic is given by

$$Z_t = \log \frac{\sup_{\boldsymbol{\theta}} L_t(\boldsymbol{\theta})}{\sup_{\boldsymbol{\theta}'\in R_t} L_t(\boldsymbol{\theta}')}, \tag{59}$$

where the likelihood is given by $L_t(\boldsymbol{\theta}) = \prod_{i=1}^K \exp\left\{N_{t,i}\eta(\theta_i)\theta_{t,i} - N_{t,i}A(\eta(\theta_i))\right\}$. To ensure the δ-correctness, we only need to identify a threshold $\gamma(t,\delta)$ such that,

$$\mathbb{P}\left(Z_t \geq \gamma(t,\delta)\right) \leq \delta. \tag{60}$$

To this end, we show a convenient closed-form formula for Z_t. First, it is clear that

$$\begin{aligned} \sup_{\boldsymbol{\theta}} L_t(\boldsymbol{\theta}) &= \sup_{\boldsymbol{\theta}} \prod_{i=1}^K \exp\left\{N_{t,i}\eta(\theta_i)\theta_{t,i} - N_{t,i}A(\eta(\theta_i))\right\} \\ &= \prod_{i=1}^K \exp\left\{N_{t,i}\eta(\theta_{t,i})\theta_{t,i} - N_{t,i}A(\eta(\theta_{t,i}))\right\}. \end{aligned} \tag{61}$$

Then Z_t can be simplified as

$$\begin{aligned} Z_t &= \log \frac{\sup_{\boldsymbol{\theta}} \prod_{i=1}^K \exp\left\{N_{t,i}\eta(\theta_i)\theta_{t,i} - N_{t,i}A(\eta(\theta_i))\right\}}{\sup_{\boldsymbol{\theta}'\in R_t} \prod_{i=1}^K \exp\left\{N_{t,i}\eta(\theta_i')\theta_{t,i} - N_{t,i}A(\eta(\theta_i'))\right\}} \\ &= -\log \sup_{\boldsymbol{\theta}'\in R_t} \prod_{i=1}^K \exp\left\{-N_{t,i}\left(A(\eta(\theta_i')) - A(\eta(\theta_{t,i})) - \theta_{t,i}(\eta(\theta_i') - \eta(\theta_{t,i}))\right)\right\} \\ &= -\log \sup_{\boldsymbol{\theta}'\in R_t} \exp\left\{-\sum_{i=1}^K N_{t,i} d(\theta_{t,i},\theta_i')\right\} \\ &= \inf_{\boldsymbol{\theta}'\in R_t} \sum_{i=1}^K N_{t,i} d(\theta_{t,i},\theta_i') \\ &= \min_{(i,j)\in\hat{\mathcal{I}}_t\times\hat{\mathcal{I}}_t^c} N_{t,i} d\left(\theta_{t,i}, \frac{N_{t,i}\theta_{t,i} + N_{t,j}\theta_{t,j}}{N_{t,i}+N_{t,j}}\right) + N_{t,j} d\left(\theta_{t,j}, \frac{N_{t,i}\theta_{t,i} + N_{t,j}\theta_{t,j}}{N_{t,i}+N_{t,j}}\right) \\ &= \min_{(i,j)\in\hat{\mathcal{I}}_t\times\hat{\mathcal{I}}_t^c} t\cdot C_{t,i,j}. \end{aligned} \tag{62}$$

Thus, for the fixed-confidence setting, at the end of each round t, the algorithm compute Z_t. If $Z_t > \gamma_{t,\delta}$, then it stops and recommend $\hat{\mathcal{I}}_t$.

G Facts about the exponential family and properties of $C_{i,j}(\cdot)$

We restrict our attention to reward distribution following an one-dimensional natural exponential family. In particular, a natural exponential family has identity natural statistic in canonical form, i.e.,

$$p(y|\eta) = b(y)\exp\{\eta y - A(\eta)\}, \quad \eta \in \mathcal{T} = \left\{\eta : \int b(y)e^{\eta y}\mathrm{d}y < \infty\right\} \tag{63}$$

where η is the natural parameter and $A(\eta)$ is assumed to be twice differentiable. The distribution is called non-singular if $\mathrm{Var}_\eta(Y_{t,i}) > 0$ for all $\eta \in \mathcal{T}^o$, where $\mathcal{T}^o$ denotes the interior of the parameter space $\mathcal{T}$. Denote by θ the mean reward $\theta(\eta) = \int yp(y|\eta)dy$. By the properties of the exponential family, we have, for any $\eta \in \mathcal{T}^o$,

$$\theta(\eta) = \mathbb{E}_\eta[Y_{t,i}] = A'(\eta), \quad \text{and} \quad A''(\eta) = \mathrm{Var}_\eta(Y_{t,i}) > 0.$$

We immediately have the following lemma.

Lemma 35. *At any $\eta \in \mathcal{T}^o$, $\theta(\eta)$ is strictly increasing and $A(\eta)$ is strictly convex.*

Lemma 35 implies that $\theta(\cdot)$ is a one-to-one mapping. Consequently, we can use θ to denote the distribution $p(\cdot|\eta)$ for convenience. Let θ_1 and θ_2 be the mean reward of the distribution $p(\cdot|\eta_1)$ and $p(\cdot|\eta_2)$. Denote by $d(\theta_1, \theta_2)$ the Kullback–Leibler (KL) divergence from $p(\cdot|\eta_1)$ to $p(\cdot|\eta_2)$. We have convenient closed-form expressions of $d(\cdot,\cdot)$ and its partial derivative.

Lemma 36. *The partial derivatives of the KL divergense of a one-dimensional natural exponential family satisfies*

$$d(\theta_1, \theta_2) = A(\eta_2) - A(\eta_1) - \theta_1(\eta_2 - \eta_1), \tag{64}$$

$$\frac{\partial d(\theta_1,\theta_2)}{\partial \theta_1} = \eta_1 - \eta_2, \quad \frac{\partial d(\theta_1,\theta_2)}{\partial \theta_2} = (\theta_2 - \theta_1)\frac{\mathrm{d}\eta(\theta_2)}{\mathrm{d}\theta_2}. \tag{65}$$

G.1 Properties of $C_{i,j}(\cdot)$

Lemma 37. *For any $(i,j) \in \mathcal{I} \times \mathcal{I}^c$, $C_{i,j}(\boldsymbol{p})$ is concave in $\boldsymbol{p}$, strictly increasing in p_i and p_j, and*

$$\frac{\partial C_{i,j}(\boldsymbol{p})}{\partial p_i} = d(\theta_i, \bar{\theta}_{i,j}) \quad \textit{and} \quad \frac{\partial C_{i,j}(\boldsymbol{p})}{\partial p_j} = d(\theta_j, \bar{\theta}_{i,j})$$

where $\bar{\theta}_{i,j} = \bar{\theta}_{i,j}(\boldsymbol{p}) = \frac{p_i\theta_i + p_j\theta_j}{p_i + p_j}$.

Proof. To see the concavity of $C_{i,j}(\boldsymbol{p})$, note that $C_{i,j}(\boldsymbol{p}) = \inf_{\tilde{\theta}}\left\{p_i d(\theta_i, \tilde{\theta}) + p_j d(\theta_j, \tilde{\theta})\right\}$, which is the minimum over a family of linear functions in $\boldsymbol{p}$. This directly implies that $C_{i,j}(\boldsymbol{p})$ is concave in $\boldsymbol{p}$ (See Chapter 3.2 of Boyd and Vandenberghe (2004)).

The partial derivatives of $C_{i,j}(\boldsymbol{p})$ are derived as follows

$$\begin{aligned}
\frac{\partial C_{i,j}(\boldsymbol{p})}{\partial p_i} &= d(\theta_i, \bar{\theta}_{i,j}) + p_i\frac{\partial d(\theta_i,\bar{\theta})}{\partial p_i} + p_j\frac{\partial d(\theta_j,\bar{\theta})}{\partial p_i} \\
&= d(\theta_i, \bar{\theta}_{i,j}) + \frac{\partial \bar{\theta}}{\partial p_i}\cdot\left[p_i\frac{\partial d(\theta_i,\bar{\theta})}{\partial \bar{\theta}} + p_j\frac{\partial d(\theta_j,\bar{\theta})}{\partial \bar{\theta}}\right] \\
&= d(\theta_i, \bar{\theta}_{i,j}).
\end{aligned}$$

The last equality comes from (51). The calculation of $\frac{\partial C_{i,j}(\boldsymbol{p})}{\partial p_j}$ follows similarly. □

Lemma 38. *For any $(i,j) \in \mathcal{I} \times \mathcal{I}^c$, the contour of $C_{i,j}(\cdot)$ does not contain any line segment.*

Proof. To prove the result, we need to basic facts.

1. We first give the Hessian matrix of $C_{i,j}(\boldsymbol{p})$ as follows

$$H_{i,j} = \begin{bmatrix} \frac{\partial^2 C_{i,j}(\boldsymbol{p})}{\partial^2 p_i} & \frac{\partial^2 C_{i,j}(\boldsymbol{p})}{\partial p_i \partial p_j} \\ \frac{\partial^2 C_{i,j}(\boldsymbol{p})}{\partial p_j \partial p_i} & \frac{\partial^2 C_{i,j}(\boldsymbol{p})}{\partial^2 p_j} \end{bmatrix} = \frac{(\theta_i - \theta_j)^2}{(p_i + p_j)^3} \frac{\mathrm{d}\eta(\bar{\theta}_{i,j})}{\mathrm{d}\theta} \begin{bmatrix} -p_j^2 & p_i p_j \\ p_i p_j & -p_i^2 \end{bmatrix},$$

where η is the natural parameter of the exponential family; see Appendix G. By Lemma 35, $\frac{\mathrm{d}\eta(\bar{\theta})}{\mathrm{d}\theta}$ is strictly positive. Note that $H_{i,j}$ has an eigenvalue 0. The other eigenvalue can be calculated correspondingly,

$$\lambda_{\min}(H_{i,j}) = -\frac{p_i^2 + p_j^2}{(p_i + p_j)^3} \frac{\mathrm{d}\eta(\bar{\theta}_{i,j})}{\mathrm{d}\theta} (\theta_i - \theta_j)^2.$$

Note that $\lambda_{\min}(H_{i,j}) < 0$ whenever p_i or p_j is greater than 0.

2. Recall that $C_{i,j}$ satisfies the partial differential equation in (10), which has a general solution in the form of

$$C_{i,j}(\boldsymbol{p}) = p_i g_{i,j}\left(\frac{p_j}{p_i}\right),$$

where $g_{i,j}(\cdot)$ is any differentiable function. This implies that the graph[14] of $C_{i,j}(\boldsymbol{p})$ is the boundary of a cone. In particular, for any point on the surface, the line segment connecting the point and the origin belongs to the graph of $C_{i,j}(\boldsymbol{p})$.

We prove the result by contradiction. Suppose that there exist a line segment in the contour of $C_{i,j}(p_i, p_j)$. Then the triangle specified by the two end points of the line segment and the origin belongs to the graph of $C_{i,j}$. This implies that at any point in the interior of this triangle, $C_{i,j}$ has a Hessian of $\mathbf{0}$. This contradicts with the fact that the Hessian of $C_{i,j}$ at any point (p_i, p_j) such that p_i or p_j is greater than 0 is semi-negative definite, with a eigenvalue strictly smaller than 0. □

Lemma 39. *For any $(i,j) \in \mathcal{I} \times \mathcal{I}^c$, $C_{i,j}(\boldsymbol{p})$ can be written as*

$$C_{i,j}(\boldsymbol{p}) = p_i g_{i,j}\left(\frac{p_j}{p_i}\right),$$

where

$$g_{i,j}(x) = d\left(\theta_i, \frac{\theta_i + x\theta_j}{1+x}\right) + xd\left(\theta_j, \frac{\theta_i + x\theta_j}{1+x}\right).$$

Furthermore, we have

$$g'_{i,j}(x) = d\left(\theta_j, \frac{\theta_i + x\theta_j}{1+x}\right) > 0.$$

Moreover, for any i, j and j' such that $\theta_i > \theta_{j'} \geq \theta_j$, we have

$$g_{i,j}(x) \geq g_{i,j'}(x) \quad \textit{for all } x.$$

Proof. We start by identifying that

$$C_{i,j}(p_i, p_j) = p_i g_{i,j}\left(\frac{p_j}{p_i}\right).$$

Fix p_i and let $x = \frac{p_j}{p_i}$,

$$\frac{\partial C_{i,j}}{\partial p_j} = p_i g'_{i,j}(x) \frac{\partial x}{\partial p_j} = g'_{i,j}(x).$$

By Lemma 37, we have

$$\frac{\partial C_{i,j}}{\partial p_j} = d\left(\theta_j, \frac{p_i\theta_i + p_j\theta_j}{p_i + p_j}\right) = d\left(\theta_j, \frac{\theta_i + x\theta_j}{1+x}\right).$$

[14] The graph of a function $f(x)$ is defined as $G(f) = \{(x, f(x)) : x \in D\}$, where D is the domain of $f(x)$.

Hence,

$$g'_{i,j}(x) = d\left(\theta_j, \frac{\theta_i + x\theta_j}{1+x}\right) > 0.$$

To this end, we fix θ_i and x and define

$$h(\alpha) = d\left(\theta_i, \frac{\theta_i + x\alpha}{1+x}\right) + xd\left(\alpha, \frac{\theta_i + x\alpha}{1+x}\right).$$

It follows that $h(\theta_j) = g_{i,j}(x)$. To complete the proof, we need to show that, for any $\theta_i > \theta_{j'} \geq \theta_j$, $g_{i,j}(x) \geq g_{i,j'}(x)$. It suffices to verify that $h'(\alpha) < 0$ when $\theta_i > \alpha$. To see this, let

$$m = \frac{\theta_i + x\alpha}{1+x}.$$

We recall the partial derivative of $d(\cdot,\cdot)$ in (65) and calculate $h'(\alpha)$ as follows.

$$\begin{aligned}
h'(\alpha) &= \frac{\partial d(\theta_i, m)}{\partial m}\frac{x}{1+x} + x\left[\frac{\partial d(\alpha, m)}{\partial \alpha} + \frac{\partial d(\alpha, m)}{\partial m}\frac{x}{1+x}\right] \\
&= (m-\theta_i)\frac{\mathrm{d}\eta}{\mathrm{d}m}\frac{x}{1+x} + x\left[\eta(\alpha) - \eta(m) + (m-\alpha)\frac{\mathrm{d}\eta}{\mathrm{d}m}\frac{x}{1+x}\right] \\
&= x(\eta(\alpha) - \eta(m)) + ((m-\theta_i) + x(m-\alpha))\frac{\mathrm{d}\eta}{\mathrm{d}m}\frac{x}{1+x} \\
&= x(\eta(\alpha) - \eta(m)).
\end{aligned}$$

Noticing that $\eta'(\theta) = 1/\theta'(\eta) > 0$ and $m - \alpha = (\theta_i - \alpha)/(1+x)$, we have

$$h'(\alpha)\begin{cases} < 0, & \text{if } \theta_i > \alpha, \\ > 0, & \text{if } \theta_i < \alpha. \end{cases} \tag{66}$$

This completes the proof. □

H Existing algorithms

KL-LUCB (Kaufmann and Kalyanakrishnan, 2013). At each round t, the algorithm computes the upper confidence bound $U_i(t)$ and lower confidence bound $L_i(t)$ of the mean reward for each arm i:

$$\begin{aligned} U_i(t) &= \max\{q \in [\theta_{t,i}, \infty) : T_{t,i} d(\theta_{t,i}, q) \leq \beta_{t,\delta}\}, \\ L_i(t) &= \min\{q \in (-\infty, \theta_{t,i}] : T_{t,i} d(\theta_{t,i}, q) \leq \beta_{t,\delta}\}, \end{aligned} \tag{67}$$

where $\beta_{t,\delta}$ is the exploration rate we specify to be $\log((\log t + 1)/\delta)$. Then it samples two critical arms u_t and l_t:

$$u_t = \operatorname*{argmax}_{i \notin \hat{\mathcal{I}}_t} U_i(t) \quad \text{and} \quad l_t = \operatorname*{argmin}_{i \in \hat{\mathcal{I}}_t} L_i(t),$$

The algorithm stops when $U_{u_t}(t) > L_{l_t}(t)$.

UGapE (Gabillon et al., 2012). At each round t, the algorithm first compute the same $U_i(t)$ and $L_i(t)$ as (67). Then it computes the gap index $B_i(t) = \max^k_{i' \neq i} U_{i'}(t) - L_i(t)$ for each arm i, where $\max^k$ denotes the operator returning the k-th largest value. Based on $B_i(t)$, it evaluates a set of k arms: $J(t) = \{j : \operatorname{argmin}^j B_j(t)\}$. From $J(t)$, it computes two critical arms u_t and l_t:

$$u_t = \operatorname*{argmax}_{i \notin J_t} U_i(t) \quad \text{and} \quad l_t = \operatorname*{argmin}_{i \in J_t} L_i(t),$$

and samples one with larger $\beta_{t,\cdot}$. The algorithm stops when $\max_{i \in J(t)} B_i(t) > 0$.

m-LinGapE (Réda et al., 2021a). At each round t, the algorithm first compute the same $U_i(t)$ and $L_i(t)$ as (67). Then it computes the gap index $B_{i,j}(t) = U_i(t) - L_j(t)$ for each arm i. It choose $J(t)$ as the set of k arms with largest empirical means. From $J(t)$, it computes two critical arms u_t and l_t:

$$u_t = \operatorname*{argmax}_{j \in J_t} \max_{i \notin J_t} B_{i,j}(t) \quad \text{and} \quad l_t = \operatorname*{argmax}_{i \notin J_t} B_{i,u_t}(t),$$

and samples one which would reduce the variance of the empirical mean difference between them. (This is so-called "greedy selection rule" in Réda et al. (2021a).) The algorithm stops when

$$B_{l_t,u_t}(t) > 0.$$

We remark that, in Réda et al. (2021a), the authors introduce a meta algorithm called "Gap-Index-Focused" Algorithm (GIFA), which includes KL-LUCB, UGapE and m-LinGapE in its framework. Refer to Algorithm 1 and Table 1 in Réda et al. (2021a).

MisLid (Réda et al., 2021b). The algorithm is designed from the lower bound maximin problem and achieves the asymptotic optimality. It maintains a learner over the probability simplex of number of arms. At each round t, it computes a gain function and updates the learner. Then it perform tracking-rule to sample an arm. We adopt AdaHedge as the learner and use empirical gain function g_t as

$$\operatorname*{argmin}_{\boldsymbol{\lambda} \in \mathrm{Alt}(\boldsymbol{\mu}^t)} \sum_{i=1}^{K} \boldsymbol{p}_i^t (\boldsymbol{\mu}_i^t - \boldsymbol{\lambda})^2$$

where $\boldsymbol{p}^t$ is probability distribution returned by AdaHedge, $\boldsymbol{\mu}_t$ is the empirical mean estimate and $\mathrm{Alt}(\boldsymbol{\mu}^t)$ defines the set of bandit model returning the top-k arm set different from $(\boldsymbol{\mu}^t)$. As suggested by Réda et al. (2021b), the empirical gain function reduced the sample complexity practically compared with the optimistic gain function suggested by theory. For the tracking rule, we adopt C-Tracking in our experiments. For the stopping rule, we adopt the same General Likelihood Ratio Test statistic with threshold $\log((\log(t) + 1)/\delta)$.

FWS (Wang et al., 2021). The algorithm track the lower bound maximin problem. At each round t, it performs forced exploration or Frank-Wolfe update. Due to the non-smoothness of objective function, it construct an r_t-sub-differential subspaces and then solve a zero-sum game. Note r_t is a tuning parameter. We choose $r_t = t^{-0.9}/K$ as suggested in Wang et al. (2021). For the stopping rule, we adopt the same General Likelihood Ratio Test statistic with threshold $\log\left((\log(t)+1)/\delta\right)$.

Lazy-Mirror-Ascent (Ménard, 2019). The algorithm treats the lower bound problem as

$$\max_{\boldsymbol{w}} F(\boldsymbol{w}).$$

It uses a sampling rule based on an online lazy mirror ascent. At each round t, it tracks the allocation weights $\boldsymbol{w}(t)$ and updates it by

$$\tilde{\boldsymbol{w}}(t+1) = \operatorname*{argmax}_{\boldsymbol{w}} \eta_{t+1} \sum_{s=K}^{t} \boldsymbol{w} \cdot \nabla F(\tilde{\boldsymbol{w}}(s); \boldsymbol{\mu}(s)) - \mathrm{kl}(\boldsymbol{w}, \boldsymbol{\pi});$$

$$\boldsymbol{w}(t+1) = (1-\gamma_t)\tilde{\boldsymbol{w}}(t+1) + \gamma_t \boldsymbol{\pi},$$

where $\boldsymbol{\mu}(s)$ is the empirical mean estimates and $\boldsymbol{\pi}$ is the uniform distribution. There exists two tuning parameter η_t and γ_t. Based on the code released by Wouter M. Koolen[15] and the suggestion in Ménard (2019), we choose $\eta_t = \sqrt{\log K/t}/F^*$ and $\gamma_t = 1/(4\sqrt{t})$. Wouter M. Koolen comments that, although setting F^* relies on the knowledge of true arm means, it is critical for the performance. In appendix of Wang et al. (2021), the authors also comment that this method is very sensitive to the learning rate η_t, the current choice is only for experimental comparison and cannot be used in real-world scenarios.

[15] https://bitbucket.org/wmkoolen/tidnabbil/src/master/